

TPS/CLM

OPC Client for Third-Party Devices

**TPS/
Communication
Link Module**

OPC Client for Third-Party Devices ***User/Configuration Manual***

**LK12-440
Release Independent
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About This Publication

This document is the User's manual for configuring the interface between the APP/CLM (Communication Link Module) using a OPC (OLE for Process Control) Client Application and one or more third-party OPC Servers.

Important! With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32 bit and 64 bit variants). Note that CLM R200.1 is only meant for Windows 2008 nodes (Experion PKS R400/R410). For more information, see section Windows 2008 - Setup and Installation.

The APP/CLM provides a standard communications platform, which allows the TPS process network (LCN) to interface with a variety of devices and networks. Using these interfaces, an operator at a TPS user station (GUS, US) and ES-T has access to data located on a third party network, in this case by using the OPC Client/Server relationship. Access to devices via OPC is transparent to the operator since the points appear to be directly connected to TPS Process Network.

This document supplements the *CLM Platform Implementation Manual* which covers the main communications platform.

This document is primarily intended for Engineers who will configure the interface between a APP/CLM using the OPC Client application and a third-party OPC Server package.

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Acronyms

AM	TPS Application Module
APP	Application Processing Platform
BIF	Builder Input File
CDS	Custom Data Segment
CL	Control Language
CLM	Communication Link Module
CLMI	Communication Link Module Interface (board)
COM	Component Object Module
DCOM	Distributed Component Object Module
DEB	Data Entity Builder
EAPP	Experion APP
EB	Exception Build
EC	Execute Command (file)
GUS	Global User Station
HM	TPS History Module
LCN	TPS Local Control Network
IE	MicroSoft Internet Explorer
NCF	Network Configuration File
NDT	Network Definition Table
OPC	OLE for Process Control
PIT	Point Interface Table
PITE	Point Interface Table Entry
PPS	Process Point Special
STSDF	Status Definition File
TDC	Total Distributed Control
TDF	Template Definition File
TPN	TPS Network (LCN)
TPS	TotalPlant Solution System
US	TPS Universal Station
ZC	File Extension – CLM Configuration File
ZI	File Extension – Builder Input File
ZS	File Extension – Detail Status Display File
ZT	File Extension – Template definition File

Section 1 – Introduction

1.1 Overview

Introduction

This manual describes how to access data from remote networks and/or devices using OPC Group and Data Items from a third-party OPC Server connection to APP Based CLM.

The OPC Client for Third-Party Devices (hereafter known as *the Client*) communicates with OPC Servers that support the *OPC Data Access Custom Interface Specification, 2.0*.

The Client can connect to OPC Servers provided by one or more vendors. Each vendor of an OPC Server determines the devices and data to which its server has access, along with the data names and the details about how the server physically accesses that data. The Client is Windows 2003 and Windows 2008 based and installs to the Honeywell Application Processing Platform (APP node using the standard APP Base Software (AM personality) and communicates to third-party networks and servers by a virtual interface board called the CLMI.

Note: With CLM R200.1, CLM is supported on Experion APP (Windows 2008 platform on both 32-bit and 64 bit variants). It does not work on Windows 2003.

LCN Working Knowledge Required

This manual assumes a working knowledge level of proficiency on the LCN and the APP and AM in particular.

What this manual does

This manual describes the methods of installation and configuration for the Client. It describes how to define and configure the OPC network devices and map groups and data items to image point parameters on the TPS and Experion process network.

What this manual does not do

This manual does not document how to write specific process applications using the data retrieved in the Data Items. Nor does it document how to install or configure the third-party OPC Server. Refer to the *CLM Platform Implementation Manual* for any additional information that is required to complete a CLM OPC Interface implementation. Refer to the appropriate vendor manuals for process application information.

Continued on next page

1.1 Overview, Continued

About the OPC Data Access Interface

An OPC Server is comprised of several objects: the server, the group, and the item. The OPC Server object maintains information about the server and serves as a container for OPC Group objects.

The OPC Group object maintains information about itself and provides the mechanism for containing and logically organizing OPC Items. The OPC Group also provides a way for clients to organize data. For example, the group might represent items in a particular operator display or report.

The client can define one or more OPC Items within each Group. The OPC Items represent connections to data sources. Note that the items are not the data sources - they are just connections to them. For example, the tags in a DCS system exist regardless of whether an OPC Client is currently accessing them. The OPC Item should be thought of as simply specifying the address of the data, not as the actual physical source of the data. Associated with each item is a Value, Quality, and Time Stamp. The Value is in the form of a Microsoft VARIANT, the Quality is current status of the value and the Time Stamp contains the time the OPC Server received the value.

Relationship between CLM Objects and OPC Objects

OPC Object	CLM Object	Comments
OPC Server	Port	This assignment is made both from the CLM page of the TPS Configuration Utility and by creating a Port Network Data Table (NDT) entry through the online CLM Builder.
OPC Group	PIT	This assignment is made by creating a Point Interface Table (PIT) entry through the online CLM Builder. When a device that contains this PIT is placed In –Service by a user, an OPC group of the same name is created in the server object.
OPC Item	PITEP	This assignment is made by parameter definition (PITEP) within the template file (.ZT) that is used to build Point Interface Table Entries (PITEs) through the online CLM Builder. When a device associated with this PITE is placed In-Service, an OPC data item is created in the server object for each PITEP defined.

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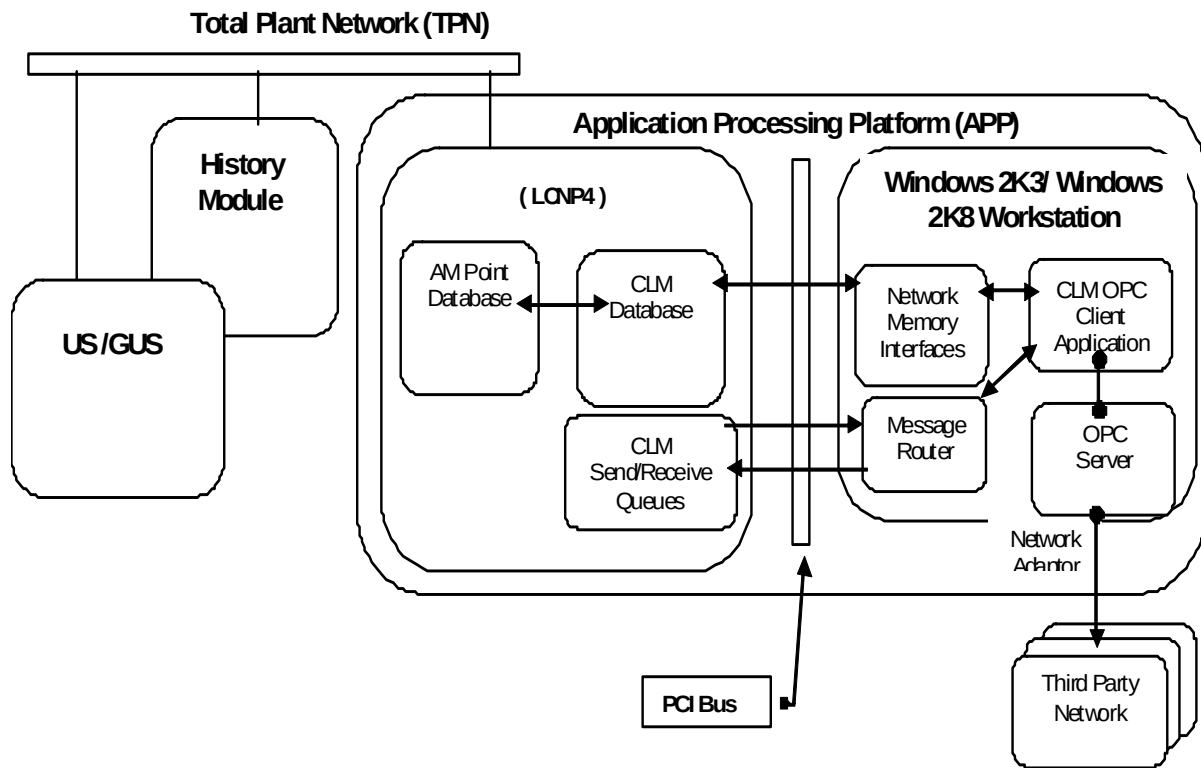
1.1 Overview, Continued

Intended audience	This document is primarily intended for engineers responsible for designing and implementing the Client connection to a third-party OPC Server.
Prerequisite knowledge	Since this manual supplements the <i>CLM Platform Implementation Manual</i>, you should become familiar with that manual, with emphasis on the following sections: • Building CLM Interface Structures - Section 5 • Defining Interface Structure Templates - Section 6 • Network Interface Design Considerations - Section 7 In addition, you should be familiar with the OPC Server and the third party network to which this application will interface.
APP/CLM is a modified APP	The APP/CLM is a modified Application Processing Platform (APP) that provides connection to third party networks from the Windows 2003 and 2008 environment. Multiple communication interfaces are supported.
AM point defines image point	The Client provides the TPS and the Experion operator access to information obtained from the third-party OPC Server. The APP/CLM uses Application Module (AM) image points, taking advantage of Custom parameters where beneficial to image data retrieved from the OPC item objects.
Parameter changes by operator	The operator can monitor and change the values of data items in the third-party network by changing the associated parameters in the AM image point. Figure 1.1-1 depicts the relationship between a TPS/Experion System, OPC Server, and devices on a third-party network.
Physical connection	The Client is connected to the third-party OPC Server using the Microsoft Component Object Model (COM).

Continued on next page

1.1 Overview, Continued

Figure 1.1-1 Overview of the Client Interface to OPC Servers



Section 2 – Client Installation

2.1 Overview

Introduction

This section covers the process to install the Client. There are two parts to the installation, the Windows side and the LCN side.

STOP!

Installation of the APP/CLM Platform is a prerequisite to installing this application. Do not continue past this point until the APP/CLM Platform package installation is completed!

2.2 Client Package Contents

Client package items - general

The OPC Client Application includes the general items listed below. In addition, the provided media includes setup and installation files.

- *TPS/CLM OPC Client Application Manual* (this document)
- Communication Protocol Software – Shared Library Module (CLMOPC.DLL)
- Control Language (CL) Object Programs to support device or network specific requirements
- Custom Data Segment (CDS) Source for custom AM/CLM image points
- Custom Display Package for device or network specific functions
- EC files to Unlink/Link CL Blocks to an image point
- Template files for the OPC supported NDT layer
- PITE template file for each image point type
- Prototype Builder Input Files (BIF) for each template type
- Windows 2003 or 2008 installation executable for OPC Client application
- AM/CLM emulated disk file for OPC Client application

Table 2.2-1

The release media contains these specific files, as listed below.

CLM OPC install.exe	Installation script file for the Windows side of the Client application.
clm_opc_install.msi	Installer package for the Windows 2008 side of the Client application
CLM OPC.lcn	GUS (emulated disk) format file containing all LCN-based Client files
O_AI.ZI	Analog Input template form for Builder Input File
O_AO.ZI	Analog Output template form for Builder Input File
O_D31.ZI	Digital Composite 3-State (1 out) template form for Builder Input File
O_D32.ZI	Digital Composite 3-State (2 out) template form for Builder Input File
O_D41.ZI	Digital Composite 4-State (1 out) template form for Builder Input File
O_D42.ZI	Digital Composite 4-State (2 out) template form for Builder Input File
O_D5.ZI	Digital Composite 5-State (3 out) template form for Builder Input File
O_DEVICE.ZI	OPC network device template form for Builder Input File
O_DIN3.ZI	Digital Input 3-State (1 input) template form for Builder Input File
O_DIN4.ZI	Digital Input 4-State (2 input) template form for Builder Input File
O_FLAG.ZI	Flag point template form for Builder Input File
O_NUMER.ZI	Numeric point template form for Builder Input File
O_PORT.ZI	Client Instance Assignment (Port) template form for Builder Input File
O_AI.ZT	Analog Input Template Definition File (TDF)
O_AO.ZT	Analog Output Template Definition File (TDF)

Continued on next page

2.2 Release Media Contents, Continued

Table 2.2-1, continued

O_D31.ZT	Digital Composite 3-State (1 out) Template Definition File (TDF)
O_D32.ZT	Digital Composite 3-State (2 out) Template Definition File (TDF)
O_D41.ZT	Digital Composite 4-State (1 out) Template Definition File (TDF)
O_D42.ZT	Digital Composite 4-State (2 out) Template Definition File (TDF)
O_D5.ZT	Digital Composite 5-State (3 out) Template Definition File (TDF)
O_DEVICE.ZT	OPC Network Device Template Definition File (TDF)
O_DIN3.ZT	Digital Input 3-State (1 input) Template Definition File (TDF)
O_DIN4.ZT	Digital Input 4-State (2 input) Template Definition File (TDF)
O_FLAG.ZT	Flag point Template Definition File (TDF)
O_NUMER.ZT	Numeric point Template Definition File (TDF)
O_PORT.ZT	Client Instance Assignment (Port) Template Definition File (TDF)
O_AI.CL	Analog Input CDS Package
O_AO.CL	Analog Output CDS Package
O_CLMI.ZC	Client Blank Configuration File for APP/CLM
CLMSTAT2.DS	OPC Client User Interface Display Source File
CLMDETD1.DS	OPC Client User Interface Display Source File
CLMDETD2.DS	OPC Client User Interface Display Source File
CLMDETD3.DS	OPC Client User Interface Display Source File
CLMDETD4.DS	OPC Client User Interface Display Source File
OPCSBUF1.DS	OPC Client User Interface Display Source File
OPCSBUF2.DS	OPC Client User Interface Display Source File
OPCSBUF2.DS	OPC Client User Interface Display Source File
OPCSBUF3.DS	OPC Client User Interface Display Source File
OPCSBUF4.DS	OPC Client User Interface Display Source File
OPCPIT1.DS	OPC Client User Interface Display Source File
OPCPIT2.DS	OPC Client User Interface Display Source File
OPCPIT3.DS	OPC Client User Interface Display Source File
OPCPIT4.DS	OPC Client User Interface Display Source File
OPCLIST1.DS	OPC Client User Interface Display Source File
OPCLIST2.DS	OPC Client User Interface Display Source File
OPCLIST3.DS	OPC Client User Interface Display Source File
OPCLIST4.DS	OPC Client User Interface Display Source File
OPCPITE1.DS	OPC Client User Interface Display Source File
OPCPITE2.DS	OPC Client User Interface Display Source File
OPCPITE3.DS	OPC Client User Interface Display Source File
OPCPITE4.DS	OPC Client User Interface Display Source File
TPSCLM OPC CLIENT USER MANUAL	This User/Configuration Manual.

2.3 Windows 2003 - Setup and Installation

Important! This section is only applicable for users who will continue to use the TPS node on Windows 2003 operating system.

Internet Explorer Required!

The Client requires certain key files and components that are installed to the Windows 2003 node during the installation of Microsoft Internet Explorer. The minimum version is IE 5.5. Installation of the Client application will fail if Internet Explorer is not installed.

Shutdown TDC Emulators Service

In the Windows 2003 environment, the Honeywell TDC Emulators Service emulates the peripheral boards that exist in a traditional TPN node chassis. In the APP/CLM, this service emulates a virtual communications board called the CLMI. Before installation of the Client application may proceed, this emulation service must be stopped.

Step	Action
2.3.1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2.3.2	From the Start menu select Settings/Control Panel/Administrative Tools.
2.3.3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 2.3-1 below.
2.3.4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Stop button.
2.3.5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message “<i>Could not stop TDC Emulators Service</i>”. When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcaserver.exe process is no longer in the active process list.

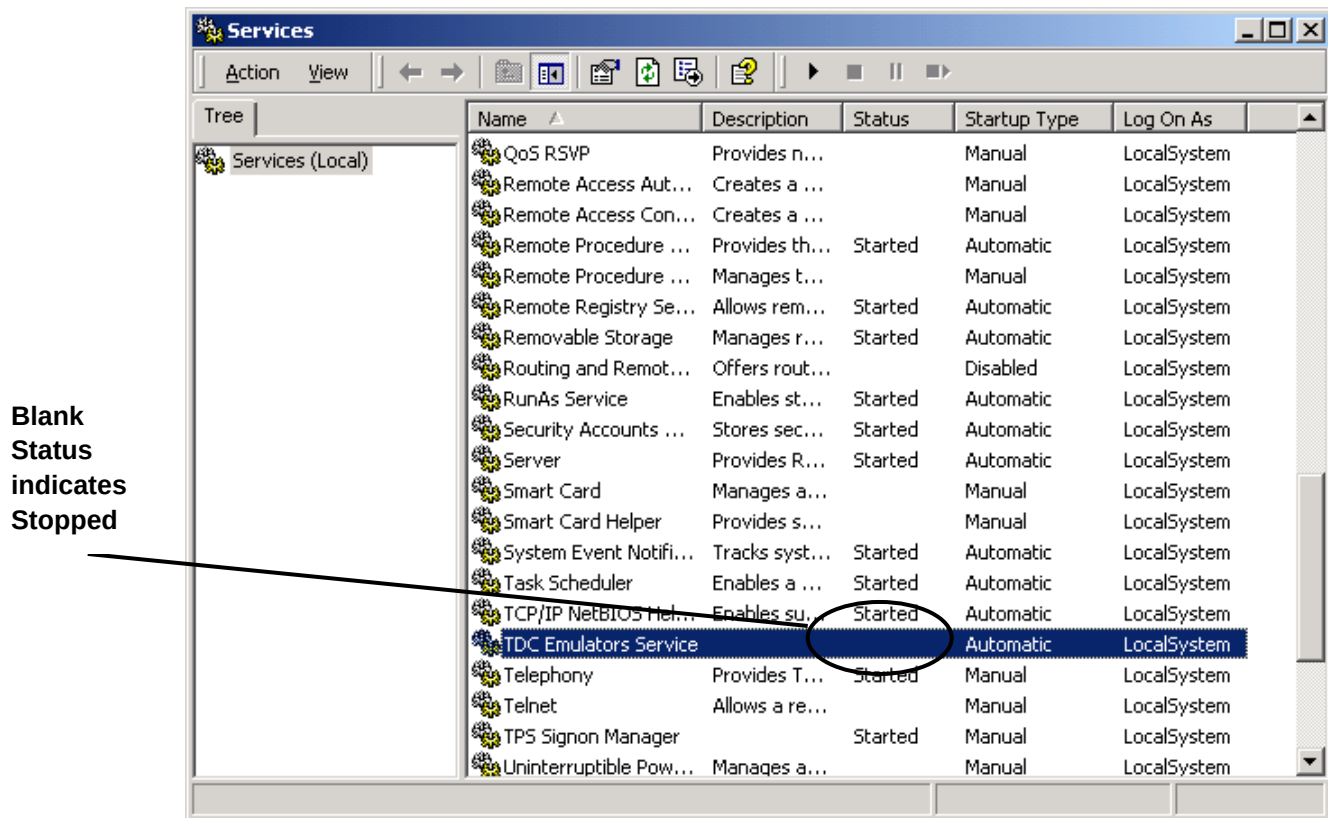


Figure 2.3-1 TDC Emulators Service Stopped

Install OPC Client Application

Now we are ready to install the Client application to the TPS APP (Windows side of the APP/CLM).

Step	Action
2.3.6	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2.3.7	Insert the Windows 2003 CLM OPC media distribution into the CD drive.
2.3.8	View the media content in the drive using Windows Explorer. Launch the installation program by double-clicking the install file 'CLM OPC INSTALL.EXE' on the release media in directory TPS.
2.3.9	Follow the instructions for installation. Note: The actual Client connection to the OPC Server is not established until the virtual Port for the Client is configured and placed In-Service. Uninstall: The install script completes by copying an executable file to the APP/CLM hard drive to be used if an uninstall is required. The file is CLM OPC Uninstall.exe and is located at \Program Files\Honeywell\TPS\CLM (or the directory where the Platform install file was stored). See Appendix E-1 for Instructions on doing an Uninstall.

2.3 Windows 2003 - Setup and Installation, Continued

Start TDC Emulators Service

Before the AM personality can be loaded on the Native Window side, the TDC Emulators Service must be started.

Step	Action
2.3.10	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2.3.11	From the Start menu select Settings/Control Panel/Administrative Tools.
2.3.12	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 2.3-1 previously.

Note: The Startup option on the TDC Emulators Service should be set to Automatic so that it will start up whenever the APP/CLM node is restarted.

Step	Action
2.3.13	Select the TDC Emulators Service entry.
2.3.14	From the drop-down menu, select Action, then Properties.
2.3.15	If the Startup Type is not Automatic, then select the drop-down menu and change it to Automatic. See Figure 2.3-2 below.
2.3.16	Select OK to accept the changes.

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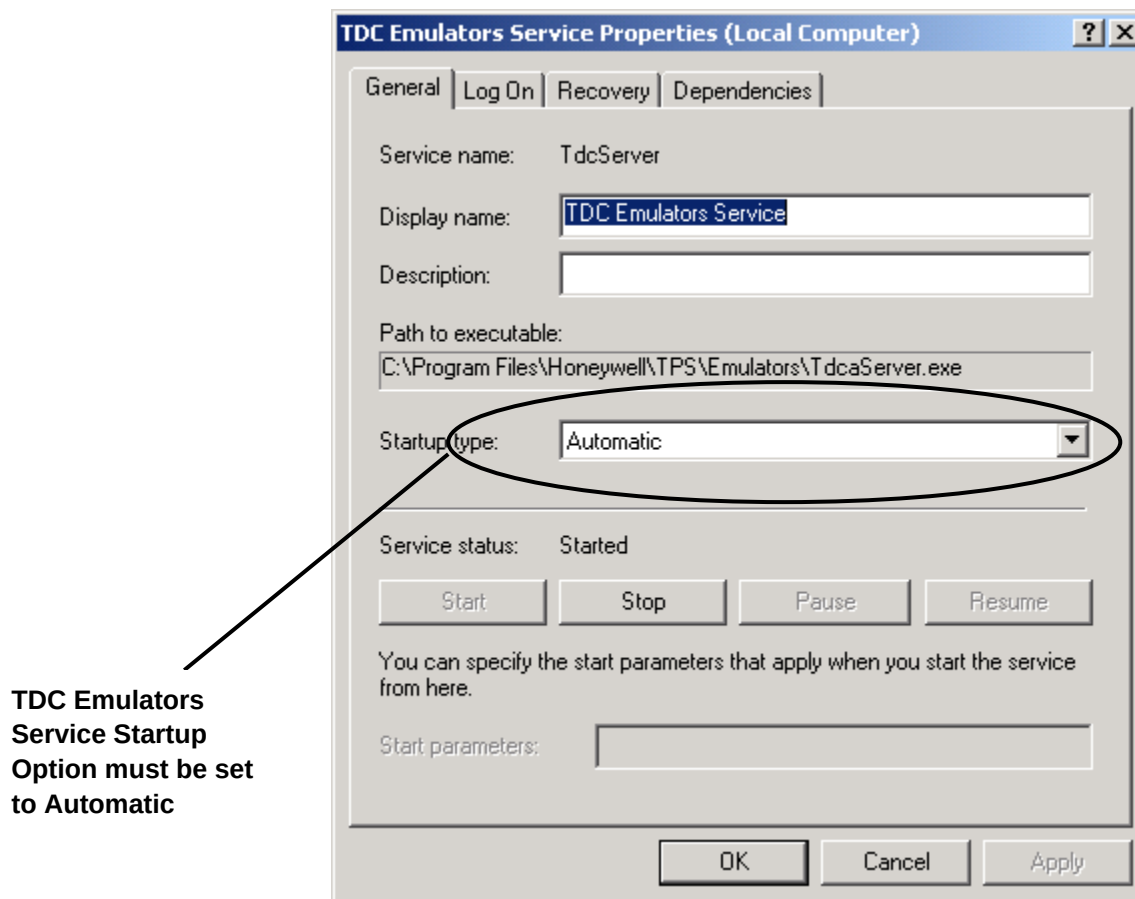


Figure 2.3-2 Automatic Startup option on TDC Emulators Service

2.3.17	If the Status for TDC Emulators Service is Stopped , select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Start button.
2.3.18	A message box appears indicating that the service is being started. This will take about 10-15 seconds. The Services display should appear as in Figure 2.3-3 below.

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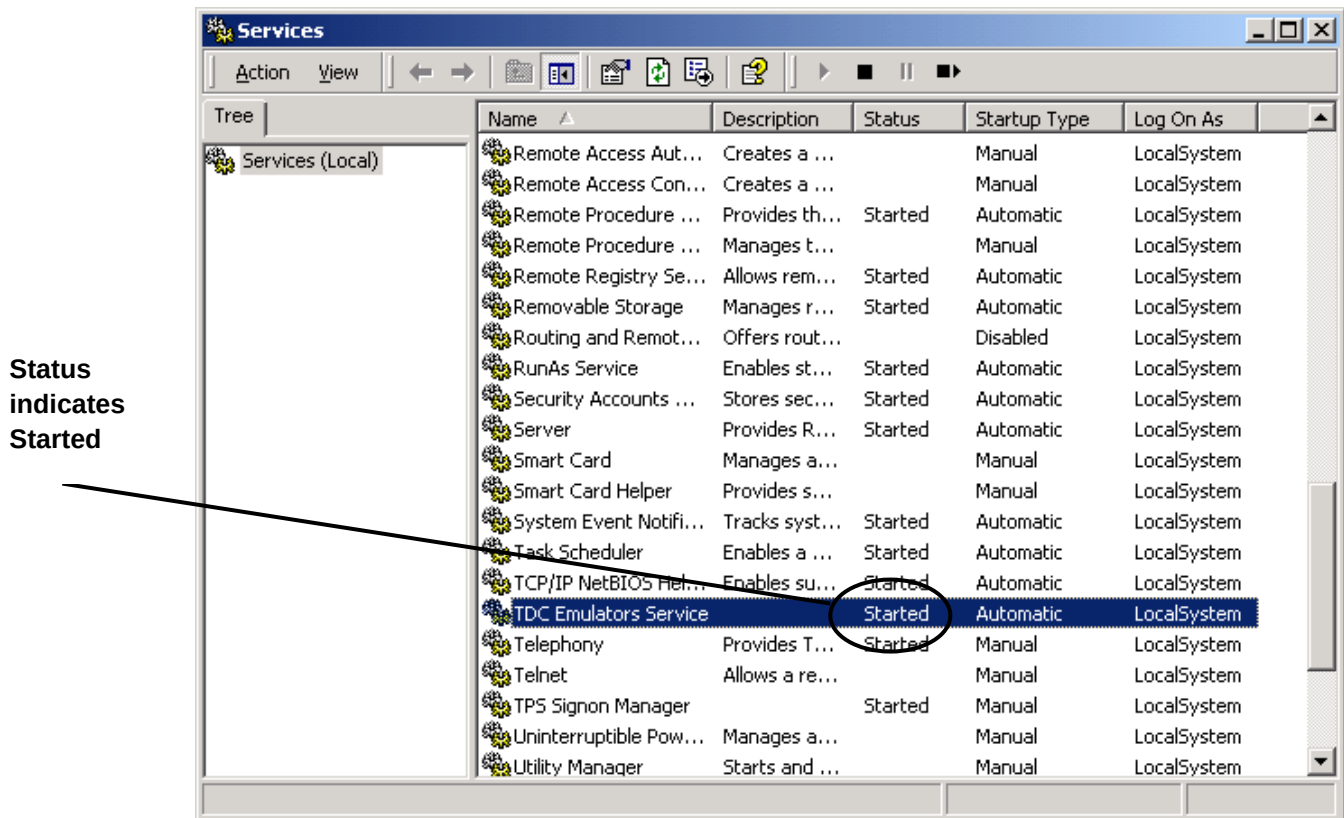


Figure 2.3-3 TDC Emulators Service Started

The next step

This concludes part 1, the installation to the Windows 2003 side of the APP/CLM.

Note:

- Verify if the firewall is turned OFF in both the Client and Remote Server node for CLM configuration.
- If the firewall is ON, ensure that an Exception has been raised in the exception list.

The next step is to restart the TPN side of the APP/CLM node. Continue with Section 2.5 to restart the APP node and load the AM personality.

2.4 Windows 2008 - Setup and Installation

Important! This section is only applicable for users who will be using the Experion node on Windows 2008 operating system (32-bit and 64-bit variants).

Shutdown TDC Emulators Service

In the Windows 2008 environment, the Honeywell TDC Emulators Service emulates the peripheral boards that exist in a traditional TPN node chassis. In the APP/CLM, this service emulates a virtual communications board called the CLMI. Before installation of the Client application may proceed, this emulation service must be stopped.

Step	Action
2.4.1	On the Experion APP Windows side of the APP/CLM), log onto Windows as Administrator.
2.4.2	From the Start menu select Control Panel/ System and Security/Administrative Tools.
2.4.3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 2.4-1 below.
2.4.4	If the Status for TDC Emulators Service is Started, select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Stop button.
2.4.5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message “<i>Could not stop TDC Emulators Service service</i>”. When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt, and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcaserver.exe process is no longer in the active process list.

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2.4 Windows 2008 – Setup and Installation, Continued

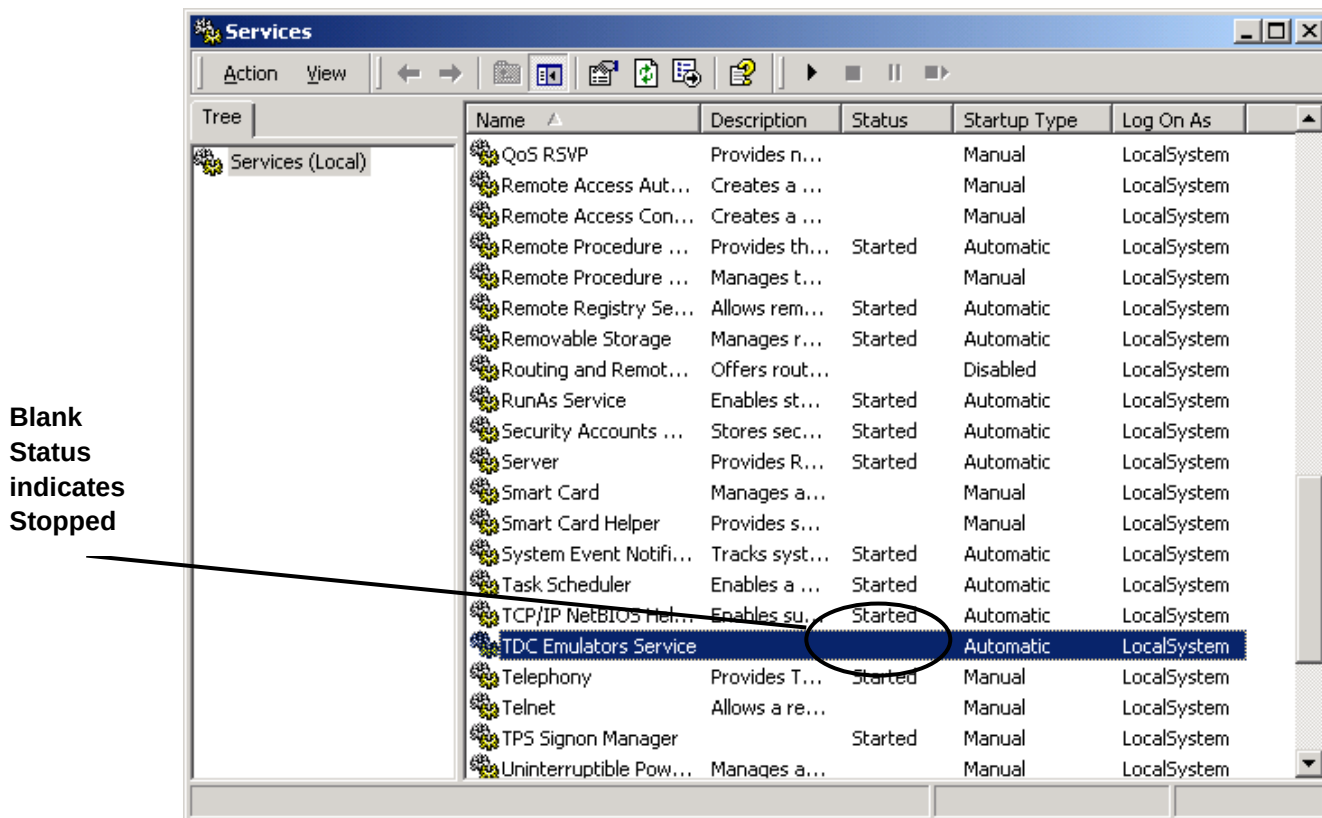


Figure 2.4-1 TDC Emulators Service Stopped

Install OPC Client Application

Now we are ready to install the Client application to the Experion APP (Windows side of the APP/CLM) of the APP/CLM.

Step	Action
2.4.6	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2.4.7	Insert the Windows 2008 CLM OPC media distribution into the CD drive.
2.4.8	View the media content in the drive using Windows Explorer. Launch the installation program by double-clicking the install file 'CLM_OPC_INSTALL.MSI' on the release media in directory TPS.
2.4.9	Follow the instructions for installation. Note: The actual Client connection to the OPC Server is not established until the virtual Port for the Client is configured and placed In-Service. Uninstall: You can uninstall the software with the help of the Add or Change a program in the Control Panel. See Appendix E-2 for more information on Uninstallation.

Continued on next page

2.4 Windows 2008 - Setup and Installation, Continued

Start TDC Emulators Service

Before the AM personality can be loaded on the TPN side of the APP, the TDC Emulators Service must be started.

Step	Action
2.4.10	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
2.4.11	From the Start menu select Settings/Control Panel/Administrative Tools.
2.4.12	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure 2.4-1 previously.

Note: The Startup option on the TDC Emulators Service should be set to Automatic so that it will start up whenever the APP/CLM node is restarted.

Step	Action
2.4.13	Select the TDC Emulators Service entry.
2.4.14	From the drop-down menu, select Action, then Properties.
2.4.15	If the Startup Type is not Automatic, then select the drop-down menu and change it to Automatic. See Figure 2.4-2 below.
2.4.16	Select OK to accept the changes.

Continued on next page

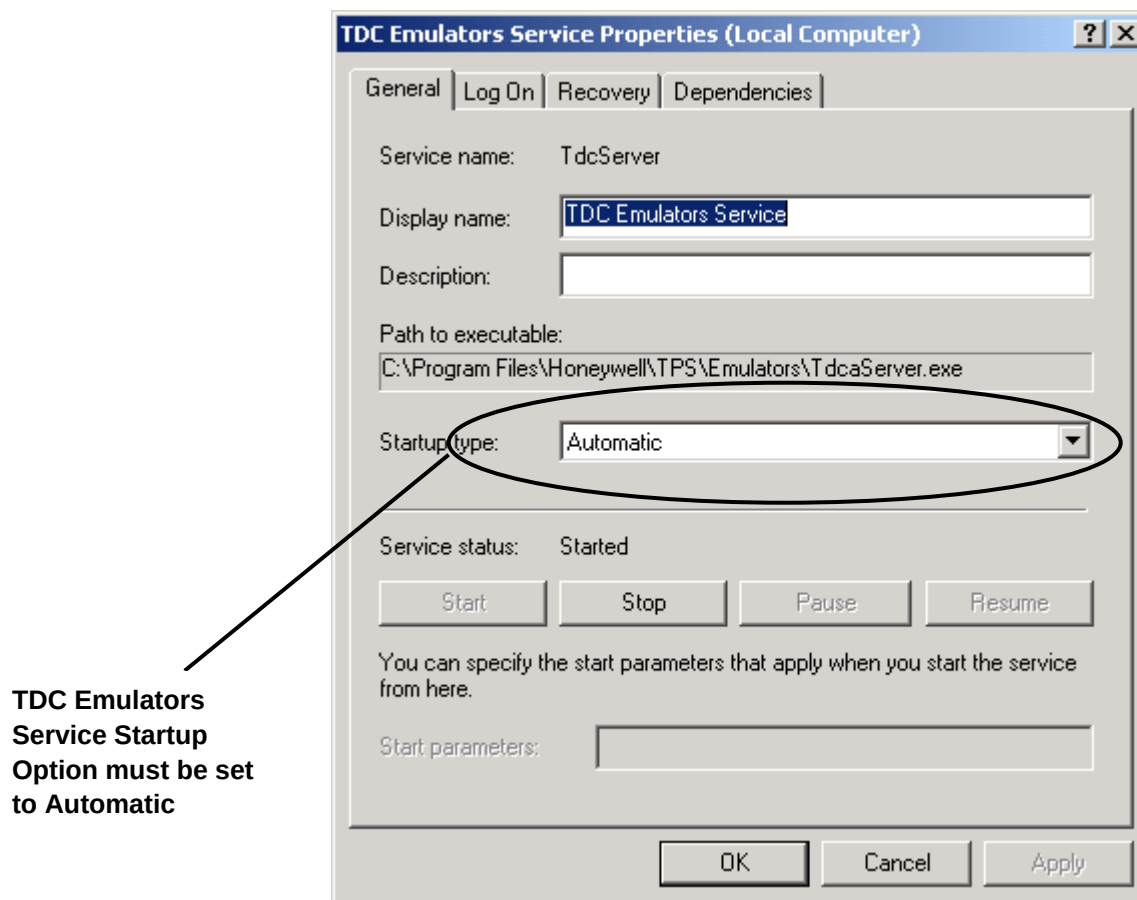


Figure 2.4-2 Automatic Startup option on TDC Emulators Service

2.4.17	If the Status for TDC Emulators Service is Stopped , select the TDC Emulators Service entry. From the drop-down menu, select Action , then the Start button.
2.4.18	A message box appears indicating that the service is being started. This will take about 10-15 seconds. The Services display should appear as in Figure 2.4-3 below.

Continued on next page

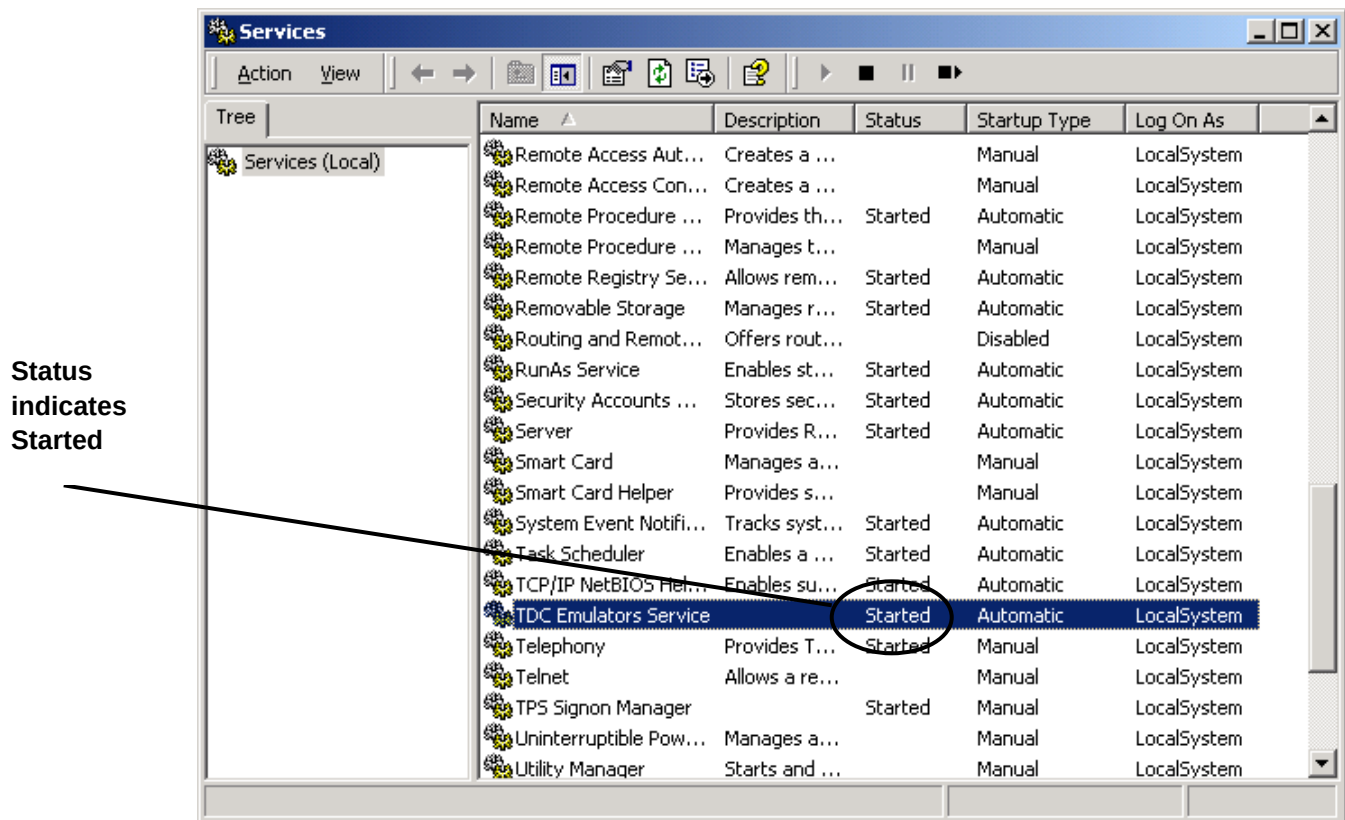


Figure 2.4-3 TDC Emulators Service Started

The next step

This concludes part 1, the installation to the Windows 2008 side of the APP/CLM.

Note:

- Verify if the firewall is turned OFF in both the Client and Remote Server node for CLM configuration.
- If the firewall is ON, ensure that an Exception has been raised in the exception list.

The next step is to restart the TPN side of the APP. Continue with section 2.5 to restart the APP node and load the AM personality.

2.5 Restart the APP and Load the AM Personality

Overview

At this point, the PC should be restarted and the AM personality loaded. The CLM Platform software that hosts the OPC Client application was installed previously and is loaded as part of the AM personality.

Step	Action
2.5.1	From the Application Module Status Display, if the CLM node is online, do a SHUTDOWN.
2.5.2	Restart the APP node PC.
2.5.3	At the APP, log on to Windows as Administrator.
2.5.4	From the START menu, select PROGRAMS and HONEYWELL TPS. Note: In Windows 2008, select Programs, Honeywell Experion PKS, System Management, and Configuration Utility from the Start menu.
2.5.5	Select LCNP STATUS.
2.5.6	Select the RESET LCNP and Yes to reset the LCNP card. See Figure 2.4-1.

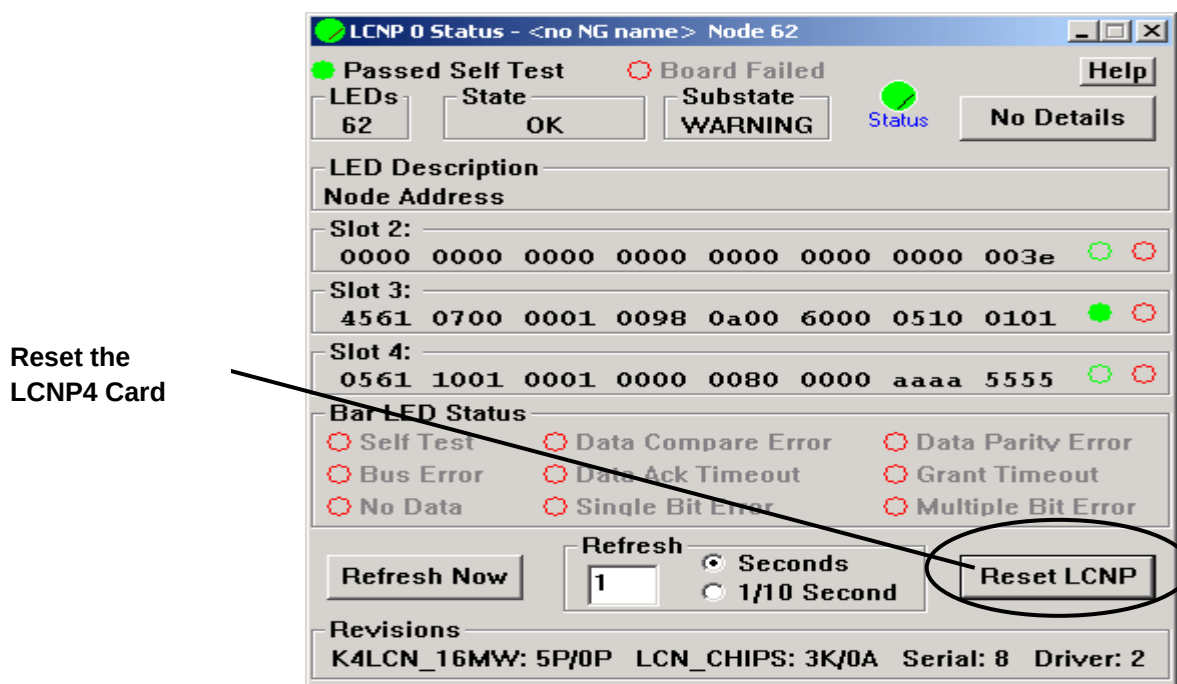


Figure 2.5-1 LCNP Status Display

Continued on next page

2.5 Restart the APP and Load the CLM Personality, Continued

Step	Action
2.5.7	Invoke the AM Status Detail Display from a GUS or US.
2.5.8	When the PWR_ON state for the CLM node is indicated on the display after the reset, use the AUTOLOAD NET function to load the AM personality.
2.5.9	Confirm that the AM/CLM loads successfully and does indicate that the node type shows as a CLM on the display.

The next step

The next step is to install the Client files to the LCN side system media. Continue on with the next Section 2.5 of the installation process.

2.6 TPN Process Network (LCN) – Setup and Installation

Overview

This section details the installation procedure that applies to the files required by the LCN side of the OPC Client application. This includes template files, CDS packages, Exception Build files (.EB) and others required to map image point parameters to the Server data items.

Step	Action
2.6.1	Make a copy of the APP/CLM OPC Client for Third-Party Devices release media.
2.6.2	From the SET PATHS function of the Engineering personality, verify and set the following paths on your History Module or system media: • CL SOURCE/OBJ - UCLM suggested. • CL CUSTOM GDF - CDSG • CL PARAM LIST - same path as the CL SOURCE/OBJ. • USER DEFLT PATH - same as the CL SOURCE/OBJ
2.6.3	Review the files in the product release which will be copied to your system to assure no conflicts (duplicate names) in naming with files already on your system. This is especially critical for CDS files, which may be overwritten.

Important!

Read the *Customer Release Guide* supplied with this package for important information that applies to your installation, including HM sizing requirements, special instructions, and reinstallation information.

Continued on next page

2.6 TPN Process Network (LCN) – Setup and Installation,

Continued

Write Files to LCN

Step	Action
2.6.4	At a GUS station, insert the OPC Client release CD.
2.6.5	From the Native Window Pull-Down menu, select Access and Mount/Dismount Emulated Disks... and follow the mount procedure to make the OPC files accessible to the LCN. The CLM OPC Client LCN emulated file, opc.lcn, is contained on the release disk in directory lcn.
2.6.6	Determine the logical device id of the emulated disk where the CD is mounted. (Hint: Use the Console Status Display)
2.6.7	From the Command function within the Engineering Personality, enter the following command to begin the install procedure: <pre>EC \$Fn>&OPC>O_INST.EC \$Fn</pre> where: \$Fn = the logical device id (e.g., \$F1) containing the OPC Client release CD
2.6.8	This action will kick off an executable command script that will copy files from the release media to the destination system media, usually the History Module. The user will be prompted for destinations on some files, while others must reside in specific directories and will be placed there with no prompting. Caution: The EC script contains commands to compile CDS packages. You should carefully review the parameter names and types of these packages to avoid conflicts with existing parameters on your system. If you are unsure if there will be conflicts, then you should select the option to “compile the CDS at a later time”. When you are ready to compile them, refer to Table 2.2-1 for a list of all of the CDS files (*.CL). Follow the prompts output by the install execution and enter the required information to continue through the procedure. A log file is generated, O_INST.LS, that contains the results of all actions and responses from the EC script. This file must be reviewed for errors. Stop! If any errors exist in O_INST.LS, they must be corrected before beginning the configuration process! If you have delayed the compile of the CDS packages, they must be examined and compiled before beginning the configuration process!

2.6 TPN Process Network (LCN) – Setup and Installation,

Continued

Compile OPC Operational Display Set

Step	Action
2.6.9	At a GUS station and the Engineering Main Menu, invoke the Picture Editor function.

Note: The remainder of the steps assume the UCLM directory on the History Module as the location for the display source. If you have copied the display source files to a different directory location, use that directory instead of UCLM in the directions and commands listed following.

2.6.10	On the command line, load the CLM display data base file (from the CLM Platform installation) by typing: LOAD NET>UCLM>CLMDDB
2.6.11	On the command line, invoke the Multiple Picture Compiler by using the provided selection file. To execute, type: MCOMP NET>UCLM>OPC_DS All OPC displays are listed here for reference: CLMSTAT2 OPCSBUF1 OPCSBUF2 OPCSBUF3 OPCSBUF4 OPCPIT1 OPCPIT2 OPCPIT3 OPCPIT4 OPCLIST1 OPCLIST2 OPCLIST3 OPCLIST4 OPCPITE1 OPCPITE2 OPCPITE3 OPCPITE4 CLMDETD1 CLMDETD2 CLMDETD3 CLMDETD4
2.6.12	Check the OPC_DS.ER file for errors when the function completes.

Client Installation complete – the next step

The installation of the OPC Client application for the APP/CLM is now complete!

Server information will be required during configuration in the following sections. If you have not already done so, install the OPC Server(s) that you will be connecting.

When you have installed the required OPC Server(s), continue on to Section 3 to begin Client configuration.

Section 3 – OPC Client Configuration

3.1 Overview

Introduction

This section describes how the Client actually defines the server structure. You will assign the Client to one of two virtual interface boards (CLMI), establish the Server objects, define remote devices, define group names and establish naming conventions for group and data items.

Note: Data items will not be defined and mapped until section 4 when the image points are built.

Getting Started

Review and understand the following information.

- *CLM Platform Implementation Manual*
- OPC Server Application Manuals (from OPC Server Vendor)
- An overview diagram of your OPC installation: Identify the address of each OPC-based device in your installation.
- A listing of the OPC Data Items to be imaged by AM points.

Introduction

All CLM interface structures are defined using the CLM Configuration Builder, which is part of the standard CLM Platform software. Elements used by the OPC Client include:

- Network Definition Table (NDT)
- Point Interface Table (PIT)
- Point Interface Table Entry (PITE)
- Point Interface Table Entry Parameter (PITEP)

A complete description of the CLM Configuration Builder and the build process is included in the *CLM Platform Implementation Manual*.

CLM Data Structures

All CLM data structures belong to a hierarchy, and must be built in the following sequence:

CLMI, PORT, DEVICE, PIT , and PITE entries

The Next Step -

To begin, we will lead you through the steps to configure the virtual interface board (CLMI) that will host the Client application.

3.2 Configure the Virtual Interface Board (CLMI)

Overview

This section defines the steps to configure the virtual interface board or CLMI. The CLMI is standard across all APP/CLM applications. It is included as part of the APP/CLM Platform package and not the OPC Client application package. The CLMI must be defined to be able to host the Client application. The CLMI gives you the ability to connect up to four Servers.

Warning!

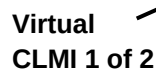
This section assumes that this is a first time installation and there is no existing application installed that would cause the configuration file and CLMI entry to already exist. If this is NOT a first time installation, skip these steps and continue with Section 3.5 to define your server object. If you are unsure, contact the Honeywell Technical Assistance Center.

Define CLMI in the Configuration Utility

First, the virtual CLMI must be defined to the APP, with regard to chassis position and logical board number. This is done through the TPS and Experion Configuration Utility, using the CLM option.

Step	Action
3.2.1	Determine the logical board number to be used, 1 or 2.
3.2.2	Determine the virtual chassis slot position for the CLMI board. This can be 2 through 4. (Position 1 is occupied by the LCNP4 card on the APP).
3.2.3	From the Start menu, select Programs, Honeywell TPS, and Configuration Utility. Note: In Windows 2008, select Programs, Honeywell Experion PKS, System Management, and Configuration Utility from the Start menu.
3.2.4	From the drop down menu, select Configure and CLM.
3.2.5	Select the Virtual Board 1 or Virtual Board 2 tab, depending on the board number you are working with. (Determined in 3.2.1)
3.2.6	Select the check box for the CLMI.
3.2.7	Select the drop down menu on the CLMI entry and pick the +Warning(EW) error control.
3.2.8	Select the radio button for the virtual chassis slot position (in 3.2.2)
3.2.9	Select the Apply button and OK to exit the CLM Configuration Utility.
3.2.10	Example: If you were to use CLMI 1 in virtual chassis slot position 3, the CLM Configuration page would look like Figure 3.2-1.

3.2 Configure the Virtual Interface Board (CLMI), Continued



Virtual
CLMI 2 of 2

Virtual Chassis
Position 3

Figure 3.2-1 CLM Configuration Page Example – Virtual CLMI

Define the LCN Configuration File (ZC)

Now the LCN side must be able to recognize and match the definitions that were entered onto the CLM Configuration Utility page. This is done by creating a configuration file that will reside on the &CUS directory of the system media, usually the History Module. This file must exist to successfully start up the OPC Client application on the CLMI.

The steps following define how to create this file.

3.2 Configure the Virtual Interface Board (CLMI), Continued

Step	Action
3.2.11	From a GUS station and the Engineering main menu, enter the COMMAND PROCESSOR function.
3.2.12	Set your path name to point to the &CUS directory Note: For the remainder of this procedure, we assume the LCN system media to be the History Module.
3.2.13	Copy the OPC Client sample configuration file (O_CLMI.ZC) to a new file, based on the APP/CLM node number. CP O_CLMI.ZC CLMI0nn.ZC -D where nn = APP/CLM node address Example: If your APP/CLM LCN node address is 58, using virtual chassis slot 3 for board 1, then your file name should be CLMI058.ZC and look as follows: CLMI 1,3 END
3.2.14	Invoke the LCN Editor on the file created in step 3.2.13.
3.2.15	Edit the CLMI directive in the file to insert the board number (from step 3.2.1) and chassis slot position (from 3.2.2).
3.2.16	Save the file and Exit the LCN Editor.

Build the CLMI

Now that we have the CLMI configured, we need to actually build it. This is done through the CLM Configuration Builder. If you need additional assistance in the steps that follow, refer to the *CLM Platform Implementation Manual*. Please follow these steps to build the CLMI virtual board entry.

Step	Action
3.2.17	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
3.2.18	Select the CLM node number.
3.2.19	When the lower half of the display appears, select CLM BUILDER .
3.2.20	When the Builder invokes, select ENTRY TYPE and enter CLMI. Data should appear on the lower half of the screen.
3.2.21	Select ENTRY NAME and enter up to an eight character name for your CLMI virtual board and press enter.
3.2.22	Select the A1_CLMI entry and enter the board number (from step 3.2.1). <u>All other entries must be left at their default values.</u> The display should now look similar to the Figure 3.3-2.

3.2 Configure the Virtual Interface Board (CLMI), Continued

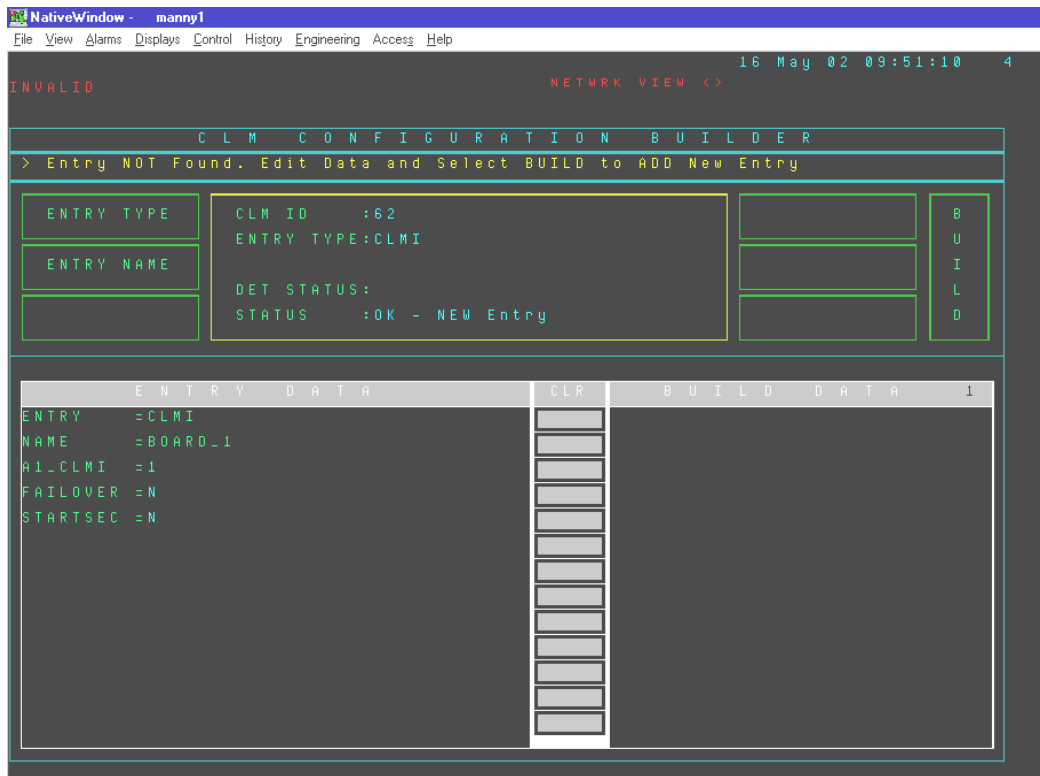


Figure 3.2-2 CLM Configuration Builder – CLMI Example

Step	Action
3.2.23	Select BUILD.
3.2.24	Verify that the entry successfully builds.
3.2.25	From the Application Modules Status Display, select the CLM node and execute the SAVE DATA function to save the changes.
3.2.26	Repeat the steps above for the second virtual board if required. Below is an example of a configuration file (.ZC) for 2 virtual boards. Example: If your APP/CLM LCN node address is 58, using virtual chassis slot 3 for board 1 and virtual chassis slot 4 for board 2, then your file name should be CLMI058.ZC and look as follows: <pre>CLMI 1,3 END CLMI 2,4 END</pre>

CLMI Configuration Complete

This completes the CLMI configuration. The next step is to define and configure the server objects. Continue on with Section 3.3.

3.3 Configure the Server Object(Port)

Overview

This section defines the steps to configure the server object. The Client will support up to four server objects on a virtual CLMI. This is accomplished by means of the APP/CLM port concept. A port definition must be configured for each server object to be connected.

Determine the Client/Server Relationship

First, determine the Client/Server connections to be established. There are several configurations that could occur.

- Single Client/Server connection
- Multiple Client/Single Server connection
- Multiple (Client/Server) connections

Single Client/Server

This configuration is the most basic and configures a single instance of the Client (i.e. a single port on the virtual CLMI) to connect to a single server object.

Multiple Client/Single Server

This configuration configures multiple instances of the Client (i.e. multiple ports on the virtual CLMI) to connect to a single server object.

Multiple (Client/Server)

This configuration configures multiple server objects against at least a single instance of the Client. For example, a single Client instance could connect to server object 1 and two Client instances could connect to server object 2.

Define Server Object in the Configuration Utility

The next thing that must be done is to define the server object to the APP and the virtual CLMI. This is done through the TPS Configuration Utility, using the CLM option.

Step	Action
3.3.1	From the Start menu, select Programs, Honeywell TPS, and Configuration Utility. Note: In Windows 2008, select Programs, Honeywell Experion PKS, System Management, and Configuration Utility from the Start menu.
3.3.2	From the drop down menu, select Configure and CLM.
3.3.3	Select the Virtual Board 1 tab for board 1 or Virtual Board 2 tab for board 2. (Determined earlier in 3.2.1)
3.3.4	Select the check box for Application 1.
3.3.5	Select the drop down menu for Application 1 and select OPC.

3.3 Configure the Server Object(Port), Continued

Step	Action
3.3.6	Assign each instance of the Client to the virtual CLMI by selecting the appropriate Port check box. For example, if you need two instances of the Client connecting to the Server, begin by checking box 1.
3.3.7	Below the application/port selection line, a pair of windows will open up for the selected port (in the example we are using, port 1). These windows are for the Server ID and the REMOTE ID node identifier, if applicable.
3.3.8	In the SERVER ID window, enter the Server ID of the OPC Server object. This entry is mandatory. Refer to the Server documentation to identify this name. This must be accurate.
3.3.9	If the Server object is located in a remote node (i.e. not internal to the Client node), then DCOM communication must be configured between the Client node and the remote server node. Enter the Network Host name and update it against the IP address in the Host file or Network Computer name of the remote node containing the server object into the REMOTE ID window. If the server object is local within the Client node, then leave this field blank.
3.3.10	<i>Repeat</i> steps 3.3.6 through 3.3.9 for ports 2 through 4, if applicable. In our example, we would need to repeat and enter data again for port 2. Note: For each port, you are prompted for the server id and remote id. Different server ids (i.e. objects) may be defined on each port.
3.3.11	Refer to the example in Figure 3.3-1.

3.3 Configure the Server Object(Port), Continued

Must be checked to use

Client Instance Assignments – Port 1 assigned

Server Object ID Name – Port 1 and Port 2

Remote Node ID (Blank if local)

The screenshot shows the CLM Configuration Utility window with the following configuration:

- Virtual Board 1** tab is selected.
- CLM Interface**: ☒ CLM, +Warning (EW) dropdown.
- Slot**: Radio buttons for 2, 3, 4. Slot 3 is selected.
- Application 1**: ☒ Application 1, OPC dropdown.
- Ports**: Port 1 and Port 2 are checked.
- Server ID**: OPC.Matikon.Modbus for Port 1 and Port 2.
- Remote ID**: Blank for Port 1 and Port 2.
- Application 2**: ☐ Application 2, Ports 1, 2, 3, 4 are unchecked.
- Application 3**: ☐ Application 3, Ports 1, 2, 3, 4 are unchecked.
- Application 4**: ☐ Application 4, Ports 1, 2, 3, 4 are unchecked.

Buttons at the bottom: OK, Cancel, Apply, Help.

Figure 3.3-1 CLM Configuration Utility – Server Object Example

Continued on next page

3.3 Configure the Server Object(Port), Continued

Build the Server Object

Now that we have defined the Server objects and the instances of the Client connections, we need to actually build the entries. In the APP/CLM, we use the Port template file to represent the connection between the Server object and the instance of the Client. The CLM Configuration Builder is used to input the appropriate data regarding naming option for the Server and assignment of the instance to the virtual CLMI board.

User Input for Server Object Build

There are several fields that are configurable when building the Server object (port). These fields are listed in Table 3.3-1.

Table 3.3-1 Server Object User Input Field Description (port)

Field	Description	Default
A1_CLMI	The virtual CLMI number (from previous step 3.2.1)	1
A2_PORT	Assigned instance on the virtual CLMI (1 through 4)	1
	The following 8 values indicate whether a condition will cause BAD to be stored to a parameter in the Image Point. If the value is Y and the LCN value corresponding to the Item is Numeric, the Image Point value will be set to BAD when the condition occurs.	
CONFIGER	Configuration Error; the Server could not locate a Data Item	Y
NOT_CONN	Not Connected; the Data Item does not currently have a value	Y
DEV_FAIL	Device Failure	Y
SENSFAIL	Sensor Failure	Y
LSTKNOWN	Last Known Value; Communications with device failed; CLM has the last known good value.	Y
COMMFAIL	Communications Failure; Communications with device failed; there is NO last known good value	Y
OUT_SERV	Out of Service; Item is Inactive	Y
UNCERTN	Item is of Uncertain Quality	Y
BAD_QUAL	Bad Quality Value returned	Y
ALLOTHER	All other potential quality return values not covered by the above.	N
NAME_OPT	Options for generating Group and Item Names for OPC Servers Range = 0 through 7. For specific naming option details, refer to Section 4.2.	0

Continued on next page

3.3 Configure the Server Object(Port), Continued

Step	Action
3.3.12	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
3.3.13	Select the CLM node number.
3.3.14	When the lower half of the display appears, select CLM BUILDER .
3.3.15	When the Builder invokes, select ENTRY TYPE and enter O_PORT. Data should appear on the lower half of the screen.
3.3.16	Select ENTRY NAME and enter up to an eight character name for your Client instance (port assignment) and press enter.
3.3.17	Using the field definitions from the previous Table 3.3-1, enter the appropriate user values into the Builder entries. Caution: The CLMI and Instance assignments must match those that were configured previously through the CLM page of the TPS Configuration Utility. The display should now look similar to the Figures 3.3-2 and 3.3-3.

The screenshot shows the 'CLM CONFIGURATION BUILDER' window. At the top, it says 'Entry NOT Found. Edit Data and Select BUILD to ADD New Entry'. Below this, there are several input fields and a status area. The 'ENTRY TYPE' is set to 'O_PORT' and 'ENTRY NAME' is 'SERVER_1'. The 'CLM ID' is '62'. The 'DET STATUS' is 'OK - NEW Entry'. On the right side, there is a vertical stack of buttons labeled 'B', 'U', 'I', 'L', 'D'. Below the main configuration area, there is a table with columns 'ENTRY DATA', '*MORE*', 'CLR', and 'BUILD DATA'. The 'ENTRY DATA' column contains the following values: 'ENTRY = O_PORT', 'NAME = SERVER_1', 'CLMI = 1', 'PORT = 1', 'CONFIGER = Y', 'NOT_CONN = Y', 'DEV_FAIL = Y', 'SENSFAIL = Y', 'LSTKNOWN = Y', 'COMMFAIL = Y', 'OUT_SERV = Y', 'UNCERTN = N', and 'BAD_QUAL = Y'. The 'CLR' column has a series of empty boxes. The 'BUILD DATA' column is empty.

Figure 3.3-2 CLM Configuration Builder – Server Object (Port) Example Page 1

Continued on next page

3.3 Configure the Server Object(Port), Continued

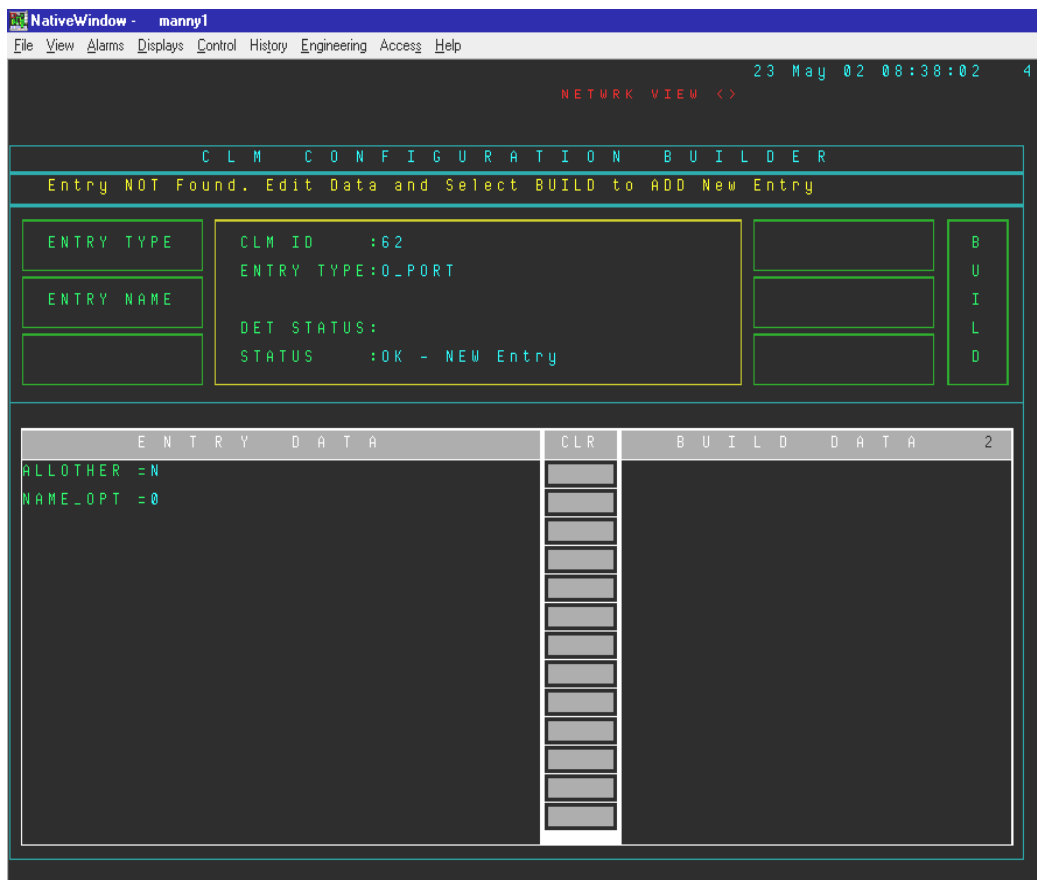


Figure 3.3-3 CLM Configuration Builder – Server Object (Port) Example Page 2

Step	Action
3.3.18	Select BUILD.
3.3.19	Verify that the entry successfully builds.
3.3.20	Repeat steps 3.3.15 through 3.3.18 for each instance of the Client defined previously through the CLM page of the TPS Configuration Utility.
3.3.21	From the Application Modules Status Display, select the CLM node and execute the SAVE DATA function to save the changes.

3.4 Configure the OPC Remote Device(s)

Overview

This section defines the steps to configure the OPC-based device. This device is recognized by the Server object as a source for group and item data, but does not contain any group or data items itself. All group and item information is contained completely within the Server object.

Remote Devices Must be Defined

In most implementations of Industrial application OPC Servers, remote devices or networks are part of the server object definition. The Client defines these remote devices by means of the APP/CLM device template files using the CLM Configuration Builder. These device entries are then used to assign and coordinate group and data item definitions for the server object.

Note: Refer to the Figure 1.1-1 in Section 1. These remote devices are part of the THIRD PARTY NETWORK connected to the server.

Exception – No Remote Devices

In the event of a server object with no remote devices, the Client will still require at least one device to be built for every Client instance assigned, although the device will physically not exist. This device must adhere to the virtual CLMI (1-2) and Client instance assignments (1-4), but would generally default to a remote physical device address of 1. You still may need to enter a specific OPC device name. See Table 3.4-1 below.

Build the Remote Device

In the APP/CLM, the Client uses the device template file to represent the remote devices connected to the Server object. The CLM Configuration Builder is used to input the appropriate data regarding physical address and specific OPC device name, if any.

User Input for Remote Device Build

There are several fields that are configurable when building the OPC remote device (device). These fields are listed in Table 3.4-1.

3.4 Configure the OPC Remote Device(s), Continued

Table 3.4-1 Remote Device User Input Field Description (device)

Field	Description	Default
A1_CLMI	The virtual CLMI number (from previous step 3.2.1)	1
A2_PORT	Assigned instance on the virtual CLMI (from previous step, depending on which Client instance assignment (1-4) that this server-defined device will be accessed.	1
A3_DEV	The Physical device address on the remote network	1
OPCDEVNM	Optional OPC Item name for the device (dash-dash (--), if unused) - Maximum 30 characters	--
REQ_UPD	Requested Server Update Rate – Requested (expected) rate at which the Server should check for updates from the remote devices. Value is in 10 mSec increments (i.e. 4=40ms, 90=900ms). 0 = No Requested update rate; Updates are per Server configuration.	100 (1 sec)
UPD_ALM	Update Alarm Differential Value – Maximum allowable difference between the requested and actual Server update rates. If this difference is greater than this value, then an application error is logged in the Event Viewer journal. Value is in 10 mSec increments. 0 = No differential checking.	50 (.5 sec)

Build Remote Devices

Follow the steps below to build remote devices from the CLM Configuration Builder.

Step	Action
3.4.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
3.4.2	Select the CLM node number.
3.4.3	When the lower half of the display appears, select CLM BUILDER .
3.4.4	When the Builder invokes, select ENTRY TYPE and enter O_DEVICE. Data should appear on the lower half of the screen.
3.4.5	Select ENTRY NAME and enter up to an eight character name for your remote device and press enter.
3.4.6	Using the field definitions from the previous Table 3.4-1, enter the appropriate user values into the Builder entries. Caution: The CLMI and Instance assignments must match those that were configured previously through the CLM page of the TPS Configuration Utility. Note: If the Server object has no remote device definition, then build only a single device here, and use the default value of 1 for the physical address. The display should now look similar to the Figure 3.4-1.

Continued on next page

3.4 Configure the OPC Remote Device(s), Continued

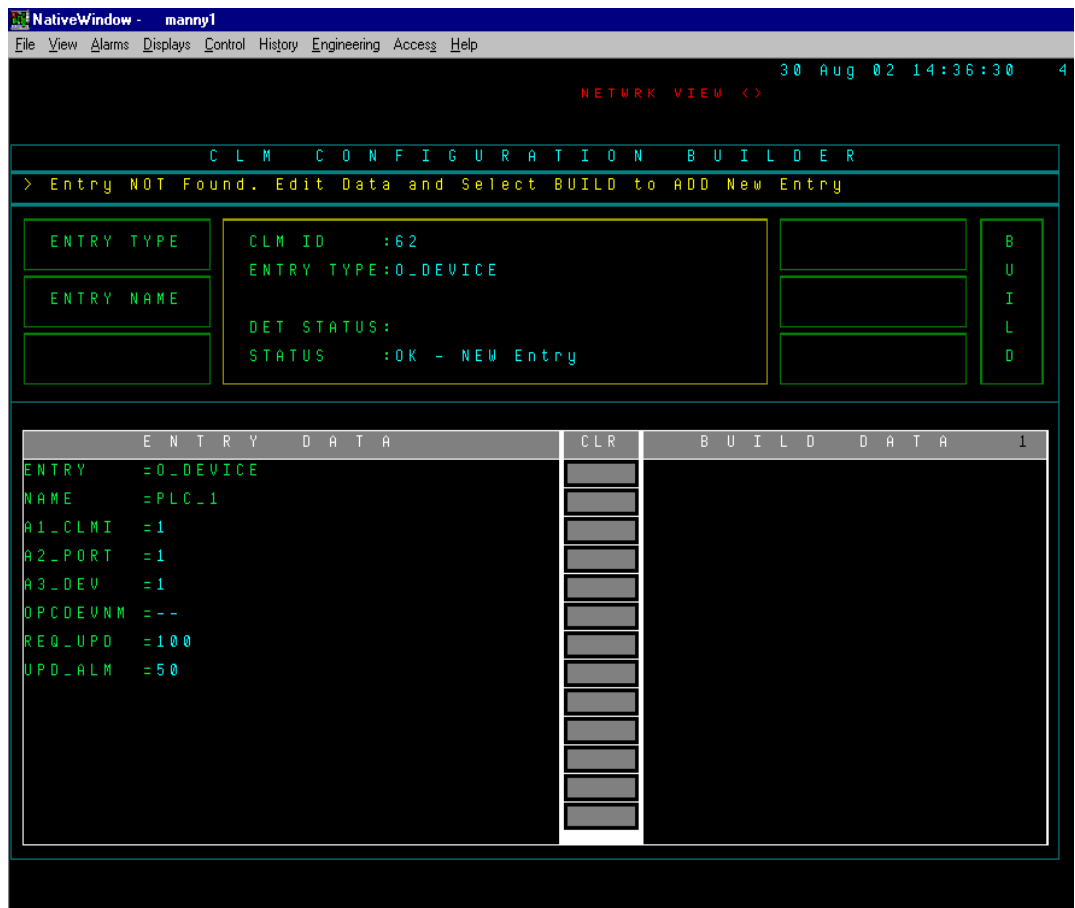


Figure 3.4-1 CLM Configuration Builder – Remote Device Example

Step	Action
3.4.7	Select BUILD.
3.4.8	Verify that the entry successfully builds.
3.4.9	<i>Repeat</i> steps 3.4.5 through 3.4.8 for each remote device defined across each Client instance assignment (port).
3.4.10	From the Application Modules Status Display, select the CLM node and execute the SAVE DATA function to save the changes.

3.5 Configure the OPC Group (PIT)

Overview

This section defines the steps to configure the OPC groups. Each group is assigned to a device that has been previously defined. The Client application defines the group by means of the APP/CLM PIT template file, using the CLM Configuration Builder.

Group Definition Naming Limitations

The OPC group is defined based on conventions established primarily by the Server object. However, the Client application restricts the naming structures to be the same as the PIT template entry. This means there is a limitation of eight characters available for the actual group name definition in the Client. Generally, the OPCDEVNM field of the associated remote device may be used as the first common thirty characters that may be concatenated with the actual group (PIT) name.

Example: In DEV_32, the OPCDEVNM field="fieldvalvenetwork" and the group (PIT) name = "pump1". Using the correct Naming option (see section 3.3) can concatenate these fields together to create

"fieldvalvenetworkpump1" , if naming option 1, or

"fieldvalvenetwork.pump1" , if naming option 5.

Build the Group

At least one group must be built for each remote device defined in the Client. The CLM Configuration Builder is used to build these entries.

User Input for OPC Group

There are several fields that are configurable when building the OPC Group (PIT). These fields are listed in Table 3.5-1.

Continued on next page

3.5 Configure the OPC Group (PIT), Continued

Table 3.5-1 OPC Group User Input Field Description (PIT)

Field	Description	Default
CYCLE	Scheduling cycle of the group. This is not normally used since data is retrieved on data change. However, there may be situations that require data to be moved to the LCN image point database simultaneously for processing, such as in some analyzer applications. The data items mapped to the image point parameters must have their INPSCHED field set to Y to take advantage of the cycle. For more information about Data Item/Image Point Parameter Configuration, see Section 4. If CYCLE used, set to a value in seconds, minutes, or hours.	-1 (Do Not Schedule)
CYC_ALM	Number of cycles behind that the APP/CLM scheduler must be before issuing an alarm. This function is not used by the Client application. Use the default.	0
SLO_DWN	Number of cycles to be scheduled before processing all data items mapped to image point parameters that have their SLO field set to Y. This function is not used by the Client application. Use the default.	5
NDT_NAME	This is the name of one of the associated remote devices that was previously built in Section 3.4. By naming that device here, you are effectively connecting this OPC group definition to that device.	<blank>
GRP_NAME	This function is not used by the Client application. Leave this entry blank. (See note 1)	<blank>

Note 1: The GRP_NAME field is a APP/CLM function and does not relate to the OPC group structure.

Continued on next page

3.5 Configure the OPC Group (PIT), Continued

Step	Action
3.5.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
3.5.2	Select the CLM node number.
3.5.3	When the lower half of the display appears, select CLM BUILDER .
3.5.4	When the Builder invokes, select ENTRY TYPE and enter PIT. Data should appear on the lower half of the screen.
3.5.5	Select ENTRY NAME and enter up to an eight character name for your OPC group name and press enter.
3.5.6	Using the field definitions from the previous Table 3.5-1, enter the appropriate user values into the Builder entries. The display should now look similar to the Figure 3.5-1.

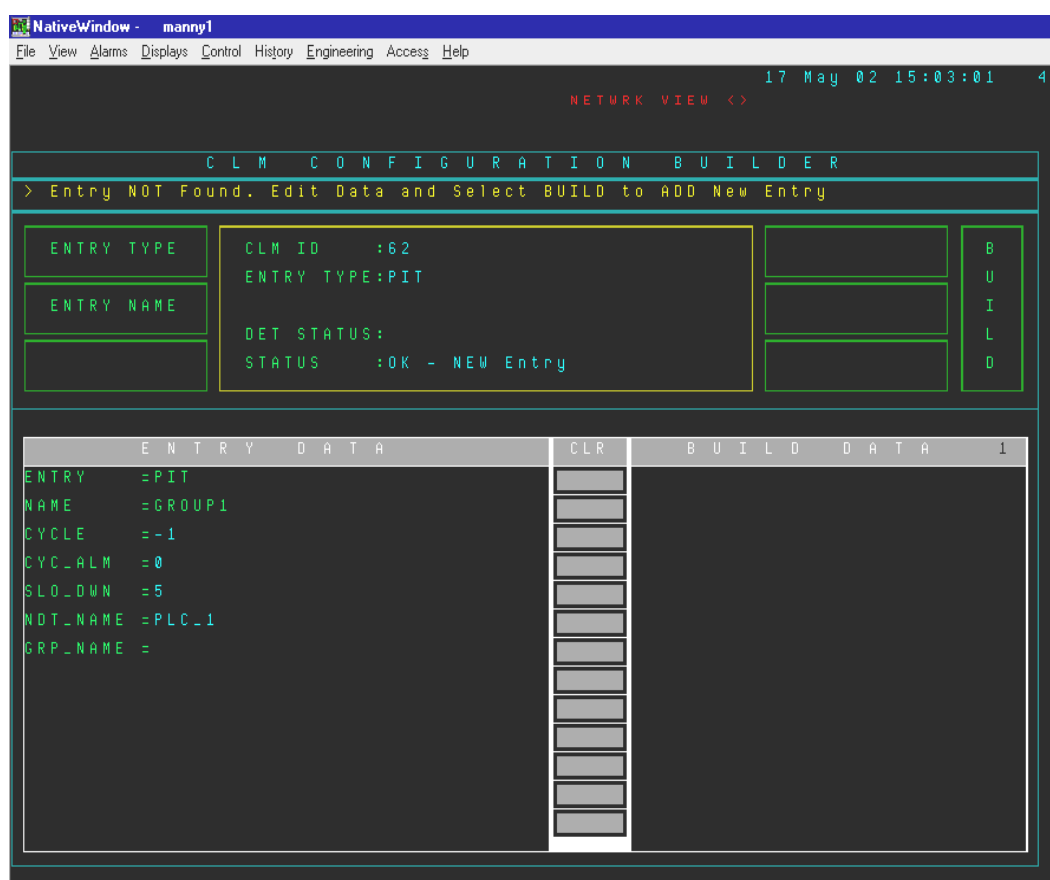


Figure 3.5-1 CLM Configuration Builder – OPC Group (PIT) Example

Continued on next page

3.5 Configure the OPC Group (PIT), Continued

Step	Action
3.5.7	Select BUILD.
3.5.8	Verify that the entry successfully builds.
3.5.9	<i>Repeat</i> steps 3.5.5 through 3.5.8 for each OPC group to be defined across each remote device.
3.5.10	From the Application Modules Status Display, select the CLM node and execute the SAVE DATA function to save the changes.

Section 4 – Data Item/Image Point Parameter Configuration

4.1 Overview

The Client application images the data in OPC items of the Server object by mapping them to LCN image point parameters. This allows bi-directional data flow between the LCN and the Server object. These mapped connections are strategically grouped together into specific point types for ease of use on the LCN side. These types include:

- Analog Input
- Analog Output
- Numeric*
- Flag (On/Off)*
- Digital Input 3-State (On/Off/Bad)
- Digital Input 4-State (Open/Close/Moving/Bad)
- Digital Composite 3-State (On/Off/Bad)*
- Digital Composite 4-State (Open/Close/Moving/Bad)*
- Digital Composite 5-State (Open/Close/Stop/Moving/Bad)*

* These image point types support both input from and output to the Server object

Prerequisite!

Note: Before configuring image points you must have completed *Section 3 – OPC Client Configuration*.

4.2 Client Group and Item Naming Options

Introduction

Data values are identified in the OPC Server object by Item Name. This section describes the various ways that Item Names can be configured in the Client application. The NAME_OPT field in the Server object definition (port) defines the configuration options for Group Names and Item Names. See section 3.2.2.

These naming options are a predefined method of establishing group and item names based on user input data through fields entered through the CLM Configuration Builder. Depending on the naming option selected, these fields may be used by themselves or concatenated together to form group and item names. These fields include:

- OPCDEVNM - From the O_DEVICE template file. Optionally used to define up to a 30 character remote device name. The default is "--" for unused.
- PIT NAME – From the PIT template file. This is the 8 character PIT entry name defined by the user. It is the default OPC group name.
- DPOINT - From the PITE template files (various). This is the optional 30 character common prefix text string that will be concatenated with all DPAR (see below) fields defined in the same PITE template file to define the item names. The default is "--" for unused.
- DPAR - From the PITE template files (various). This is the required (up to) 30 character text string that defines the item name in the server object that maps to the image point parameter.

Name_Opt Range

The Valid values for the NAME_OPT field in the O_PORT entry are 0-7, with 4 and 6 not used. Values of 4, 6, or outside the given range will result in a runtime configuration error and logged message in the Event Viewer.

Caution! The CLM Configuration Builder does not range check the NAME_OPT user entry values!

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Naming Option 0

Group name will be the same as the name of the PIT definition built in the CLM Configuration Builder.

Item names will be formed by combining the DPOINT field in the common part of the template and the DPAR field in each parameter section of the template.

The generic format of the group and item names for naming option 0 is as follows:

Group	= <PIT Name>
Item	= <DPOINT DPAR>
	(= concatenate)

Example:

A data item PUMPSCOMPRESSOR1 is part of OPC group LINE1 in the Server object. It is used to represent the current running state of a pump. CLM Builder entries to generate these groups and data items would be as follows:

PIT:ENTRY NAME = LINE1

A100:DPOINT = PUMPS

A100:IT_PV = COMPRESSOR1

(Note that IT_PV is the O_FLAG text-specific DPAR entry)

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Naming Option 1

Group name will be the same as the PIT definition built in the CLM Builder, but prefixed with the remote device name (device).

Item names will be formed by combining the DPOINT field in the common part of the template and the DPAR field in each parameter section of the template, but by prefixed with the remote device name (device).

The generic format of the group and item names for naming option 1 is as follows:

Group = <OPCDEVNM><PIT Name>
Item = <OPCDEVNM><DPOINT||DPAR>
 (|| = concatenate)

Example:

A data item PLC5VALVESMOV1 is part of OPC group PLC5LINE1 of remote device PLC5 in the Server object. It is used to represent the OPEN or CLOSE state of the valve. CLM Builder entries to generate these groups and data items would be as follows:

DEVICE:OPCDEVNM = PLC5

PIT:ENTRY NAME = LINE1

A100:DPOINT = VALVES

A100:IT_D1 = MOV1

(Note that IT_D1 is the O_DIN3 text-specific DPAR entry)

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Naming Option 2

Group name will be formed the same as in option 0.

Data items will be formed the same as in option 0, *except* that the item name will be prefixed by the group name.

The generic format of the group and item names for naming option 2 is as follows:

Group = <PIT Name>
Item = <Group><DPOINT||DPAR>
 (|| = concatenate)

Example:

A data item LINE1METERSTEMPERATURE1 is part of OPC group LINE1 in the Server object. CLM Builder entries to generate these groups and data items would be as follows:

PIT:ENTRY NAME = LINE1

A100:DPOINT = METERS

A100:IT_INPUT = TEMPERATURE

(Note that IT_INPUT is the O_AI text-specific DPAR entry)

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4.2 Client Group and Item Naming Options, Continued

Naming Option 3

Group name will be formed as in option 1.

Data items will be formed as in option 1, *except* that the item name will be prefixed by the group name (as in option 0, PIT name only).

The generic format of the group and item names for naming option 3 is as follows:

Group = <OPCDEVNM><PIT Name>
Item = <PIT Name><OPCDEVNM><DPOINT||DPAR>
 (|| = concatenate)

Example:

A data item LINE1PLC5METERSTEMPERATURE2 is part of OPC group PLC5LINE1 in remote device PLC5 in the Server object. CLM Builder entries to generate these groups and data items would be as follows:

DEVICE:OPCDEVNM = PLC5

PIT:ENTRY NAME = LINE1

A100:DPOINT = METERS

A100:IT_INPUT = TEMPERATURE2

(Note that IT_INPUT is the O_AI text-specific DPAR entry)

Naming Option 4

Not Used.

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Naming Option 5

Group name is formed the same as in option 1, *except* that a period (“.”) is used to separate the fields.

Data items are formed the same as they are in option 1, *except* that a period (“.”) is used to separate the fields.

The generic format of the group and item names for naming option 5 is as follows:

Group	= <OPCDEVNM>.<PIT Name>
Item	= <OPCDEVNM>.<DPOINT DPAR> (= concatenate)

Example:

A data item PLC5.PUMPSCOMPRESSOR1 is part of OPC group PLC5.LINE1 of remote device PLC5 in the Server object. This item is used to represent the running on/off state of a pump. CLM Builder entries to generate these groups and data items would be as follows:

```
DEVICE:OPCDEVNM = PLC5
PIT:ENTRY NAME = LINE1
A100:DPOINT = PUMPS
A100:IT_PV = COMPRESSOR1
(Note that IT_PV is the O_FLAG text-specific DPAR entry)
```

Naming Option 6

Not Used.

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Naming Option 7

Group name is formed as in option 5.

Data items are formed the same as they are in option 3, *except* that a period (".") is used to separate the fields.

The generic format of the group and item names for naming option 7 is as follows:

Group = <OPCDEVNM>.<PIT Name>
Item = <PIT Name>.<OPCDEVNM>.<DPOINT||DPAR>
 (|| = concatenate)

Example:

A data item LINE1.PLC5.VALVESMOV1 is part of OPC group PLC5.LINE1 in remote device PLC5 in the Server object. This data item is used to represent the open/closed state of a valve. CLM Builder entries to generate these groups and data items would be as follows:

DEVICE:OPCDEVNM = PLC5
PIT:ENTRY NAME = LINE1
A100:DPOINT = VALVES
A100:IT_D1 = MOV1

(Note that IT_D1 is the O_DIN3 text-specific DPAR entry)

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Device Names

Many OPC Servers require that devices be configured in the OPC Server so that OPC Groups and OPC Items can be associated with the devices. These devices typically provide a means to configure attributes of the third party network that are not covered by the OPC Data Access Interface (i.e. communications port name/address, baud rate, device address on the third party network).

The Client Application provides the following ways to specify Device Names for OPC Server objects. If the OPC Server does not require or support devices, then no remote device name need be configured, although at least one entry must still be built.

The OPC remote Device Name is determined as follows. In the Device NDT entry (O_DEVICE):

If OPCDEVNM key field is undefined (i.e. "--"), the OPC Device Name is undefined and therefore the Builder entry name is used,

ELSE

Use the value in OPCDEVNM as the OPC remote Device Name.

DPAR Aliases

Each image point type has one or more user changeable parameters that is mapped to a data item in the Server object. The generic field name for this data item, as defined in the naming options is the DPAR field. For clarity to the user, the DPAR field for each parameter in each point type supported by the Client application has been assigned an alias name. The point type and assigned aliases are listed here.

<u>Point Type</u>	<u>DPAR Alias Name(s)</u>
O_AI	IT_INPUT
O_AO	IT_SP
O_D31, O_D32	IT_D1, IT_OP1, IT_FB1, IT_OP2, IT_FB2
O_D41, O_D42	IT_D1, IT_D2, IT_OP1, IT_FB1, IT_OP2, IT_FB2, IT_OP3, IT_FB3
O_D5	IT_D1, IT_D2, IT_OP1, IT_FB1, IT_OP2, IT_FB2, IT_OP3, IT_FB3
O_DIN3	IT_D1
O_DIN4	IT_D1, IT_D2
O_FLAG	IT_PV
O_NUMER	IT_PV

Continued on next page

4.2 Client Group and Item Naming Options, Continued

Configuration of OPC Item Names

DPOINT allows an ASCII string to be specified for a point name in the Server object. The DPOINT field is used as a common text string prefix that is concatenated to all DPAR fields defined within the template to form longer item names. If the value of DPOINT is "--" (i.e., 2 dashes), no point name (or prefix) is required to form an OPC Item Name and the item name is generated entirely from the DPAR field. For example: if the OPC item to be read is POSITION and it is mapped to a flag point using the O_FLAG template file, the following values apply:

Item name entries for O_FLAG

Field	Value	Comments
DPOINT	--	Optional portion of the OPC Item name. In this case since it is "--", the DPOINT value is not used to form the OPC Item name.
IT_PV*	POSITION	The OPC Item name.

If the OPC Item to be read is VAT1.TEMPERATURE and it is mapped to an AM regulatory point using the O_AI template file, the following values apply:

Item name entries for O_AI

Field	Value	Comments
DPOINT	VAT1.	Optional portion of the OPC Item name.
IT_INP*	TEMPERATURE	The OPC Item name.

If the OPC Items to be read are two limit switches of a valve VALVE1OPEN and VALVE1CLOSED and it is mapped to an AM Switch point using the O_DIN4 template file, the following values apply:

Item name entries for O_DIN4

Field	Value	Comments
DPOINT	VALVE1	Optional portion of the OPC Item name.
IT_D1*	OPEN	The OPC Item name for Bit D1
IT_D2*	CLOSED	The OPC Item name for Bit D2

* Remember that these are alias strings for the DPAR field.

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4.3 Data Item Conversion Codes

Introduction

Data items ‘input from’ and ‘output to’ OPC Servers may be converted between OPC and LCN data types. Parameter conversion definitions are used to define these data conversions.

LCN vs. OPC Data Types

The data type of values received from (and sent to) an OPC Server may be the same as the LCN data type. Following are the default OPC types for corresponding LCN types.

LCN vs. OPC Data Types

LCN Data Type	OPC Data Type
Analog value	An IEEE 4-byte real value (VT_R4).
Logical value	A Boolean (True/False) value is specified. A value of 0xFFFF (all bits 1) indicates True; a value of 0 (all bits 0) indicates False (VT_BOOL).
String value	A wide double-byte (Unicode) string (VT_BSTR).
LCN Enumeration	An unsigned 2-byte integer (VT_UI2).

Continued on next page

4.3 Data Item Conversion Codes, Continued

OPC Client - supported conversions

A Parameter Conversion Definition is specified in two parts that are separated by a dash. Following is the general form:

lcn type{, **length**}-{**opc type**}, {**scale**}, {**start**, **stop**}{**deadband**}{**options**}

The fields before the dash make up the LCN Conversion Code - Table 4.3-1.

The fields after the dash make up the OPC Conversion Code - Table 4.3-2.

Fields enclosed in brackets (i.e., { }) may not be required for all Conversion Definitions.

Image point parameters on the LCN may be mapped to send data to, or receive data from, items in the Server object.

Some of the templates for the Client application may be configured to not display the Conversion Definition values.

For more information, refer to the definition of the Pnn_ICNV and Pnn_OCNV in the PITE template definition section of the *CLM Platform Implementation Manual* and Appendix B of this manual.

Conversion Code – LCN side

The LCN conversion code is the first part of the Parameter Conversion Definition and determines the data type of an LCN parameter value.

lcn type{, **length**}-{**opc type**}, {**scale**}, {**start**, **stop**}{**deadband**}

Table 4.3-1 LCN/CLM Conversion Codes

LCN type, Length	Data Type
2	Analog value, IEEE 32-bit Floating Point
3	Logical value, ON or OFF
4,N	String value, N is the length of the string.
5,N	LCN Enumeration, N = the maximum number of states.

Continued on next page

4.3 Data Item Conversion Codes, Continued

Conversion Code – OPC side

The OPC conversion code is the second part of the Parameter Conversion Definition, and determines how the data received from (or sent to) the OPC Server is handled.

lcn type{, length}-{opc type},{scale}, {start, stop}{deadband}{options}

If the OPC value contains fields or bits that need to be extracted, start and stop bits may be specified in the conversion code, but the data type of the OPC value must be an Unsigned Integer (VT_UI2 or VT_UI4).

Values received from (and sent to) the Server object may be scaled. For example, if a value is multiplied by 10 on input (scale = 6), the value should be divided by 10 on output (scale = 5).

Values from the Server object may also have a deadband applied. There are four types of deadband that may be applied to received values.

- Minimum change, where the received value must deviate by GREATER THAN the specified deadband from the current value (> db)
- Maximum change, where the received value must be LESS THAN the specified deadband from the current value (< db)
- Internal Change-Only, where the received value is passed on only if it differs from the current value ()
- Server deadband, where the Server does all of the deadband checking and the Client passes through ALL values received, regardless of the current value (*)

Caution!

The conversion codes available to the OPC Client are designed to allow the user a lot of flexibility. It is the user's responsibility to be sure that values exchanged fit the expected format.

Incorrect "data typing" can lead to loss of data accuracy, data loss, or possible loss of view to the OPC network.

Continued on next page

4.3 Data Item Conversion Codes, Continued

Conversion Code - OPC Type

lcn type{, length}-{**opc type**},{scale}, {start, stop}{deadband}{options}

Table 4.3-2 Client Supported OPC Conversions

OPC type can be any VARIANT type. Following are the valid VARIANT types and whether they are supported by the Client application:

Type	Description	Supported in Client
0	Unused	--
2	Signed 16-bit Integer (VT_I2)	Yes
3	Signed 32-bit Integer (VT_I4)	Yes
4	32-bit Floating Point (VT_R4)	Yes
5	64-bit Floating Point (VT_R8)	Yes
6	Currency (VT_CY)	No
7	Date (VT_DATE)	No
8	String (VT_BSTR)	No
11	Boolean (VT_BOOL)	Yes
17	Unsigned Character (VT_UI1)	Yes
18	Unsigned 16-bit Integer (VT_UI2)	Yes
19	Unsigned 32-bit Integer (VT_UI4)	No

Conversion Code - Scaling

lcn type{, length}-{opc type},{**scale**}, {start, stop}{deadband}{options}

Table 4.3-3 Conversion Code Scaling Options

Scale is optional unless **start** and **stop** are used. This is the scale factor to be applied to the value which can be one of the scale factors listed below.

0	No scaling
1	Divide by 100,000
2	Divide by 10,000
3	Divide by 1000
4	Divide by 100
5	Divide by 10
6	Multiply by 10
7	Multiply by 100
8	Multiply by 1000
9	Multiply by 10,000
10	Multiply by 100,000
11	Invert the value (logicals and enumerations (Flags) only)

Continued on next page

4.3 Data Item Conversion Codes, Continued

Conversion Code - Start/Stop

Lcn type{, length}-{opc type},{scale}, {**start, stop**}{deadband}{options}

Start and Stop Bit Rules

- Only OPC types of unsigned integer (VT_UI2 and VT_UI4) may be used when extracting a specific range of bits.
- Default range bits, if not specified are 0-15 and 0-31 for VT_UI2 and VT_UI4 respectively.
- Bit ranges are validated at run-time.

Conversion Code - Deadband

Lcn type{, length}-{opc type},{scale}, {start, stop}{**deadband**}{options}

The deadband can be implemented by four different means, as described following.

Symbol	Description
> db	Minimum Change Required – If the new value received from the OPC server varies from the current value by an absolute amount that is GREATER THAN the specified deadband (db), then the new value will be stored to the entry, and ultimately to the image point parameter.
<db	Maximum Change Allowed – If the new value received from the OPC server varies from the current value by an absolute amount that is LESS THAN the specified deadband (db), then the new value will be stored to the entry, and ultimately to the image point parameter. This effectively “clips” all extreme values that do not fall within the normal change patterns.
*	Server Deadband Pass-through – This mode assumes that all deadband functionality is done entirely in the Server. ALL values received from the OPC server will be passed to the entry, and ultimately to the image point parameter, REGARDLESS OF WHETHER THE DATA HAS CHANGED.
{blank}	Internal Change-Only deadband (Default) – If the new value received from the OPC server IS NOT EQUAL TO the current value, then the new value will be stored to the entry, and ultimately to the image point parameter. Values that do not change will not be passed.

4.3 Data Item Conversion Codes, Continued

Conversion Code - options

lcn type{, length}-{opc type},{scale}, {start, stop}{deadband}{options}

The options can be implemented by appending the letter directly to the end of the existing input or output conversion code. Refer to following examples.

Symbol

Description

N

Convert the input value to NAN when the input value is 1e12.

The input value is converted to NAN if the value received by the client from the server is 1e12 and the letter N is appended to the input conversion code. The OPC server variant must be double precision (5).

Convert the output value to NAN when the output value is 1e12.

The output value is converted to NAN if the value sent to the server from the client is 1e12 and the letter N is appended to the output conversion code. The OPC server variant must be double precision (5).

O

Convert the input value to NAN when the input value is -1e12.

The input value is converted to NAN if the value received by the client from the server is 1e12 and the letter O is appended to the input conversion code. The OPC server variant must be double precision (5).

Convert the output value to NAN when the output value is -1e12.

The output value is converted to NAN if the value sent to the server from the client is -1e12 and the letter O is appended to the output conversion code. The OPC server variant must be double precision (5).

Continued on next page

4.3 Data Item Conversion Codes, Continued

Examples The following table contains examples of some commonly used Conversion Codes.

Table 4.3-4 Client Conversion Code Examples

LCN Type	OPC Type	ICNV or OCNV	Description
Analog Value	Single Precision Floating Point	2-4	LCN analog (2) – OPC floating point (4), no scaling, change-only deadbanding
Logical	Boolean	3-11	LCN logical (3) - OPC Boolean (VT_BOOL), change-only deadbanding
Logical	Boolean	3-11,11	LCN logical (3) - OPC Boolean (VT_BOOL), Logically inverted (11), change-only deadbanding
Analog Value	Signed 16-bit Integer	2-2 or 2-2,0	LCN analog (2) – OPC Signed 16-bit Integer (VT_I2), no scaling, change-only deadbanding
Enumeration – Flag	Boolean	5,2-11	LCN enumeration (5), (2) states - OPC Boolean (VT_BOOL), change-only deadbanding
Analog Value	Single Precision Floating Point	2-4,7	LCN analog (2) - OPC Floating point value (VT_R4), scaled @ * 100 (7), change-only deadbanding
Enumeration - Flag (Note 1)	Unsigned 16-bit Integer	5,2-18,0,1,1	LCN enumeration (5), (2) states – OPC unsigned 16-bit integer (18), no scaling (0), start bit (1), stop bit (1) (i.e. extract single bit), change-only deadbanding
String	String	4,10-8	LCN string (4), (10) characters – OPC String (VT_BSTR), change-only deadbanding
Analog Value	Unsigned 16-bit Integer	2-18	LCN analog (2) – OPC Unsigned 16-bit Integer value (VT_UI2), no scaling, change-only deadbanding
Analog Value (deadband)	Single Precision Floating Point	2-4,0>.1 or 2-4>.1	LCN analog (2) – OPC Floating Point value (VT_R4), no scaling, minimum deadband of 0.1
Analog Value (Note 2)	Unsigned 16-bit Integer	2-18,0,0,3	LCN analog (2) – OPC Unsigned 16-bit Integer (VT_UI2), no scaling, start bit 0, stop bit 3 (i.e. 4 bits: value range = 0-15), change-only deadbanding
Analog Value (deadband)	Signed 32-bit Integer	2-3<10.0 or 2-3,0<10.0	LCN analog (2) – OPC Signed 32-bit integer, no scaling, maximum deadband of 10.0

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4.3 Data Item Conversion Codes, Continued

Logical (deadband)	Boolean	3-11*	LCN logical (3) – OPC Boolean (11), with pass-through deadband (* = always send to parameter)
Analog Value	Double Precision Floating Point	2-5M	Analog Value converted to NAN when analog value is 1e12.

Continued on next page

4.3 Data Item Conversion Codes, Continued

The following notes reference specific examples above.

Note 1: The unsigned Integer value determines the ordinal element selected in the enumeration. Values are zero-based.

Note 2: Bit ranges are all zero-based (i.e. 0-15) and should be entered into the conversion code as such.

The following notes are general and reference conversion codes in general.

- Scaling on input of a value will normally require reverse scaling prior to output.
 - Bit ranges are defined from bit 0, with bit 0 being the Least Significant Bit (LSB).
 - The internal Change-Only deadband is the default, with the other three options available. The GREATER THAN and LESS THAN deadband options apply only to numeric values.
 - Start and Stop bit fields are optional, They allow fields to be extracted from a range of bits in a 16-bit or 32-bit Unsigned Integer.
-

4.4 Parameter/Data Item Mapping Options

Introduction

Mapping level options provide control over input and output data flow. These options are included in the CLM Configuration Builder template file for each mapped image point parameter on all available point types.

Input Mapping Options (IFNC)	
INPSCHED	Update the image point parameter only when the PIT is scheduled. Note: Data is still retrieved and collected from the OPC server on data change, but not stored to the image point parameter until the PIT actually executes. This can be useful if a higher level application needs to process several pieces of data at the same time.
SLO	Update the image point parameter only every nth time, based on the SLO_DWN field of the associated PIT.
ECP	Not used by the OPC Client application
PPS	Update the image point parameter that are have the INPSCHED field set to Y when the PPS is initiated for the AM/CLM image point.
SYN	Synchronization when the device is placed in service. This is required to establish the data item input connection from the server.
CN	Not used by the OPC Client application
SB	Not used by the OPC Client application
RBI	Not used by the OPC client application

4.4 Parameter/Data Item Mapping Options, Continued

Output PITE Parameter Options (OFNC)	
OUTSCHED	Output the image point parameter to the OPC server when the PIT is scheduled. Note: Unlike the ECP function below, the image point parameters may reside off-node.
SLO	Output the image point parameter only every nth time, based on the SLO_DWN field of the associated PIT.
ECP	For exception processing from the image point parameter to the server. This may be on demand by the operator from the detail display or graphic, or may be initiated from CL execution by use of the CLM_OUT parameter. Refer to the <i>CLM Platform Implementation Manual</i> .
PPS	Output the image point parameters that have their OUTSCHED field set to Y when the PPS is initiated for the AM/CLM image point.
SYN	Synchronization when the device is placed in service. This is required to establish the data item output connection from the server. Caution! Data will be sent to the server on synchronization!
RB	Not used by the OPC Client application.
CBO	Not used by the OPC Client application.

Section 5 – Client Image Point Database

5.1 Image Point Build Overview

Overview

This section describes the steps necessary to build and configure the connection between the image points on the LCN side and the OPC items in the Server object.

Point configuration examples

In the details of the point types described in this section, not all parameters and fields available on the LCN or CLM builder forms are identified. Only parameters required to properly setup the required structures are shown. Other parameters may be entered by the user to satisfy their specific implementation.

Client template design

The supplied OPC template files are designed in general as follows:

- The AM image point is processed after input by always setting the point option PPS to yes.
 - A conversion code must be entered by the user for each OPC address. With multiple options for both discrete and analog data types, it is generally not possible to predetermine the conversion strings.
-

5.2 Analog Input (AM Regulatory Point Type)

Introduction

The Analog Input is an AM point that images a single numeric value from an OPC Item in the Server object. This section describes how to create and configure this point type. Note that all parameters below are not necessarily shown unless they are required or may be utilized by the design engineer.

Theory of operation

The Analog Input point operates by mapping the OPC item directly to the INPUT custom parameter in the attached O_AI Custom Data Segment (CDS) package. The image point is configured to have no scheduling period, but is PPS'd by the CLM Platform software when new data is available from the OPC interface. The point then executes causing the Data Acquisition (DAQ) algorithm to move the raw data to the PV of the point. Use of trip limits and alarming is available since the image point is AM-based. If communications errors occur, the INPUT parameter may be set to BAD by the CLM Platform software, thus propagating a BAD value to the PV.

Point configuration

Analog Input Point Configuration

Using the LCN Data Entity Builder, an AM image point must be built, using the table following as a template for data entry. Entries left blank are either optional or user defined.

Parameter	Description	Entry	Notes
NAME	POINT NAME		
UNIT	UNIT ID		
PTDESC	POINT DESCRIPTOR		
EUDESC	E.U. DESCRIPTOR		
KEYWORD	POINT KEYWORD		
PRIMMOD	PRIMARY MODULE POINT ID	----	
PTDISCL	POINT DISCLOSURE	FULL	
PVALGID	PV ALGORITHM	DATAACQ	
CTLALGID	CONTROL ALGORITHM	NULL	
OVERVAL	OVERVIEW DISPLAY VALUE		
SUPPIO	I/O DISPLAY SUPPRESSION	OUTSUPPR	
PERIOD	POINT EXECUTION PERIOD	NOPERIOD	CLM S/W will PPS

Continued on next page

5.2 Analog Input (AM Regulatory Point Type), Continued

Point configuration

Analog Input Point Configuration, continued

Parameter	Description	Entry	Notes
NOPINPTS	NO. OF INPUT CONNECTIONS	1	From O_AI CDS package
PISRC1	SOURCE POINT ID.PARAMETER 1	*.INPUT	
PIDSTN1	DESTINATION PARAMETER	P1	
PIACTSTS1	DEST CONN ACTIVITY STATUS 1	ACTIVE	
CLSLOTS	NUMBER OF CL BLOCKS	0	Req'd on all CLM points
NOPKG	NO. OF CUSTOM DATA PACKAGES	2	
NUMSWTCH	NUMBER OF SWITCHES	0	
NOGINPTS	NUMBER OF GENERAL INPUTS	0	
NOGOPTS	NUMBER OF GENERAL OUTPUTS	0	
PKGNAME(1)	SOURCE FILE 1 NAME	O_AI	
PKGNAME(2)	SOURCE FILE 2 NAME	CLM_DMD3	

CDS parameters

Analog Input Point CDS Parameters

Parameter	Description	Entry	Notes
O_AI	CDS Parameters on PV Algorithm Page		
INPUT	POINT VALUE		Updated from OPC data
CLM_DMD3	CDS Parameters on Custom Page		
CLM_PPS	STORE TO PROCESS POINT PITE		Special Applications
CLM_OUT	STORE TO PROCESS OUTPUT PITE		Special Applications
DEV_STA	DEVICE IN_SERV (ON/OFF)		Updated by CLM S/W
PT_FAIL	DEVICE COMPONENT FAILURE		Not used for OPC
HDW_REF	HARDWARE REFERENCE		Not used for OPC
CLM_PORT	NAME OF SERVER OBJECT (PORT)		Updated by CLM Builder
CLM_DEV	NAME OF DEVICE		Updated by CLM Builder
CLM_PIT	NAME OF OPC GROUP (PIT)		Updated by CLM Builder

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5.2 Analog Input (AM Regulatory Point Type), Continued

Builder input file format

Analog Input Point Builder Input File Format

Using the CLM Builder, the OPC data for the Analog Input type point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_AI	Template Name (.ZT)
NAME		LCN Analog Input point name
PIT_NAME		PIT name (OPC Group Name)
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_INP		OPC Item Name (Required)
INPSCHED	Y or N	Update the image point only when the associated PIT executes. (Default = N, update on data change)
SB_OOS	Y or N	Set the value BAD when the associated device is placed out of service (Analog only) (Default = Y, set value BAD on out of service)
INP_ICC		Input conversion code (Default = "2-4", LCN analog to Floating Point)
END		End of current entry

Example - O_AI

Analog Input point AI100.Input maps to Group GRP1 and Item VALUE1. The item VALUE1 is a 16-bit signed integer that needs to be read in with no scaling, no deadband, and on data change only. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_AI
NAME = AI100
PIT_NAME = GRP1
DPOINT = --
IT_INP = VALUE1
INPSCHED = N
SB_OOS = Y
INP_ICC = 2-3
END
```

5.3 Analog Output (AM Regulatory Point Type)

Introduction

The Analog Output is an AM point that images an OPC item as an analog value and is used to output a value to that OPC item. This section describes how to create and configure this point type. Note that all parameters below are not necessarily shown unless they are required or may be utilized by the design engineer.

Theory of Operation

The Analog Output point maps the SP of a Regulatory AM point (Null PV and CL Control Algos) to the desired numeric OPC item. Feedback is used to validate the completed store by mapping a CDS parameter OUTF to the OPC item (normally the same as the SP item). The point must be in AUTO mode in order to output a SP value. The SP engineering unit range (SPEULO, SPEUHI) must be set up, as should the CV range since it is defined as Engineering Units type.

With the point in AUTO, the operator changes the SP. This value is sent by “exception processing” to the OPC item. When the value changes in the data item, the feedback is detected and sent back. A General Input (GI) moves the OUTF data to the CV and causes the OP to update with the proper range percentage value.

Note: Outputs are not sent out in the MAN mode.

5.3 Analog Output (AM Regulatory Point Type), Continued

Point configuration

Analog Output Point Configuration

Using the LCN Data Entity Builder, an AM image point must be built, using the table following as a template for data entry. Entries left blank are either optional or user defined.

Parameter	Description	Entry	Notes
NAME	POINT NAME		
UNIT	UNIT ID		
PTDESC	POINT DESCRIPTOR		
EUDESC	E.U. DESCRIPTOR		
KEYWORD	POINT KEYWORD		
PRIMMOD	PRIMARY MODULE POINT ID	----	
PTDISCL	POINT DISCLOSURE	FULL	
PVALGID	PV ALGORITHM	NULL	
CTLALGID	CONTROL ALGORITHM	CL	
OVERVAL	OVERVIEW DISPLAY VALUE		
SUPPIO	I/O DISPLAY SUPPRESSION	INPSUPPR	
PERIOD	POINT EXECUTION PERIOD	NOPERIOD	
NMODE	POINT'S NORMAL MODE	AUTO	
NMODATTR	NORMAL MODE ATTRIBUTE	OPERATOR	
MODEPERM	MODE PERMISSIVE	PERMIT	
EXTSWOPT	EXT MODE SWITCHING OPTION	NONE	
SPFORMAT	SP DECIMAL POINT FORMAT		
SPEUHI	SP EU HIGH RANGE in EUs		
SPEULO	SP EU LOW RANGE in EUs		
SPHILM	SP HIGH LIMIT		
SPLOLM	SP LOW LIMIT		
SP	SET POINT		
SPOPT	SET POINT OPTION	NONE	

Continued on next page

5.3 Analog Output (AM Regulatory Point Type), Continued

Point configuration

Analog Output Point Configuration, continued

Parameter	Description	Entry	Notes
CVTYPE	TYPE OF CV	ENGRUNIT	
INITTYPE	INITIALIZATION TYPE	EXT	
ARWDI	ANTI-RESET WINDUP DIRECTION	DIRECT	
NOCOPTS	NUMBER OF OUTPUT CONNECTIONS	0	
CVEUHI	CV ENGINEERING UNITS HIGH		
CVEULO	CV ENGINEERING UNITS LOW		
OP	CONTROL OUTPUT		
OPHILM	OP HIGH LIMIT VALUE IN %		
OPLOLM	OP LOW LIMIT VALUE IN %		
OPMCHLM	OP MINIMUM CHANGE IN %		
CLSLOTS	NUMBER OF CL BLOCKS	0	
NOPKG	No. OF CUSTOM DATA PACKAGES	2	
NUMSWTCH	NUMBER OF SWITCHES	0	
NOGINPTS	NUMBER OF GENERAL INPUTS	1	
NOGOPTS	NUMBER OF GENERAL OUTPUTS	0	
PKGNAME(1)	SOURCE FILE 1 NAME	O_AO	
PKGNAME(2)	SOURCE FILE 2 NAME	CLM_DMD3	

Continued on next page

5.3 Analog Output (AM Regulatory Point Type), Continued

Point configuration Analog Output Point Configuration, continued

Parameter	Description	Entry	Notes
CDS Parameters (O_AO) on the Control Algorithm Page			
OUTP	CURRENT OUTPUT VALUE		Updated from OPC data
CDS Parameters (CLM_DMD3) on Custom Page			
CLM_PPS	STORE TO PROCESS POINT PITE		Special Applications
CLM_OUT	STORE TO PROCESS OUTPUT PITE		Special Applications
DEV_STA	DEVICE IN_SERV (ON/OFF)		Updated by CLM S/W
PT_FAIL	DEVICE COMPONENT FAILURE		Not used for OPC
HDW_REF	HARDWARE REFERENCE		Not used for OPC
CLM_PORT	NAME OF SERVER OBJECT (PORT)		Updated by CLM Builder
CLM_DEV	NAME OF DEVICE		Updated by CLM Builder
CLM_PIT	NAME OF OPC GROUP (PIT)		Updated by CLM Builder

Builder input file format

Analog Output Point BIF (.ZI) Format

Using the CLM Builder, the OPC data for the Analog Output type point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_AO	Template Name (.ZT)
NAME		LCN Analog Output point name
PIT_NAME		PIT Name (OPC Group Name)
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_SP		OPC Item Name (Required)
SP_OCC		SP Output Conversion code Default = 2-4 (OPC floating point)
END		End of current entry

5.3 Analog Output (AM Regulatory Point Type), Continued

Example - O_AO

Analog Output point AO100.SP maps to Group GRP1 and Item OUTPUT1. The item OUTPUT1 is a 32-bit floating point that needs to be output to with no scaling and on operator demand exception processing. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_AO  
NAME = AO100  
PIT_NAME = GRP1  
DPOINT = --  
IT_SP = OUTPUT1  
SP_OCC = 2-4  
END
```

5.4 Numeric Point (AM Numeric Point Type)

Introduction

The AM Numeric point images a single OPC Item as an analog value in the Server object. This section describes how to create and configure this point type. Note that all parameters below are not necessarily shown unless they are required or may be utilized by the design engineer.

Theory of operation

Input and output may be accomplished for this point directly to or from the PV of the image point. The PV is mapped to the OPC Data Item and is updated from the Client.

Point configuration

Numeric Point Configuration

Using the LCN Data Entity Builder, an AM image point must be built, using the table following as a template for data entry. Entries left blank are either optional or user defined.

Parameter	Description	Entry	Notes
NAME	POINT NAME		
UNIT	UNIT ID		
PTDESC	POINT DESCRIPTOR		
KEYWORD	POINT KEYWORD		
ASSOCDSP	Schematic Invoked Using the "ASSOC DISP" key		
PRIMMOD	PRIMARY MODULE POINT ID	----	
EUDESC	E. U. Descriptor		
RANGEHI	PV Range Hi		
RANGELO	PV Range Lo		
PVFORMAT	PV Decimal Point Format		
PV	PV		

5.4 Numeric Point (AM Numeric Point Type), Continued

Builder input file format

Numeric Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Numeric point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_NUMERIC	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_PV		OPC Item Name (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
SB_OOS	Y or N	Set the value BAD when the associated device is placed out of service (Analog only) (Default = Y, set value BAD on out of service)
PV_ICC		PV Input Conversion code (Default = 2-4, floating point)
PV_OCC		PV Output Conversion code (Default = --, no output)
END		End of current entry

Note: CDS packages cannot be attached to AM Numeric Point Types.

Example – O_NUMERIC

AM Numeric point NUM100.PV maps to Group GRP1 and Item VALUE1. The item VALUE1 is a 16-bit signed integer that needs to be read in with (x10) scaling, a 0.5 deadband, and on data change only. There is to be no output capability from the numeric point. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_NUMERIC
NAME = NUM100
PIT_NAME = GRP1
DPOINT = --
IT_PV = VALUE1
INPSCHED = N
SB_OOS = Y
PV_ICC = 2-3,6>0.5
PV_OCC = --
END
```

5.5 Digital Input (AM Switch Point Type)

Introduction

The Digital Input point uses the AM Switch point type to image one or two logical OPC Items in the OPC Server object.

CDS parameters that are a function of the number of states are included in the separate CLM Platform software as packages DIN3CDS and DIN4CDS.

Theory of Operation

The OPC Client application supports the three and four state Digital Input points as defined within the *CLM Platform Implementation Manual*.

Logical applications in the OPC-based device that are defined as a single on/off state (or bit) are mapped using the 3-state Digital Input.

Logical applications in the OPC-based device that are defined using two bits to describe the logical state (e.g. the limit switches on a motor operated valve (MOV)) are mapped using the 4-state Digital Input.

This section only contains information for mapping the data for an application specific device.

Point Configuration

The Digital Input point type is Platform specific and therefore not defined in this manual. Please refer to the *CLM Platform Implementation Manual* for assistance in building these image points.

Custom Data Segments

The OPC-based Digital Input point type does not use any additional CDS packages. Only the standard CLM platform packages apply.

5.5 Digital Input (AM Switch Point Type), Continued

Builder input file format – 3 State DI

3-State Digital Input Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for Digital Input point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_DIN3	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		NDT PIT Table Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = --", not used)
IT_D1		OPC Item Name (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code (Default = 3-11, LCN logical to OPC Boolean)
END		End of current entry

Example - O_DIN3

AM Digital Input point PUMP100 maps to Boolean data item PUMPSTATUS in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_DIN3
NAME = PUMP100
PIT_NAME = GRP1
DPOINT = --
IT_D1 = PUMPSTATUS
INPSCHED = N
D1_ICC = 3-11
END
```

5.5 Digital Input (AM Switch Point Type), Continued

Builder input file format – 4 State DI

4-State Digital Input Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for Digital Input point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_DIN4	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		NDT PIT Table Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code (Default = 3-11, LCN logical to OPC Boolean)
IT_D2		OPC Item Name – Bit 2 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D2_ICC		D2 (Bit 2) Input Conversion code (Default = 3-11, LCN logical to OPC Boolean)
END		End of current entry

Example - O_DIN4

AM Digital Input point MOV100 maps to Boolean data items MOVOPEN and MOVCLOSE in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_DIN4
NAME = MOV100
PIT_NAME = GRP1
DPOINT = --
IT_D1 = MOVOPEN
INPSCHED = N
D1_ICC = 3-11
IT_D2 = MOVCLOSE
INPSCHED = N
D2_ICC = 3-11
END
```

5.6 Digital Composite (AM Switch Point Type)

Introduction

The Digital Composite point uses the AM Switch point type to collect up to two logical OPC Item values and output a combination of up to three logical OPC Item values in a single image point.

CDS parameters that are a function of the number of states are included in the separate CDS packages within the CLM Platform.

Theory of Operation

The OPC Client application supports the three, four and five state Digital Composite points as defined within the *CLM Platform Implementation Manual*.

Logical applications in the OPC-based device that use a single logical bit (1 or 0) for feedback are mapped using the 3-state Digital Composite. This point type will support either one or two required logical outputs.

Logical applications in the OPC-based device that use a two logical bits (1-0, 1-1, 0-1, 0-0) for feedback are mapped using the 4-state or 5-state Digital Composite. This point type will support either one, two or three required logical outputs.

Logical applications in the OPC-based device that use a two logical bits (1-0, 1-1, 0-1, 0-0) for feedback AND require three logical outputs (e.g. MOV w/STOP capability) are mapped using the 5-state Digital Composite.

This section only contains information for mapping the data for an application specific device.

Point Configuration

The Digital Composite point type is Platform specific and therefore not defined in this manual. Please refer to the *CLM Platform Implementation Manual* for assistance in building these image points.

Template Files are Configuration - Specific

The template files for the Client application have been tailored to account for several different configurations of digital composite points. These include:

- O_D31 – Digital Composite 3-State (1 input, 1 output)
 - O_D32 – Digital Composite 3-State (1 input, 2 output)
 - O_D41 – Digital Composite 4-State (2 input, 1 output)
 - O_D42 – Digital Composite 4-State (2 input, 2 output)
 - O_D5 – Digital Composite 5-State (2 input, 3 output)
-

5.6 Digital Composite (AM Switch Point Type), Continued

Builder input file
format –
O_D31

Use this format when the digital composite point has one input and only one output to be mapped.

Digital Composite Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Digital Composite point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_D31	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_OP1		OPC Item Name – Output Bit 1 (Required)
IT_FB1		OPC Item Name - Feedback Bit 1 (Required)
END		End of current entry

**Example -
O_D31**

AM Digital Composite point PUMP100 maps to an input Boolean data item PUMPSTATUS and an output control data item STARTSTOP in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_D31
NAME = PUMP100
PIT_NAME = GRP1
DPOINT = --
IT_D1 = PUMPSTATUS
INPSCHED = N
D1_ICC = 3-11
IT_OP1 = STARTSTOP
IT_FB1 = STARTSTOP
END
```

5.6 Digital Composite (AM Switch Point Type), Continued

Builder input file
format -
O_D32

Use this format when the digital composite point has one input and two outputs to be mapped.

Digital Composite Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Digital Composite point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_D32	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_OP1		OPC Item Name – Output Bit 1 (Required)
IT_FB1		OPC Item Name - Feedback Bit 1 (Required)
IT_OP2		OPC Item Name – Output Bit 2 (Required)
IT_FB2		OPC Item Name - Feedback Bit 2 (Required)
END		End of current entry

**Example -
O_D32**

AM Digital Composite point PUMP200 maps to an input Boolean data item PUMPSTATUS. The output control for start and stop of the pump is through two data items, PUMPSTART and PUMPSTOP. All items are in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_D32
NAME = PUMP200
PIT_NAME = GRP1
DPOINT = PUMP    (Note the use of the item name common prefix)
IT_D1 = STATUS
INPSCHED = N
D1_ICC = 3-11
IT_OP1 = START
IT_FB1 = START
IT_OP2 = STOP
IT_FB2 = STOP
END
```

5.6 Digital Composite (AM Switch Point Type), Continued

Builder input file format - O_D41

Use this format when the digital composite point has two inputs and one output to be mapped.

Digital Composite Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Digital Composite point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_D41	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_D2		OPC Item Name – Bit 2 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D2_ICC		D2 (Bit 2) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_OP1		OPC Item Name – Output Bit 1 (Required)
IT_FB1		OPC Item Name - Feedback Bit 1 (Required)
END		End of current entry

Example - O_D41

AM Digital Composite point MOV200 maps to input Boolean data items OPENED and CLOSED. The output control for opening and closing of the valve is through a single data item CONTROL. All items are in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_D41
NAME = MOV200
PIT_NAME = GRP1
DPOINT = --
IT_D1 = OPENED;      INPSCHED = N;      D1_ICC = 3-11;
IT_D2 = CLOSED;      INPSCHED = N;      D2_ICC = 3-11;
IT_OP1 = CONTROL
IT_FB1 = CONTROL
END
```

5.6 Digital Composite (AM Switch Point Type), Continued

Builder input file
format -
O_D42

Use this format when the digital composite point has two inputs and two outputs to be mapped.

Digital Composite Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Digital Composite point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_D42	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_D2		OPC Item Name – Bit 2 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D2_ICC		D2 (Bit 2) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_OP1		OPC Item Name – Output Bit 1 (Required)
IT_FB1		OPC Item Name - Feedback Bit 1 (Required)
IT_OP2		OPC Item Name – Output Bit 2 (Required)
IT_FB2		OPC Item Name - Feedback Bit 2 (Required)
END		End of current entry

Example -
O_D42

AM Digital Composite point MOV200 maps to input Boolean data items OPENED and CLOSED. The output control for opening and closing of the valve is through two separate data items, OPEN and CLOSE. All items are in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_D42
NAME = MOV200
PIT_NAME = GRP1
DPOINT = --
IT_D1 = OPENED;      INPSCHED = N;      D1_ICC = 3-11;
IT_D2 = CLOSED;      INPSCHED = N;      D2_ICC = 3-11;
IT_OP1 = OPEN;        IT_FB1 = OPEN;
IT_OP2 = CLOSE;       IT_FB2 = CLOSE;
END
```

5.6 Digital Composite (AM Switch Point Type), Continued

Builder input file
format -
O_D5

Use this format when the digital composite point has two inputs and three outputs to be mapped.

Digital Composite Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Digital Composite point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_D5	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_D1		OPC Item Name – Bit 1 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D1_ICC		D1 (Bit 1) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_D2		OPC Item Name – Bit 2 (Required)
INPSCHED	Y or N	Update the PV only when the PIT executes. (Default = N, update on data change)
D2_ICC		D2 (Bit 2) Input Conversion code. (Default = 3-11, LCN logical to OPC Boolean)
IT_OP1		OPC Item Name – Output Bit 1 (Required)
IT_FB1		OPC Item Name - Feedback Bit 1 (Required)
IT_OP2		OPC Item Name – Output Bit 2 (Required)
IT_FB2		OPC Item Name - Feedback Bit 2 (Required)
IT_OP3		OPC Item Name – Output Bit 3 (Required)
IT_FB3		OPC Item Name - Feedback Bit 3 (Required)
END		End of current entry

5.6 Digital Composite (AM Switch Point Type), Continued

Example – O_D5

AM Digital Composite point MOV300 maps to input Boolean data items OPENED and CLOSED. The output control for opening, closing, and stopping the valve is through three separate data items, OPEN, CLOSE, and STOP. All items are in GRP1 in the Server object. There is no inversion required. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_D5
NAME = MOV300
PIT_NAME = GRP1
DPOINT = --
IT_D1 = OPENED;      INPSCHED = N;      D1_ICC = 3-11;
IT_D2 = CLOSED;      INPSCHED = N;      D2_ICC = 3-11;
IT_OP1 = OPEN;       IT_FB1 = OPEN;
IT_OP2 = CLOSE;      IT_FB2 = CLOSE;
IT_OP3 = STOP;       IT_FB3 = STOP;
END
```

5.7 Flag Point (AM Flag Point Type)

Introduction

The AM Flag point images a single OPC Item as a logical value in the OPC Server object. This section describes how to create and configure this point type. Note that all parameters below are not necessarily shown unless they are required or may be utilized by the design engineer.

Theory of operation

The Flag point images an OPC Data Item that is either logically “ON” or “OFF”. The PV of the Flag point is mapped to the logical value. The Flag point can be configured to output a change in logical value or be used as a read-only indication. Logical inversion of the value may also be configured in the conversion codes.

Point configuration

Flag Point Configuration

Using the LCN Data Entity Builder, an AM image point must be built, using the table following as a template for data entry. Entries left blank are either optional or user defined.

Parameter	Description	Entry	Notes
NAME	POINT NAME		
UNIT	UNIT ID		
PTDESC	POINT DESCRIPTOR		
KEYWORD	POINT KEYWORD		
PRIMMOD	PRIMARY MODULE POINT ID	--	
PVSTATES0	UPPER BOX TEXT	<As desired>	
PVSTATES1	LOWER BOX TEXT	<As desired>	
UBOXCLR	UPPER BOX COLOR	<As desired>	
LBOXCLR	LOWER BOX COLOR	<As desired>	
OFFNRMAL	OFF NORMAL ALARMING	OFF ON	
PVNORMAL	PV NORMAL STATE		If OFFNRMAL selected

Note: The AM Flag Point Type does not allow CDS packages to be attached.

5.7 Flag Point (AM Flag Point Type), Continued

Builder input file format

Flag Point Builder Input File (.ZI) Format

Using the CLM Builder, the OPC data for the Flag point must be mapped from the LCN parameters to the OPC Data Items.

Prompt	Entry	Description
ENTRY	O_FLAG	Template Name (.ZT)
NAME		LCN Point Name
PIT_NAME		PIT Name
DPOINT		OPC Item Name Common Prefix (Optional) (Default = "--", not used)
IT_PV		OPC Item Name (Required)
INPSCHED	Y or N	Update PV only when the associated PIT executes. (Default = N, update on data change)
PV_ICC		Flag PV Input Conversion code. (Default = "5,2-11", LCN enumeration to OPC Boolean)
PV_OCC		Flag PV Output Conversion code. (Default = "--", no output)
END		End of current entry

Example - O_FLAG

AM Flag point FLG100.PV maps to Item PUMP1 in OPC Group GRP1. The item PUMP is a Boolean value that needs to be read on data change only, but the value needs to be inverted for display on the LCN. The pump is to be able to be controlled through the PUMP1 item as well. Assume Naming option 0.

Your CLM Builder entries might look like:

```
ENTRY = O_FLAG
NAME = FLG100
PIT_NAME = GRP1
DPOINT = --
IT_PV = PUMP1
INPSCHED = N
PV_ICC = 5,2-11,11
PV_OCC = 5,2-11,11
END
```


Section 6 - OPC Client Status Displays

6.1 Overview

Introduction

The OPC Client display set gives access to specific device data statistics regarding configuration, communication and data item quality values.

There are Status display layouts for the virtual CLMI, Server object assignment (port), and remote device in the Client application. These displays provide dynamic data (4 second refresh) to the user about the system process and are accessed from the CLM Network Detail Display. The format of these displays is defined within a file of the same template name and a “.ZS” extension on the History Module.

These displays are all accessible from the OPC Remote device detail display. The following steps may be performed to access these displays.

Step	Action
6.1.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
6.1.2	Select the CLM node number.
6.1.3	At the bottom of the display, select the desired port number.
6.1.4	Select the DETAIL STATUS target.
6.1.5	When the device detail display invokes, select the specific device you are interested in. Functional targets will appear at the bottom of the display.
6.1.6	Select the DETAIL STATUS target to invoke the detail status display.

6.1.1 CLMI Detail Status Display (CLMI.ZS)

CLMI Detail Status Display

The CLMI Detail Status Display is a APP/CLM Platform function and is not Client or application specific. The data on this page is updated by the Platform function. Data is tracked for the average number of parameters (both input and output) per second on a 4 second window and a 5 minute window. Another critical piece of data tracked is the number of PPS's being issued to the image points by Client processing whenever data item values are input that meet deadband change requirements.



Figure 6.1-1 CLMI Detail Status Display (CLMI.ZS)

6.1.2 Server Object (Port) Detail Status Display (O_PORT.ZS)

**Server Object (Port)
Detail Status Display**

The server object detail status display gives the user dynamic information for Client/Server communications and quality values from data items. The number of items to and from the server object is tracked and displayed on this screen. The Client Release version and revision identifier can be found on this display.

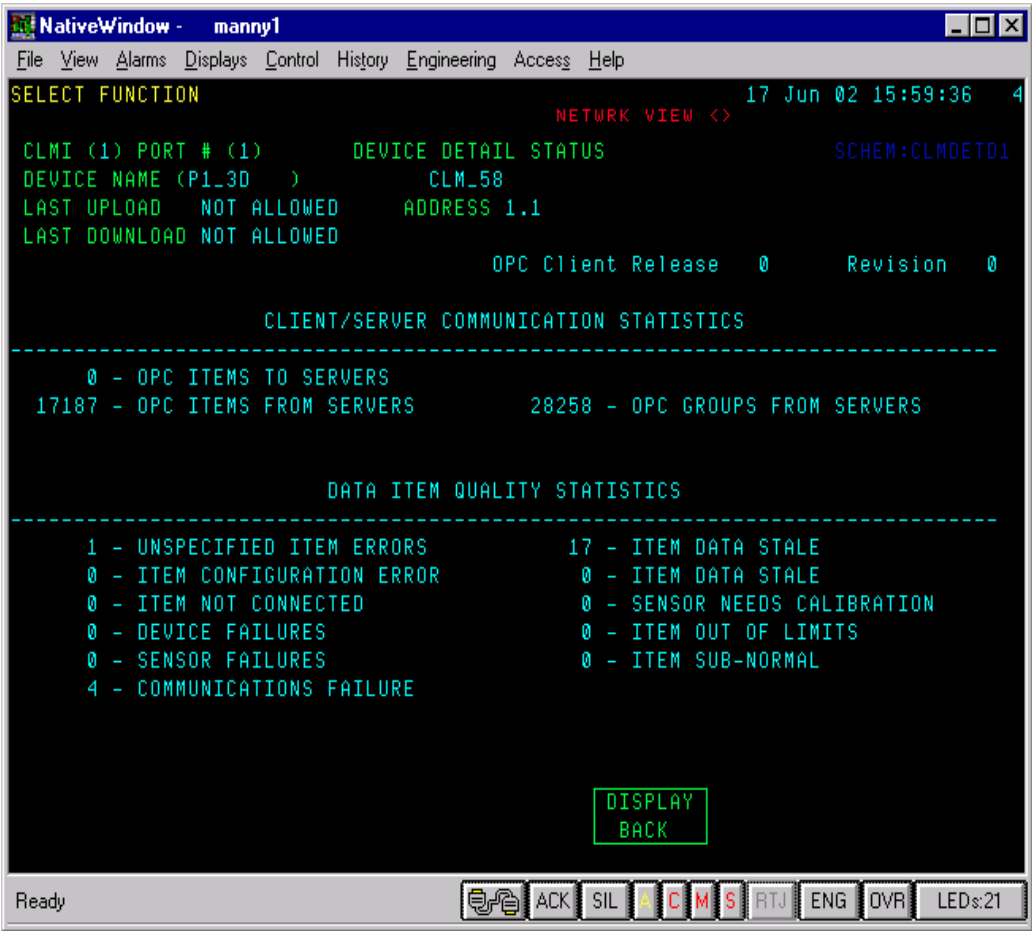


Figure 6.1-2 Server Object (Port) Detail Status Display (O_PORT.ZS)

6.1.3 Device Detail Status Display (O_DEVICE.ZS)

Device Detail Status Display

The device detail status display gives the user dynamic information for Client/Server communications, Data item quality values, and device item configuration. The number of items to and from the server object for this device is tracked and displayed on this screen. The total number of items configured on this device is available on this display.

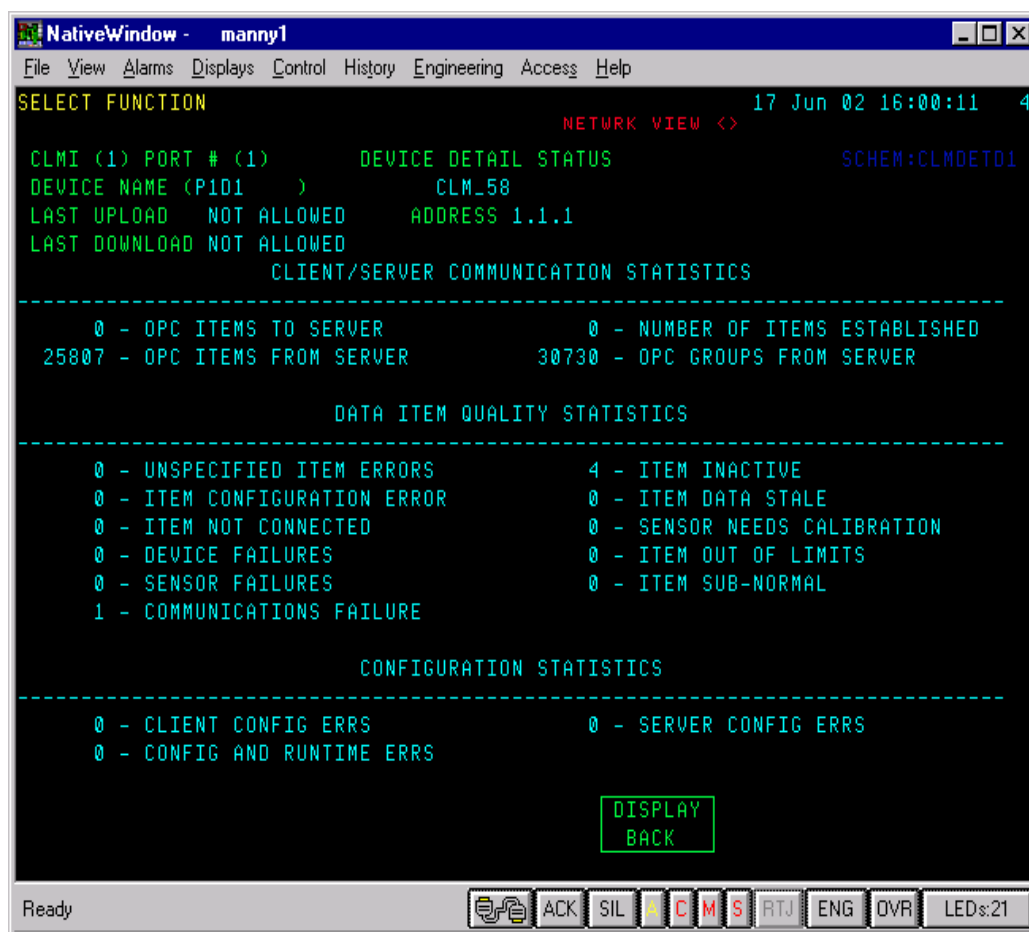


Figure 6.1-3 OPC Device Detail Status Display (O_DEVICE.ZS)

Section 7 – Client Display Operations

7.1 Starting Up the First Time

Establish Communications

After you have completed your configuration, you are ready to establish Client/Server communications and begin the validation of data received from the Server object group and items.

The steps following will guide you through this process. Please note that encountering errors may require you to go back into the configuration process to make changes. If this is the case, start this process again from the beginning.

Continued

Start the Server(s)

The first thing that needs to be done is to place the virtual CLMI in service and confirm that the server(s) startup.

Step	Action
7.1.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.1.2	Select the CLM node number.
7.1.3	When the lower half of the display appears, select the virtual CLMI target on the screen that was configured earlier in step 3.3.21. The target will highlight and functional targets will appear at the bottom of the display.
7.1.4	Select the CHANGE STATE target. Functional targets for IN SERVICE and OUT OF SERVICE will appear to the left.

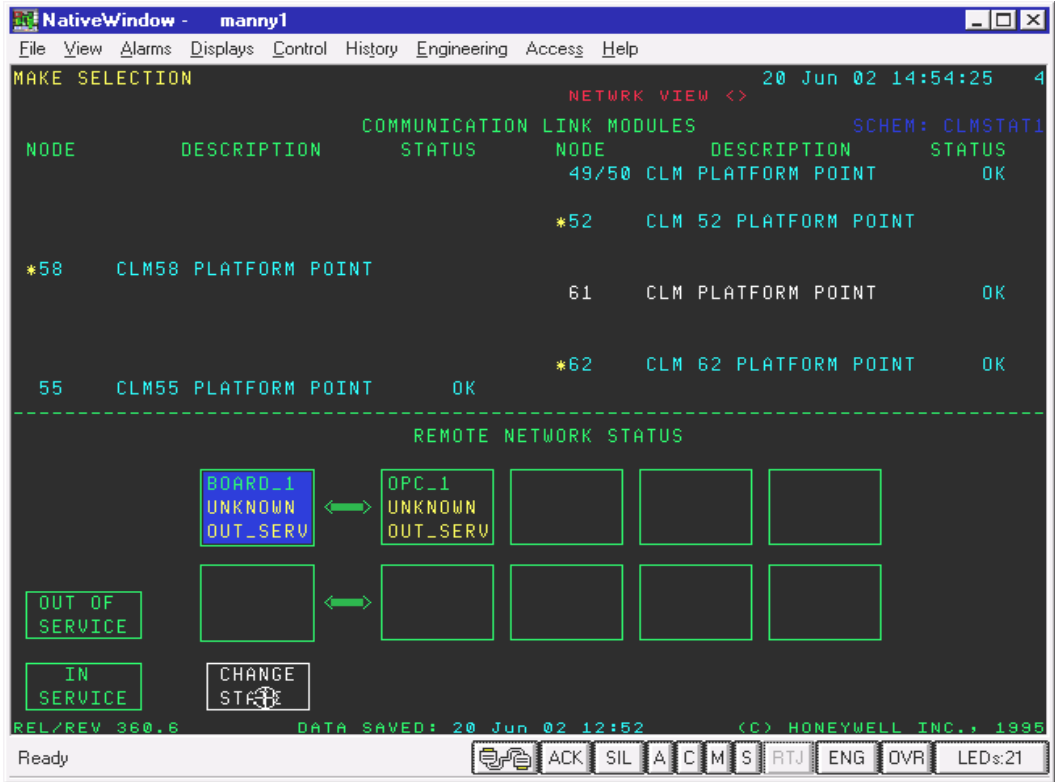


Figure 7.1-1 CLMI Change State from Out of Service

Continued on next page

7.1 Starting Up the First Time, Continued

Step	Action
7.1.5	Select the IN SERVICE target and press ENTER. After a few moments, the CLMI should change from UNKNOWN/OUT_SERV to OK/IN_SERV.

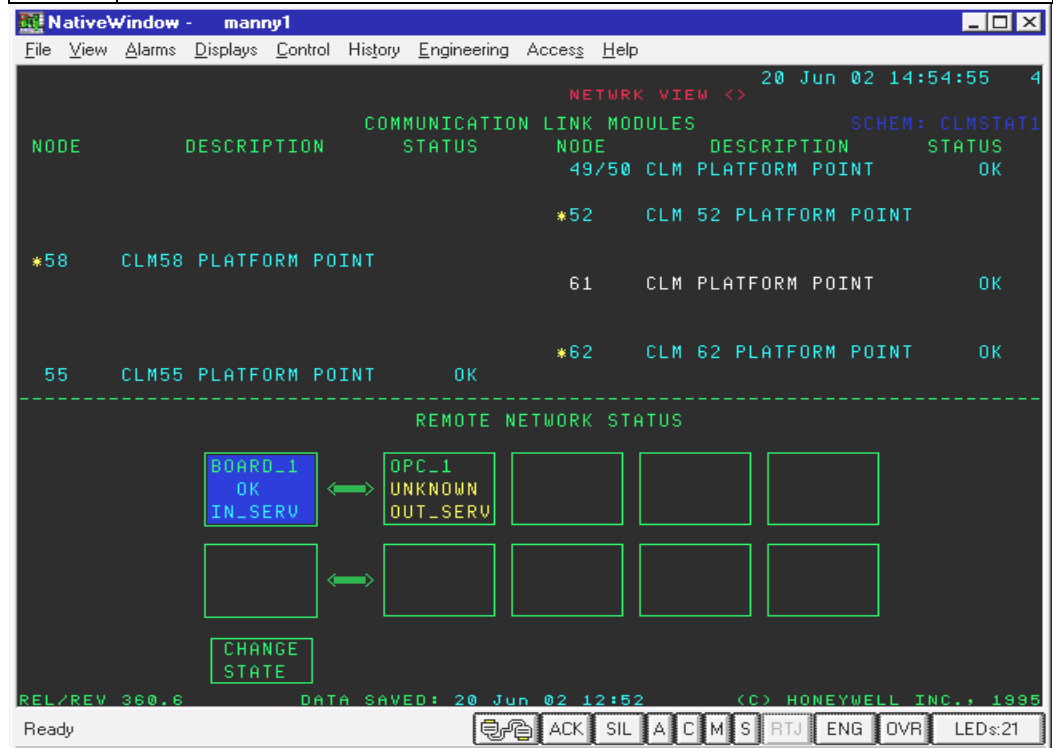


Figure 7.1-2 CLMI Change State to In Service

7.1.6	At this point, the server(s) should have been started in the APP/CLM node. At the PC, confirm that they are indeed active and running. This may be accomplished, depending on the server, by a visual indication such as an icon, or checking the services for execution. Important! Do not proceed beyond this point until you are satisfied that your server(s) is running properly. If it is not, the Event Viewer under the Administrative Tools should be examined for information on resolving the problem. Important! If you are using DCOM to start up your server in a remote node, confirm the startup of the Server in that node. There may not be a visible indication at the APP/CLM node.
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7.1 Starting Up the First Time, Continued

Step	Action
7.1.7	Once the server(s) is confirmed to be operational, each of the Client instances will need to be placed in service.
7.1.8	Select the server instance (port) target that connects to the Server object. The target will highlight and the functional target for CHANGE STATE will again appear.
7.1.9	Select the CHANGE STATE target. Functional targets for IN SERVICE and OUT OF SERVICE will appear to the left.

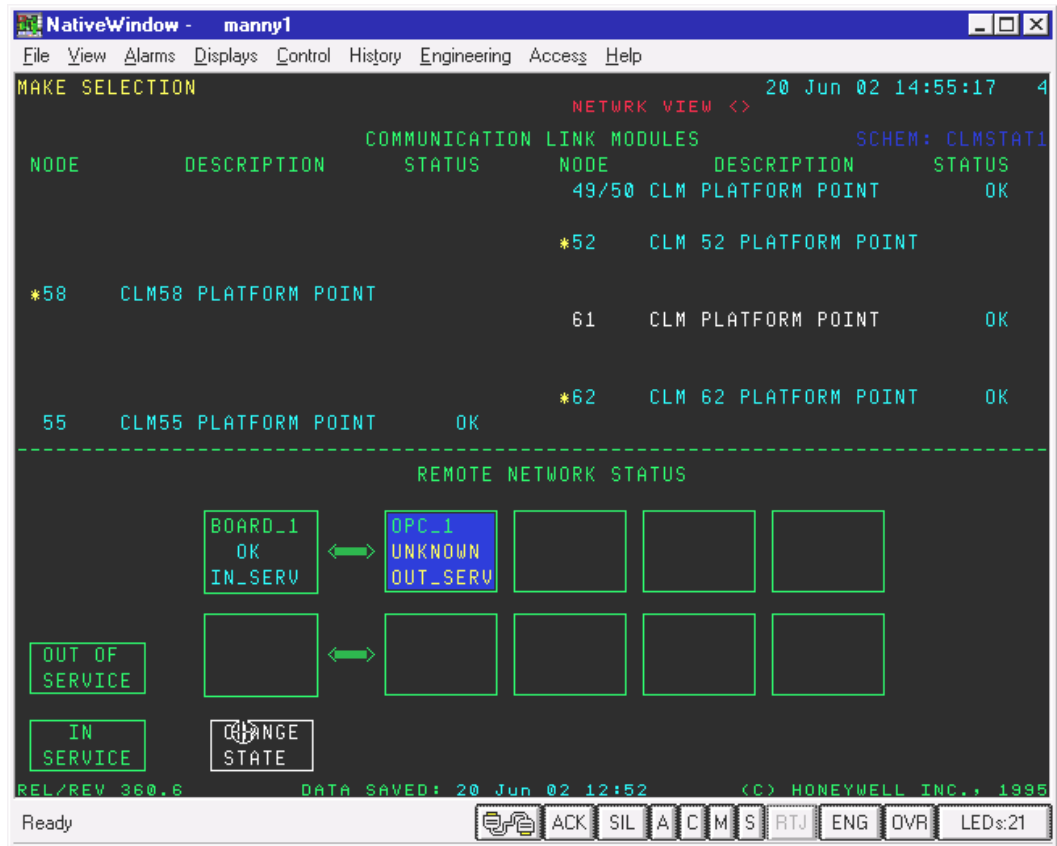


Figure 7.1-3 Client instance (port) Change State from Out of Service

Continued on next page

7.1 Starting Up the First Time, Continued

Step	Action
7.1.10	Select the IN SERVICE target and press ENTER. After a few moments, the Server instance (port) should change from UNKNOWN/OUT_SERV to OK/IN_SERV.

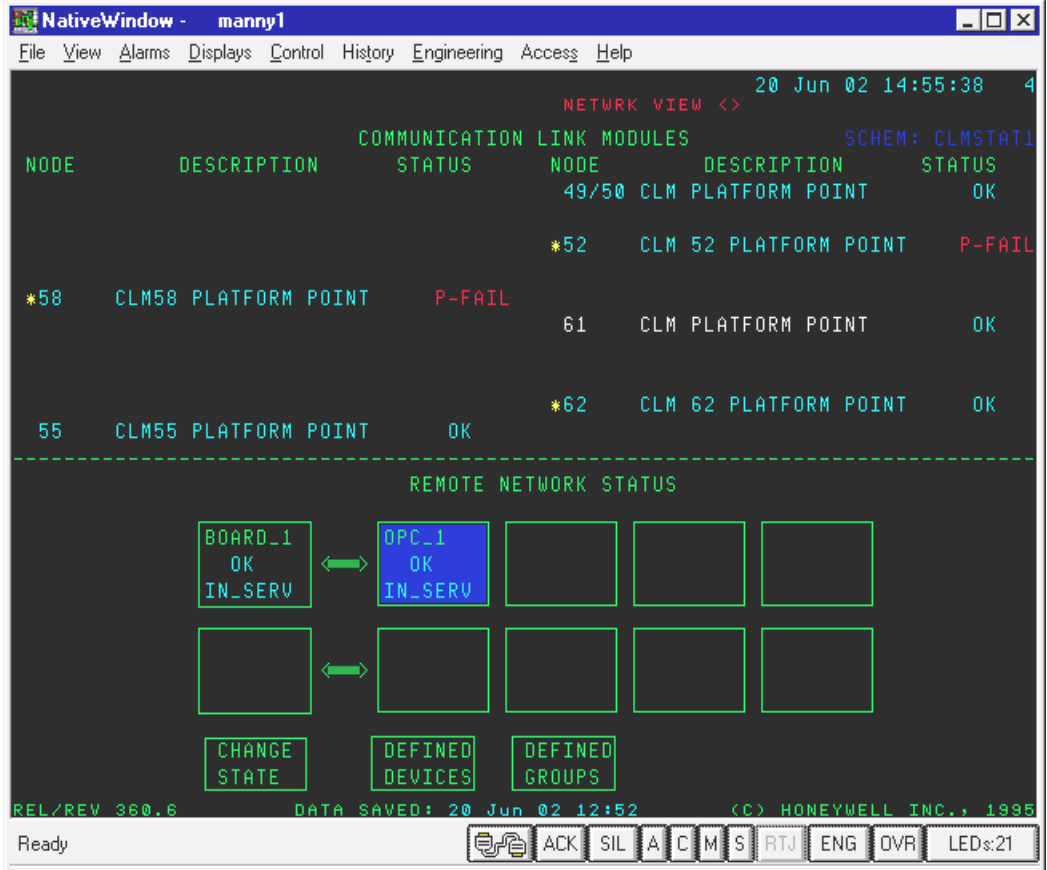


Figure 7.1-4 Client instance (port) Change State to In Service

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7.1 Starting Up the First Time, Continued

Step	Action
7.1.11	After verifying that the port has successfully gone into Service, select the DETAIL STATUS target at the bottom of the display to invoke the Device Detail Status display.
7.1.12	When the display invokes, select the device assigned to the OPC group and data items to be checked out. Reminder! Even if the Server object does not support a device concept, a device will have been built to support the group and item mapping configurations in the APP/CLM.
7.1.13	When the device target is selected, the functional target for CHANGE STATE will again appear at the bottom of the display.
7.1.14	Select the CHANGE STATE target. The functional targets for IN SERVICE and OUT OF SERVICE will appear to the left.

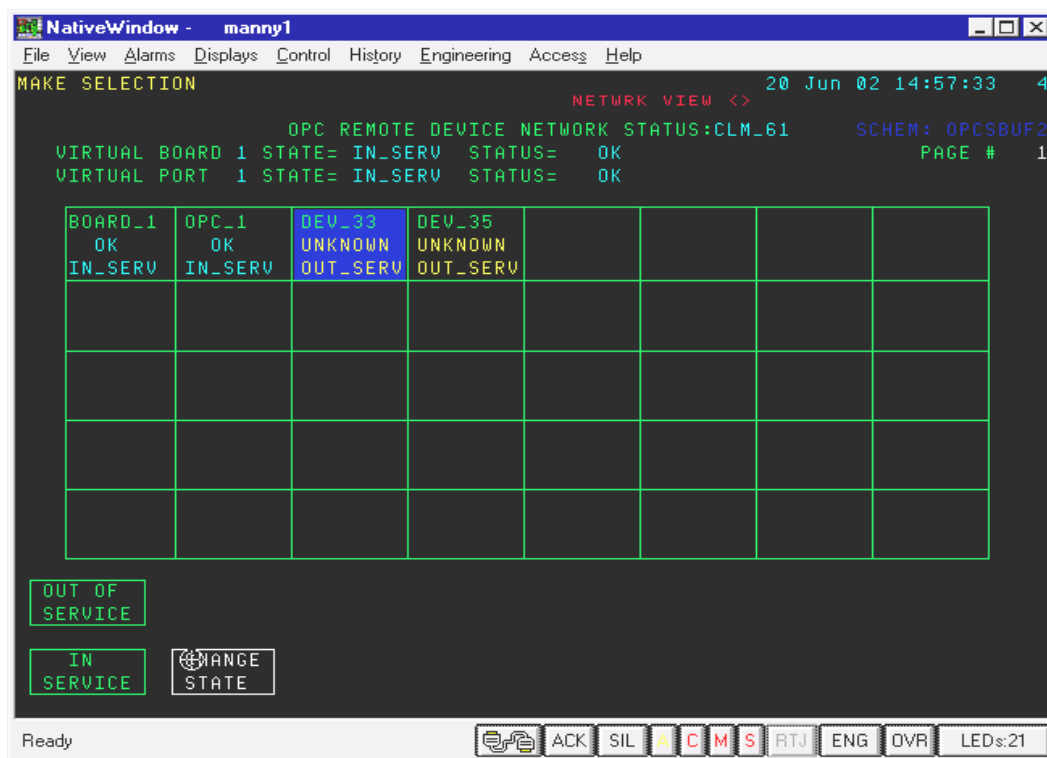


Figure 7.1-5 Device Change State from Out of Service

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7.1 Starting Up the First Time, Continued

Step	Action
7.1.15	Select the IN SERVICE target and press ENTER. The device should change from UNKNOWN/OUT_SERV to OK/SYNC and finally to OK/IN_SERV. Note: Depending on the number of data items that must be synchronized, you may or may not see the SYNC state transition. Note: The Status of the device may change to P_FAIL/IN_SERV in some cases. This indicates that there were some errors in synchronizing the data items, but general data acquisition may be available. You will want to access the Application log of the Event Viewer for specific details on the errors.

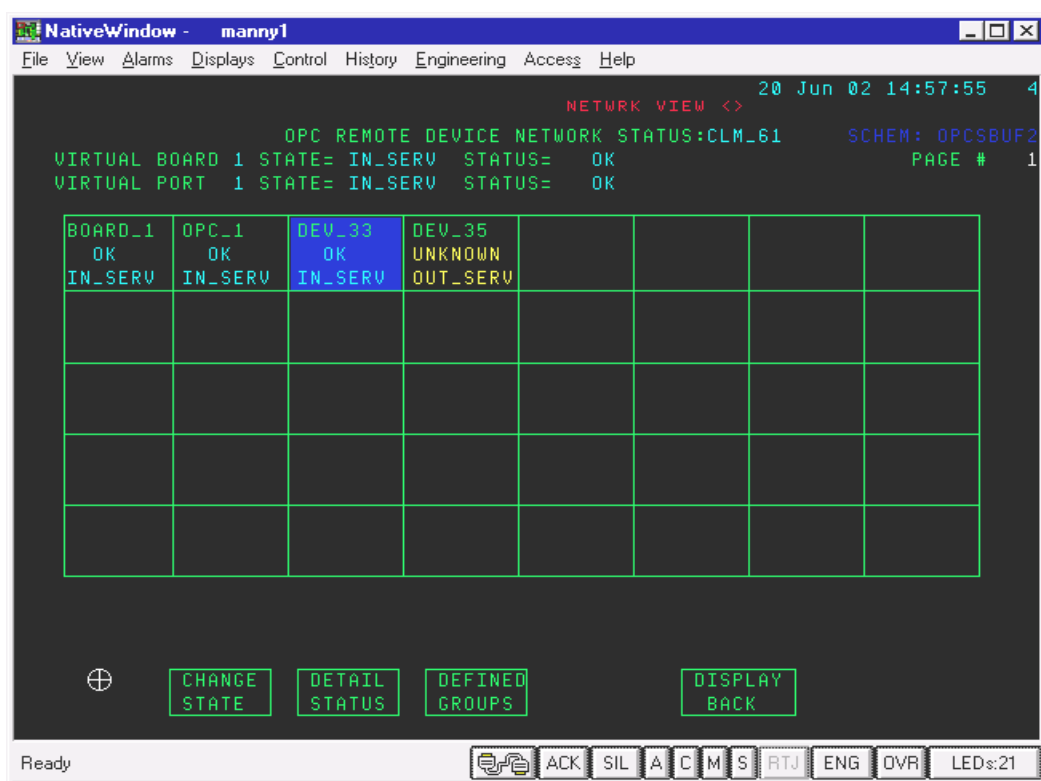


Figure 7.1-6 Device Change State to in Service

Step	Action
7.1.16	Repeat these steps for all other server instances and devices that need to be brought on line for Client/Server data acquisition.
7.1.17	From the AM Status Display, perform a SAVE DATA to save all of the changes that have been made.

7.2 Take a device Out of Service

Out of Service

There may be times when you have need to logically place a remote device out of service for various reasons. This will have the effect of ceasing all data acquisition and allowing no outputs to groups and data items assigned through that device.

This action is performed by taking the following steps.

Step	Action
7.2.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.2.2	Select the CLM node number.
7.2.3	When the lower half of the display appears, select the virtual CLMI target on the screen that was configured earlier in step 3.3.21. If the CLMI is the device to be taken out of service, go to 7.2.7
7.2.4	Select the Client instance (port) desired. If the port is the device to be taken out of service, go to 7.2.7
7.2.5	Select the DETAIL STATUS target at the bottom of the display.
7.2.6	When the display invokes, select the desired device.
7.2.7	At the bottom of the display, select the CHANGE STATE target. The functional targets IN SERVICE and OUT OF SERVICE will appear to the left.
7.2.8	Select the OUT OF SERVICE target and press ENTER. The device and all subordinate devices will now cease data acquisition functions.

7.3 Place a device In Service

In Service

You will need to place a device logically in service to be able to begin data acquisition and allow outputs to groups and data items assigned through that device. Creation of the OPC groups and items in the Server object actually takes place at the time the device is placed in service as part of the SYNCH process.

This action is performed by taking the following steps.

Step	Action
7.3.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.3.2	Select the CLM node number.
7.3.3	When the lower half of the display appears, select the virtual CLMI target on the screen that was configured earlier in step 3.3.21. If the CLMI is the device to be placed in service, go to 7.3.7
7.3.4	Select the Client instance (port) desired. If the port is the device to be placed in service, go to 7.3.7
7.3.5	Select the DETAIL STATUS target at the bottom of the display.
7.3.6	When the display invokes, select the desired device.
7.3.7	At the bottom of the display, select the CHANGE STATE target. The functional targets IN SERVICE and OUT OF SERVICE will appear to the left.
7.3.8	Select the IN SERVICE target and press ENTER. The device will now synchronize and begin data acquisition functions.

7.4 Check an Item Quality Error Code

BAD Value at image Point Parameter

At some point, a particular mapped image point parameter may be set to BAD by the Client application. This is generally an indication that a quality value has been returned from the server that indicates a problem with the specific data item. The following steps will walk you through the process of obtaining the details of the bad quality code value so that corrections can be made.

Step	Action
7.4.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.4.2	Select the APP/CLM node number. This will bring up the Network Status details on the lower half of the display.
7.4.3	At the bottom of the display, select the Client instance (port) where the device and group reside that contains the item that is mapped to the image point parameter. This will bring up functional and detail targets at the bottom of the display. Note: With the exception of the Flag and Numeric point types, port and PIT information may be obtained from the CLM_DMD3 CDS package attached to the image point containing the BAD value.
7.4.4	Select the DEFINED GROUPS target at the bottom of the display.
7.4.5	When the display invokes, select the Group (PIT) name containing the item mapped to the image point parameter. This will invoke functional and detail targets at the bottom of the display.
7.4.6	Select the MAPPED LCN PNTS target at the bottom of the display.
7.4.7	When the display invokes, locate and <i>select</i> the image point name from the list shown. Note: This may require using the PAGE_FWD target or button on the console. This will bring up functional and detail targets at the bottom of the display.
7.4.8	Select the DEFINED ITEMS target at the bottom of the display.

Continued on next page

7.4 Check an Item Quality Error Code, Continued

Step	Action
7.4.9	When the display invokes, all of the mapped image point parameters and data items will be shown for that point. If the data item is input, the error code will be located under the ITEM column under the (INP) half of the ERROR CODES section. If the data item is output, the error code will be located under the ITEM column under the (OUT) half of the ERROR CODES section.. Make a note of the error code.
7.4.10	Refer to Appendix B of this manual for high-level information on the error code, or press the HELP key for on-line information.
7.4.11	For detailed information on the error code, go to the APP and log on as Administrator.
7.4.12	From the Start menu, go to Settings\Control Panel\Administrative Tools and invoke the Event Viewer.
7.4.13	Select the Application Log and do a Refresh.
7.4.14	Keying off of the image point name, use the Search function to look through the entries for detailed information on the error. There should normally be a matching entry in the application log for every quality error code that is posted to a mapped image point parameter.
7.4.15	Make corrections as necessary. Note: These corrections may require a re-synch of the item database, which could require taking the device out of service and back in service.

7.5 Scheduling A Group for Data Collection

Scheduled vs. Subscription

There may be occasion to process multiple pieces of data together at the LCN side at the same time on a periodic basis. This can be accomplished by scheduling an OPC group for data collection. In this process, the group and item data within is still collected by subscription and stored to the Network Memory entry. However, the data is not stored to the image point parameter at that time. When the OPC group is scheduled to run on its periodic basis, the item data within the group will be stored to all mapped image point parameters.

The steps below describe the process to schedule an OPC group.

Step	Action
7.5.1	From the Native Window of a GUS select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.5.2	Select the APP/CLM node number. This will bring up the Network Status details on the lower half of the display.
7.5.3	At the bottom of the display, select the Client instance (port) where the device and group reside that contains the item that is mapped to the image point parameter. This will bring up functional and detail targets at the bottom of the display. Note: With the exception of the Flag and Numeric point types, port and PIT information may be obtained from the CLM_DMD3 CDS package attached to the image point containing the BAD value.
7.5.4	Select the DEFINED GROUPS target at the bottom of the display.
7.5.5	When the display invokes, select the Group (PIT) name containing the item mapped to the image point parameter. This will invoke functional and detail targets at the bottom of the display. The display will look similar to Figure 7.5-1 below.

Continued on next page

7.5 Scheduling A Group for Data Collection, Continued

Group data is subscription only – not polled

Schedule an OPC Group for Data Collection

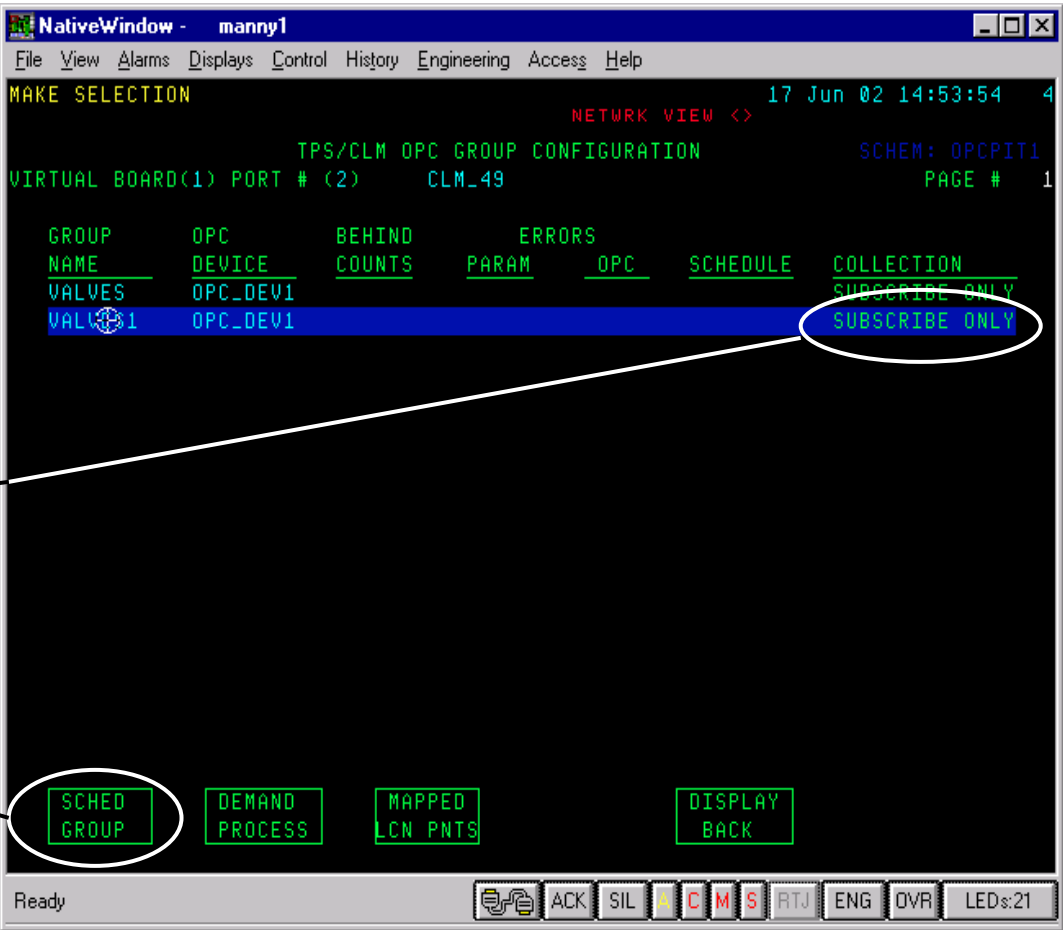


Figure 7.5-1 Group Configuration Display – Subscribed Data

Collection

Step	Action
7.5.6	Select the SCHED GROUP target and press ENTER.
7.5.7	Enter the numeric cycle value (0-2048). This is the value in seconds, minutes, or hours that is the schedule time.
7.5.8	Enter the cycle type code: S = seconds, M = minutes, H = hours.
7.5.9	Confirm that the display updates the selected group to show the schedule and indicate the POLLED/SUBSCRIBED data collection. See Figure 7.5-2 below.

Continued on next page

7.5 Scheduling A Group for Data Collection, Continued

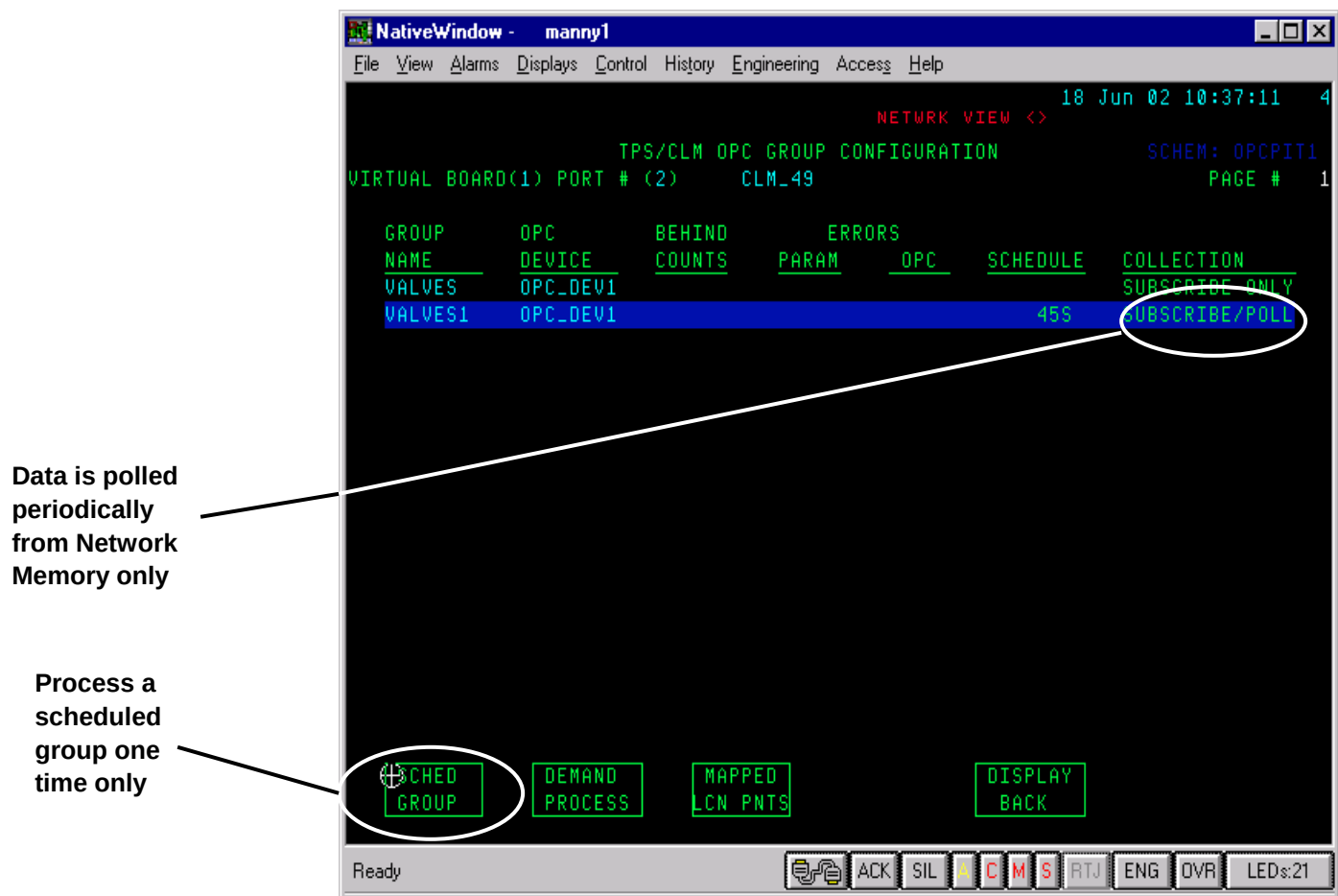


Figure 7.5-2 Group Config Display – Subscribed/Polled Data

Collection

Step	Action
7.5.10	The Group will now process on a periodic basis at the entered time schedule. Data will be retrieved from the item entries in Network Memory and stored to the mapped image point parameters for all assigned points. Note! It is important to note that the Client <u>does not</u> make a new call to the Server at this point. Data continues to be collected on a subscription basis and updated into the Network Memory entries. The scheduled execution of the group goes <u>only</u> to those entries to get the most recent data for updating the image point parameters.

7.6 Demand Process a Scheduled Group

Process a Scheduled group One Time

You may have a need to process a scheduled group a single time and not wait for the next periodic update. This could be the situation when scheduled updates are at longer durations (many minutes or possibly even hours). Selecting this option will process all parameters (only marked with INPSCHED = Y, see appendix A) of all mapped image points assigned to the selected group one time only. This means that the image point parameters will be updated and the point processed one time only.

Step	Action
7.6.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.6.2	Select the APP/CLM node number. This will bring up the Network Status details on the lower half of the display.
7.6.3	At the bottom of the display, select the Client instance (port) where the device and group reside that contains the item that is mapped to the image point parameter. This will bring up functional and detail targets at the bottom of the display. Note: With the exception of the Flag and Numeric point types, port and PIT information may be obtained from the CLM_DMD3 CDS package attached to the image point containing the BAD value.
7.6.4	Select the DEFINED GROUPS target at the bottom of the display.
7.6.5	When the display invokes, select the Group (PIT) name containing the item mapped to the image point parameter. This will invoke functional and detail targets at the bottom of the display. The display will look similar to Figure 7.6-1 below.

Continued on next page

7.6 Demand Process a Scheduled Group, Continued

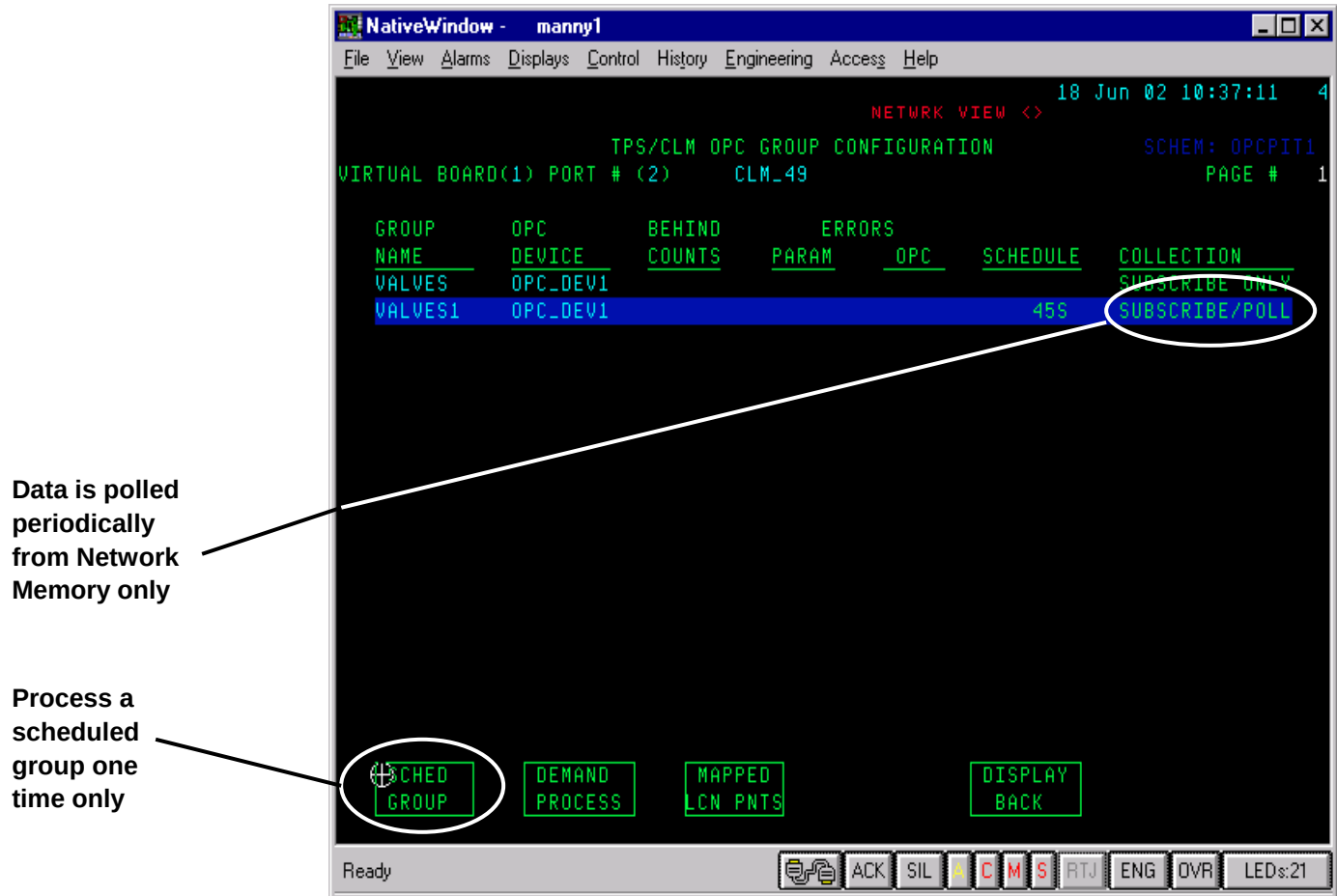


Figure 7.6-1 Group Configuration Display – Demand Process

Step	Action
7.6.6	Select the DEMAND PROCESS target and press ENTER. The Group will process exactly one time. Note: Execution of this demand function does not affect the normally scheduled process update.

7.7 Using the Mapped Data Item Display

Overview	Once you have transitioned through the OPC Client display set to select a particular image point, you will want to view the defined data items and mapped image point parameters.
Runtime Errors	This display will give you information on errors encountered for both input and output on reads and writes to the image point parameter and the data items themselves.
Dynamic Data Update Zone	Selecting a specific image point parameter on the display invokes a dynamic collection update zone at the bottom of the screen that gives the actual current parameter value, the current Network Memory entry value, and the last output value, if applicable.
Ability to expand Items to full size	The display automatically shows the last 40 (most significant) characters of the data item name for viewing. If the item name exceeds 40 characters, a “+” indication is displayed to the immediate left of the item name text. Selecting the item name will invoke an update zone at the bottom of the screen that expands the item name to its full maximum length (100 characters, with name option 7).
On-Line HELP	The display has a built in Help function that may be invoked by selecting the HELP key from the GUS/US console. This temporarily replaces the parameter/data item information with text describing the most common configuration and runtime error codes. Selecting the HELP key again returns the display to normal.

7.7.1 Invoke the Mapped Data Item Display

Mapped Data Item Display

To invoke the OPC Client mapped data item display, perform the following steps.

Step	Action
7.7.1.1	From the Native Window of a GUS, select SCHEM and enter CLMSCHEM. This will invoke the CLM Status Display.
7.7.1.2	Select the APP/CLM node number. This will bring up the Network Status details on the lower half of the display.
7.7.1.3	At the bottom of the display, select the Client instance (port) where the device and group reside that contains the item that is mapped to the image point parameter. This will bring up functional and detail targets at the bottom of the display. Note: With the exception of the Flag and Numeric point types, port and PIT information may be obtained from the CLM_DMD3 CDS package attached to the image point containing the BAD value.
7.7.1.4	Select the DEFINED GROUPS target at the bottom of the display.
7.7.1.5	When the display invokes, select the Group (PIT) name containing the item mapped to the image point parameter. This will invoke functional and detail targets at the bottom of the display.
7.7.1.6	Select the MAPPED LCN PNTS target.
7.7.1.7	When the image point overview display is invoked, select the image point you wish to view. Note: This may require the use of the PAGE FWD button.
7.7.1.8	At the bottom of the display, select the DEFINED ITEMS target.
7.7.1.9	This will invoke the Mapped Data Item display for the selected image point. The display will look similar to Figure 7.7-1 below.

Continued on next page

7.7.1 Invoke the Mapped Data Item Display, continued

Indicates Item
Name exceeds
40 characters
shown

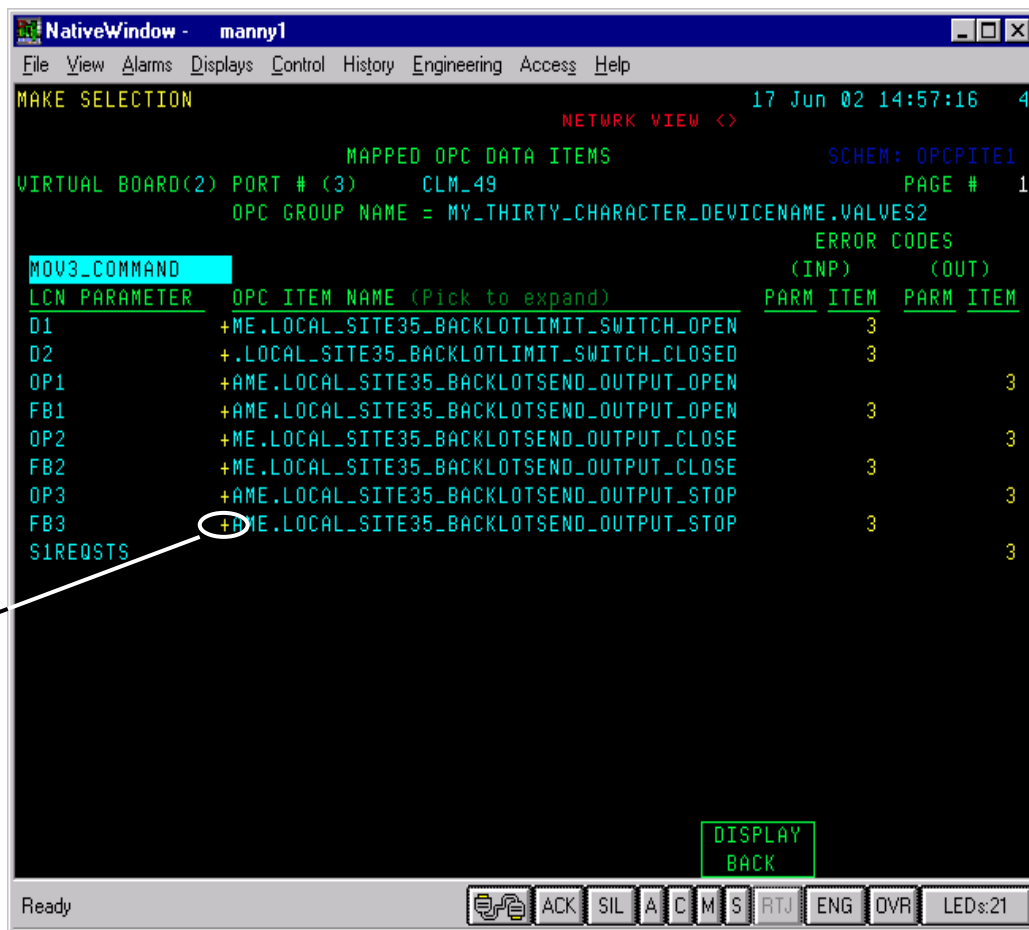


Figure 7.7-1 Mapped Data Item Display

7.7.2 Collect the Current Parameter/Item Data

Dynamic Data Update To check the current values of the mapped image point parameter/data item, perform the following steps.

Note: These steps continue from Section 7.1.1.

Step	Action
7.7.2.1	From the Mapped Data Item display, select the image point parameter that you wish to view. Important! You must select the image point parameter text string for this functionality. Selecting the data item text string will invoke the data item expansion function, discussed in the next section.
7.7.2.2	At the bottom of the screen, the current data for the image point parameter, Network Memory entry value, and last output value will be displayed. The values update every 4 seconds. See Figure 7.7-2 below.

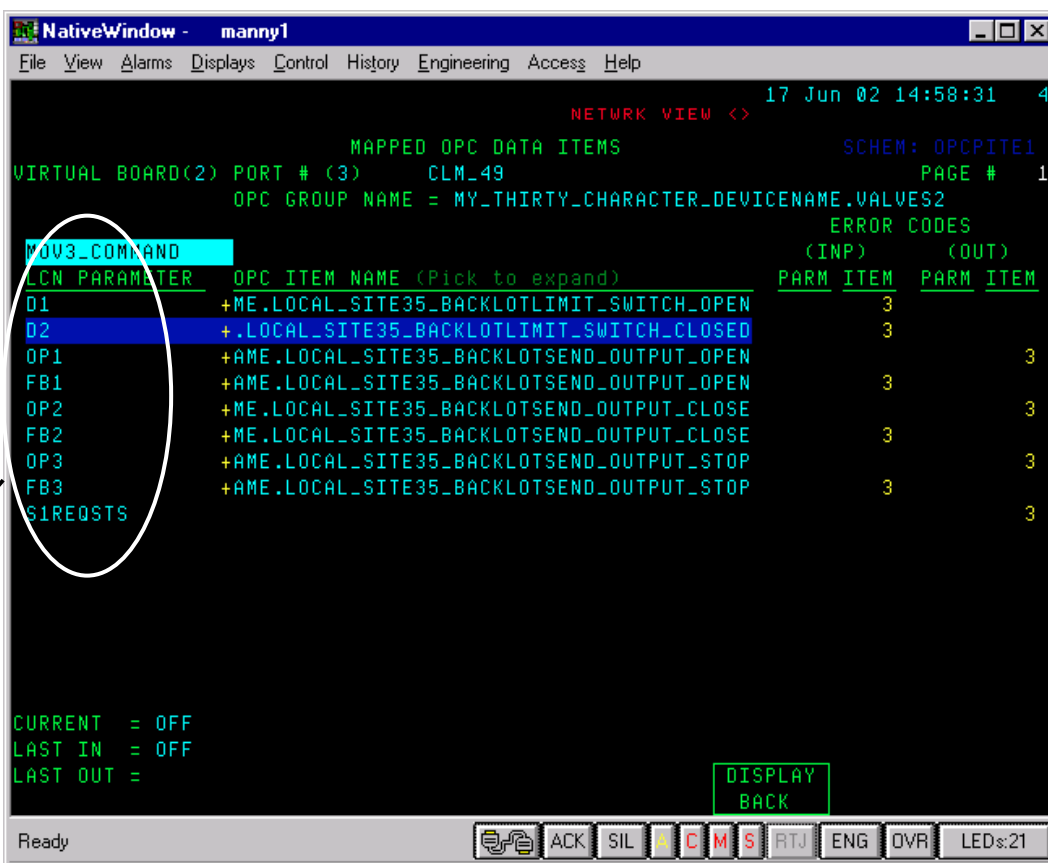


Figure 7.7-2 Mapped Data Item Display – Item Collection

7.7.3 Expand the Selected Data Item Name

Data Item Name can be up to 100 characters

Only the most significant (right-most) 40 characters are initially shown on the display. To expand the displayed 40 characters, perform the following steps below.

Note: These steps continue from Section 7.1.1.

Step	Action
7.7.3.1	From the Mapped Data Item display, select the data item that you wish to expand for viewing. Important! You must select the data item text string for this functionality. Selecting the image point parameter text string will invoke the data collection function, discussed in the previous section.
7.7.3.2	At the bottom of the screen, the full data item name is displayed (maximum of 100 characters). It is displayed in two rows, from top-right to bottom. See Figure 7.7-3 below.

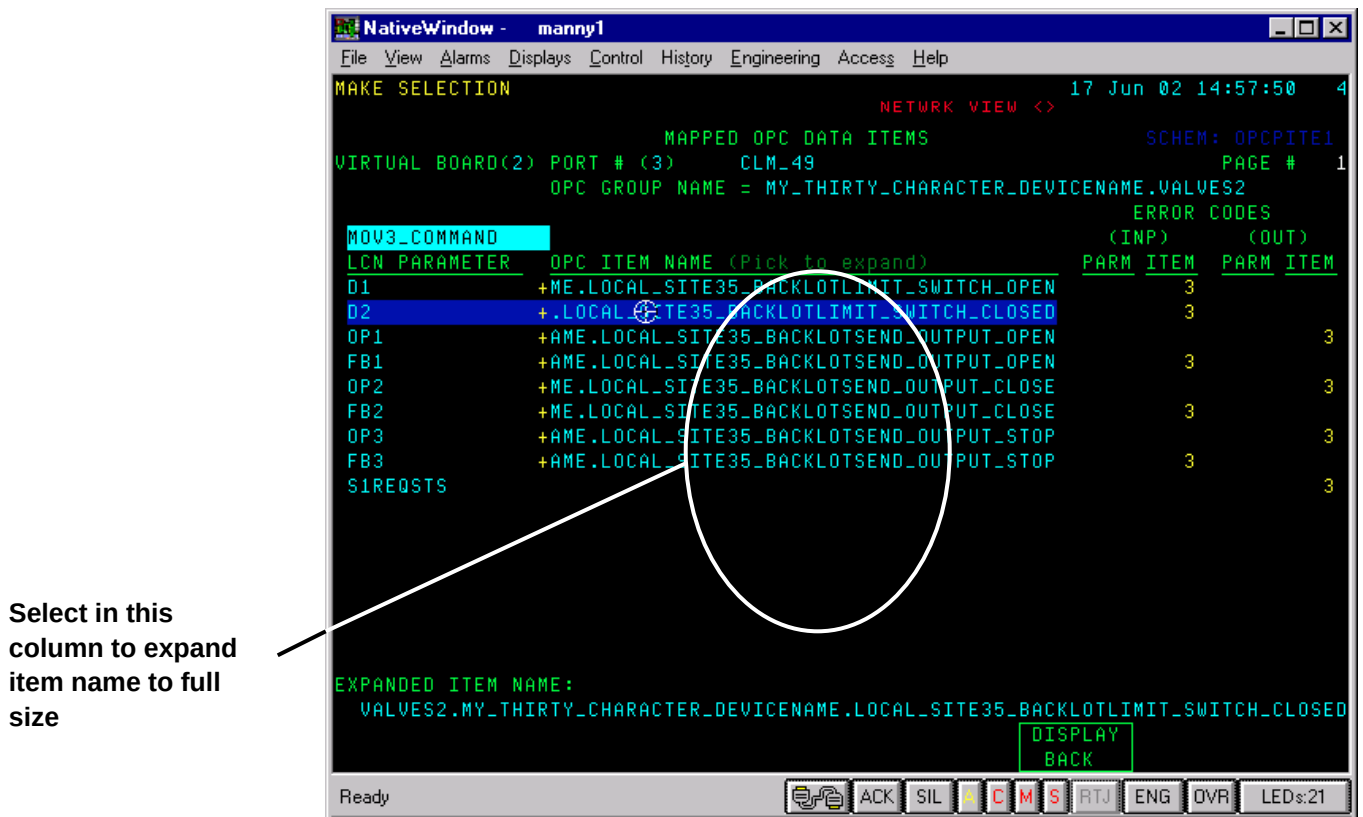


Figure 7.7-3 Mapped Data Item Display – Expanded Item Name

7.8 Understanding the Device Status and State

Status

The device status is the current application view of the communications to and from the indicated device. The status is dynamic and updated as it changes by the application. The status is the middle indication of the data representing the device on the OPC Device Detail Display.

< NAME >
< STATUS >
< STATE >

There are several status values that may appear on the specified device on the OPC Device Detail Display.

- OK** - Communications is fine. There are no errors and the device state is **In Service** by the Operator.
- P_FAIL** - Partially Failed. Communications and/or configuration is suspect. The device may or may not be communicating, but there is at least one error to report. The device state is **In Service** by the Operator.
- PRI_UNAV** - Primary Device is Unavailable. The next higher level device is unavailable due to a failure or being placed **Out of Service** by the Operator, so the current status of this device is not available. The device state is **In Service** by the Operator.
- UNKNOWN** - Unknown Status. The current status of this device is unknown because the Operator has placed this device state **Out of Service**.
This status may also occur when the device is placed **In Service** by the operator and there is no group (PIT) data assigned to the device.

STATUS
field

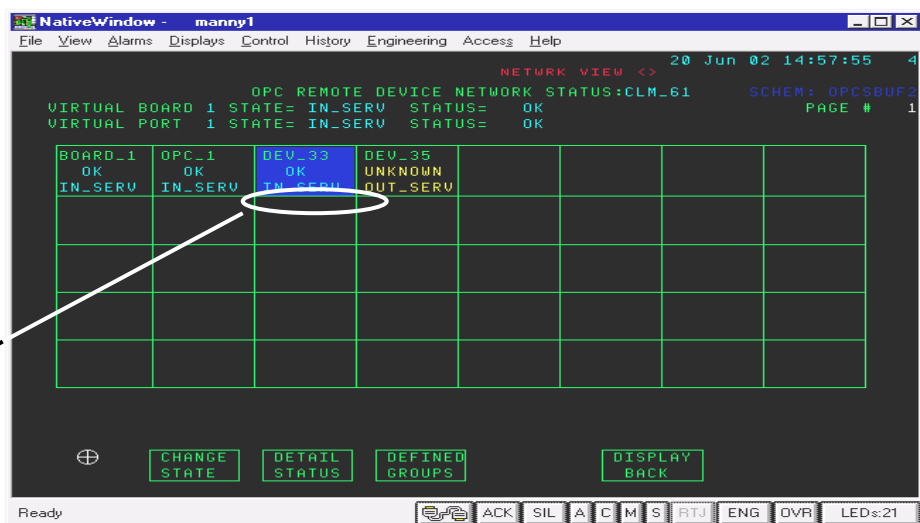


Figure 7.8-1 Device Network Status Display – Status Field

Continued on next page

7.8 Understanding the Device Status and State, Continued

State

The device operator state is the current operational state of the indicated device. This state is completely controlled by the Operator and indicated the ability to do data acquisition from the device. The operator state is the bottom indication of the data representing the device on the OPC Device Detail Display.

< NAME >
< STATUS >
< STATE >

There are several operator state values that may appear on the specified device on the OPC Device Detail Display.

IN_SERV - The device has been made available for data acquisition by Operator selection (see *Place a Device In Service*).

OUT_SERV - The device has been made unavailable for data acquisition by Operator selection (see *Place a Device Out of Service*). Data Values on mapped image point parameters may be set to BAD.

SYNC - The device has been requested to be placed In Service by the Operator and the Client is in the process of synchronizing the data between the mapped image point parameters and OPC groups and data items. When completed, the state will transition to IN_SERV.

STATE
field

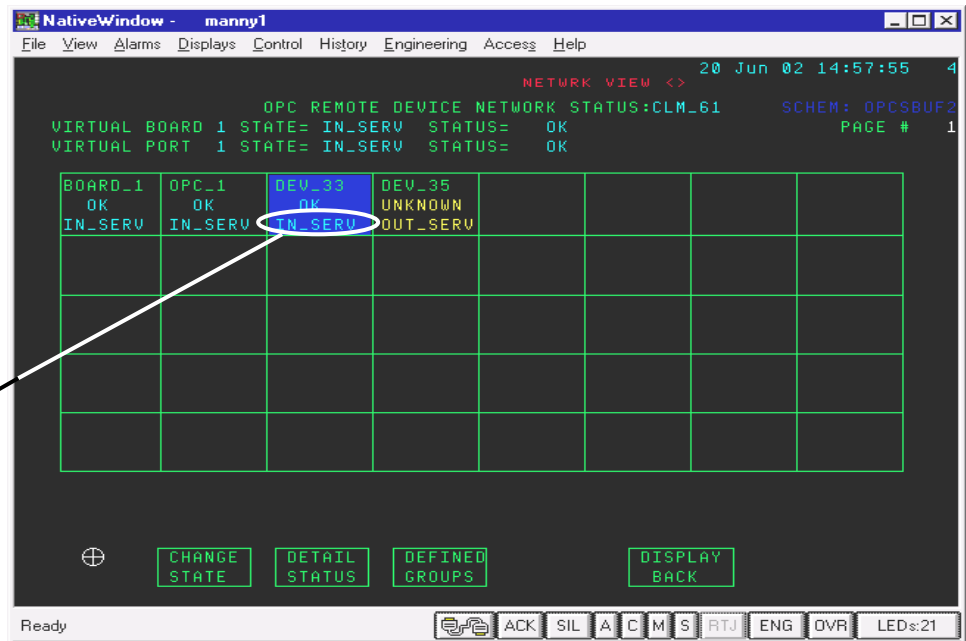


Figure 7.8-2 Device Network Status Display – State Field

Section 8 – Client Performance Expectations

8.1 Overview

The Client has been tested to establish a baseline set of performance criteria. This data is given to be the minimum reasonable performance that may be expected from the Client application on the APP/CLM node under normal operating conditions.

This section describes the test setup and results of the performance testing accomplished for the Client application.

8.2 Test Setup for Windows 2003

Server Application	Matrikon OPC Server for Modbus Devices (Modbus/TCP) - Version 3.3.1.248
LCN Release	LCN Release R630.2.1, TPS R200.1
APP Node	Refer to the APP Specification and Technical Data, TP03100B, for running Windows 2003.
Remote PLCs	Four (4) PLCs running simulator software using Ethernet communications via Modbus/TCP
Database	Writing to LCN from PLCs: 1000 - 32-bit Reals 1000 - 16-bit Integers 3000 - Booleans to Flags 3000 - Packed Integers to Flags 8000 - Total Point Count
Higher-Level Applications	The Client application was the only valid function running in the node. There was no higher level application such as CL blocks or schedule image points running during the performance test procedure. Adding higher level applications to the APP/CLM node will reduce the performance level (i.e. LCN stores/second) of the Client!

8.3 Test Setup for Windows 2008

Server Application	Matrikon OPC Server for Modbus Devices (Modbus/TCP) - Version 3.3.1.248
LCN Release	LCN Release R65X
APP Node	Refer to the APP Specification and Technical Data, TP03100B, for running Windows 2008.
Remote PLCs	Four (4) PLCs running simulator software using Ethernet communications via Modbus/TCP

Test Setup for Windows 2008, continued

Database	Writing to LCN from PLCs: 1000 - 32-bit Reals 1000 - 16-bit Integers 3000 - Booleans to Flags 3000 - Packed Integers to Flags 8000 - Total Point Count
Higher-Level Applications	The Client application was the only valid function running in the node. There was no higher level application such as CL blocks or schedule image points running during the performance test procedure. Adding higher level applications to the APP/CLM node will reduce the performance level (i.e. LCN stores/second) of the Client!

8.4 Test Physical Layout

Test Layout

The following figure shows the physical layout of the performance test configuration.

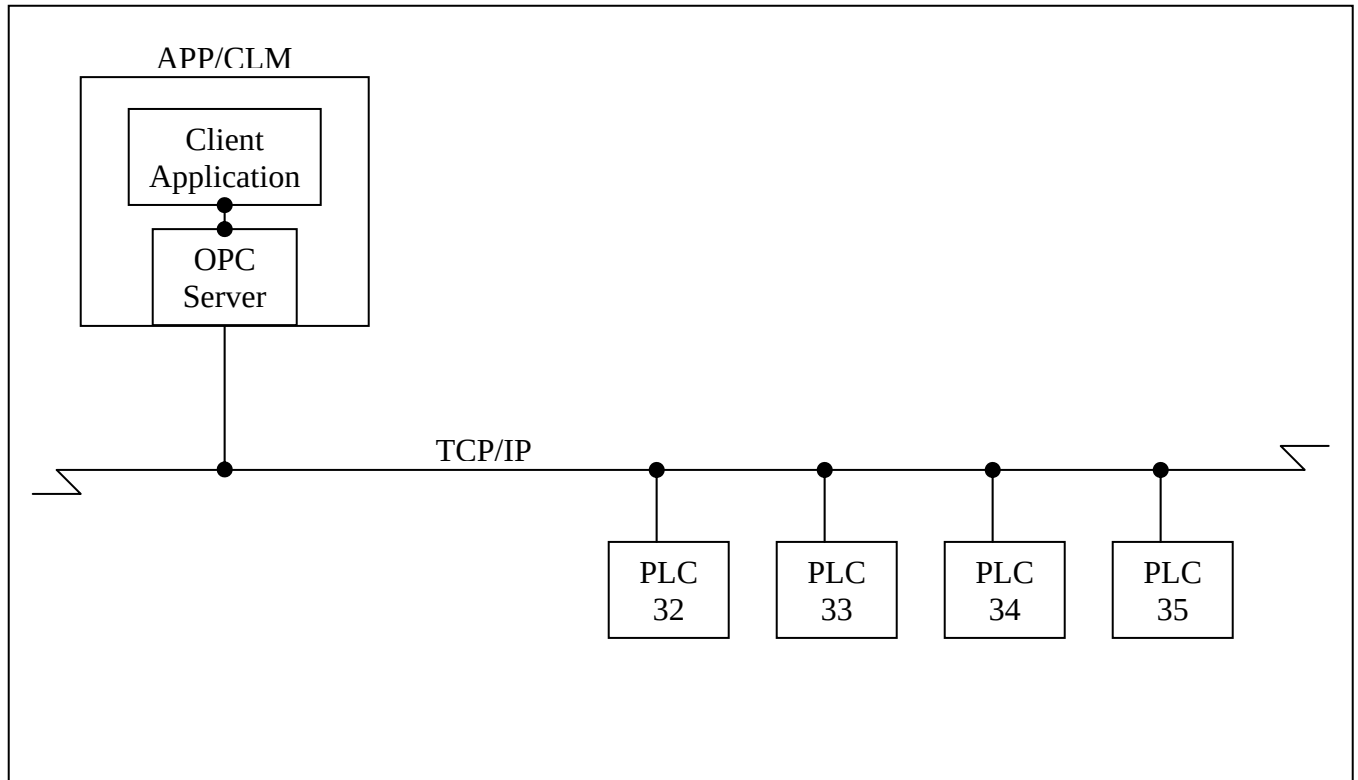


Figure 8.3-1 Performance Test Physical Layout

8.5 Test Logical Connections Layout

Test Layout

The following figure shows the logical connections layout of the performance test configuration.

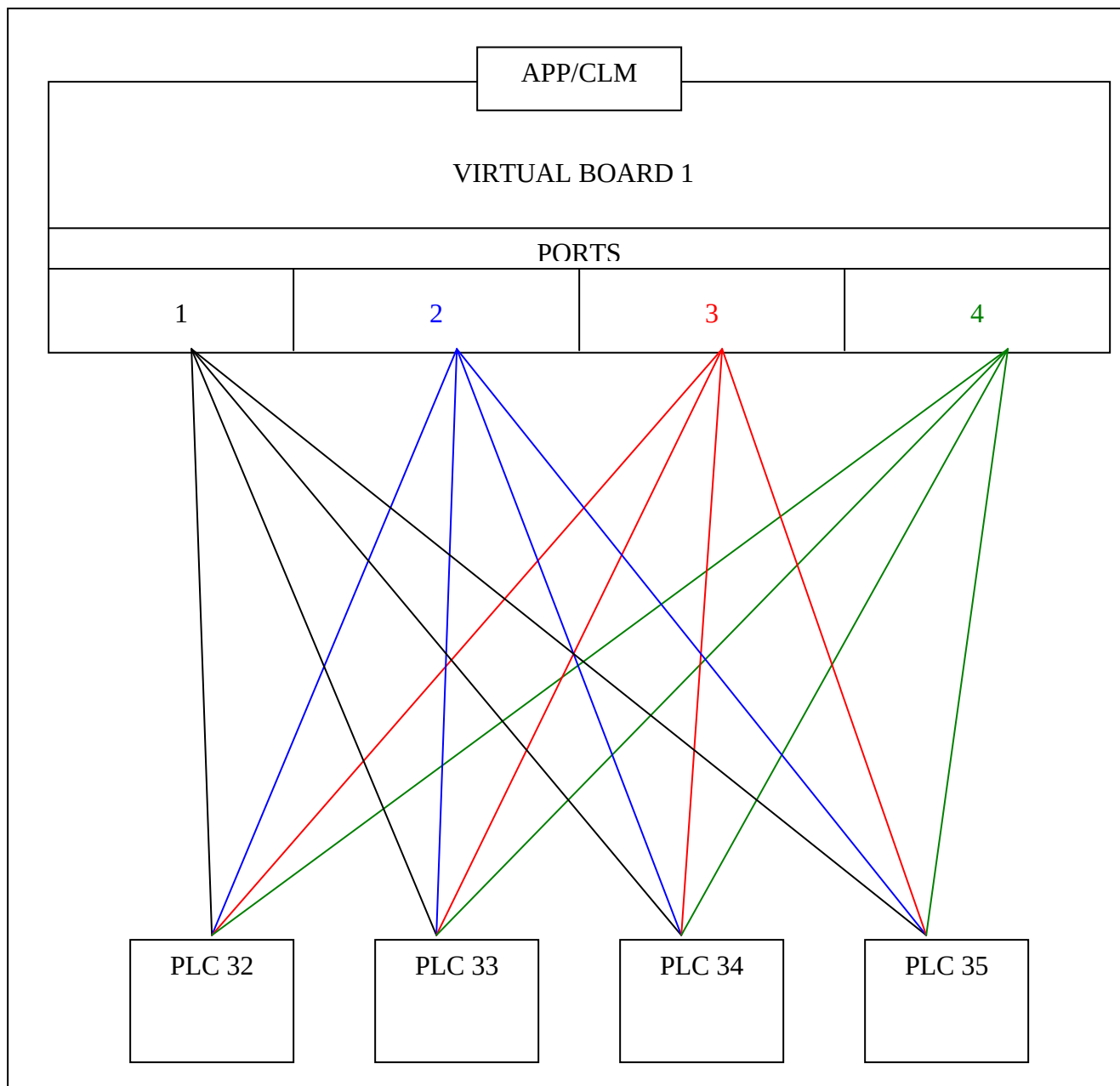


Figure 8.4-1 Performance Test Connection Layout

Each server object instance is associated with each of the 4 physical PLCs across the Ethernet. The database group and data item count is spread across the server object instances and the associated devices.

8.6 Performance Test Results

Results

The results of the performance testing are listed in this section.

Single Group

500 writes to the LCN per second on a steady-state basis.

This rate was obtained using a single configured group to bring in the mapped data items for LCN store.

This rate was seen to go as high as 600 parameter writes per second when the change rate at the PLCs was significantly increased, but sustained only on a “burst” or “upset” condition (i.e. not a steady-state rate of input).

The CPUFREE value was monitored at approximately 50%.

Multiple Groups (4)

700 writes to the LCN per second on a steady-state basis.

This rate was obtained using a four configured groups to bring in the mapped data items for LCN store.

This rate was seen to go as high as 780 parameter writes per second when the change rate at the PLCs was significantly increased, but sustained only on a “burst” or “upset” condition (i.e. not a steady-state rate of input).

The CPUFREE value was monitored at approximately 20%.

Multiple Groups - 4 or More

Adding a larger number of groups caused a large variation of performance that could not be adequately documented in terms of parameter writes per second. When the change rate is spread out among the groups, we found that the performance actually started to degrade due to the fragmentation of the messages being sent back from the application to the LCN side. These messages contained less data, but there were more of them. However, the processing horsepower required for each message remains approximately the same.

We recommend keeping the number of groups to a minimum if performance is a factor due to a high expected change rate from the Server object.

Caution!

These rates were established based on these specifically defined test conditions and meant only to assist in determination of predicted performance on given Client/Server applications. Many variables will come in to play and these performance numbers may vary, depending on the specific configuration.

Appendix A – Template Definition Files

A.1 Image Point Parameter Interface Templates

Template Files (.ZT)

All data that is retrieved from or sent to the Server object must be “mapped” into an image point parameter. This is accomplished by means of the APP/CLM Point Interface Table Entry, or PITE. The PITE is defined by a template file and used in the CLM Configuration Builder process. Each of the template files has a name with a .ZT extension. For example, the name of the file containing the Analog Input point template is O_AI.ZT. These files are normally placed in the NET>&CUS directory during the OPC Client Application installation. Refer Table 2.2-1 for the list of templates. The structure of the template file consists of a main body, or common part and contiguous individual sections for each mapped image point parameter for that particular template type (i.e. analog input).

Template File - Main Fields

Following is a definition of the fields in the main body of the PITE. The Client Analog Input point type (O_AI) is used here as an example.

Field Name	Visible	Description
ENTRY	Yes	Name of the template file, “O_AI”
TBL_TYPE	No	Defined table type (PITE)
ENT_SIZ	No	The system-generated memory size of this entry
NAME	Yes	AM image point name containing the referenced parameters
PT_ID	No	System generated: The internal entity id of the image point
PIT_NAME	Yes	The OPC group name (PIT)
DPOINT	Yes	OPC Item Name Common Prefix. This is the <i>optional</i> string data that may be appended to each of the DPAR item string values in the parameters sections of this template to create longer data item names. Unused = “_”
DPT_AD1	No	Not used.
DPT_AD2	No	Not used.
NDT_PO	No	System-generated page and offset pointer to the device entry in memory
RB_TIME	No	Not Used.
PT_OPT PPS	No	PITE options: Y = Issue a PPS command to process the AM image point whenever data is received for the mapped parameters of this point type.
NUM_KEYS	No	Number of key parameters defined in this template
NUM_PARS	No	Number of image point parameters defined in this template

A.1 Image Point Parameter Interface Templates, Continued

Parameter Addressing

The user may enter OPC Item Names and conversion strings for each PITE parameter. The conversion strings are entered as ICNV (input) and OCNV (output). Refer to Section 4.3 for details.

Template File - Parameter Section Fields

Each parameter that is “mapped” is defined with certain fields. This set of fields is repeated for each image point parameter in the template file.

Field Name	Visible	Description
Pnn_PAR	No	AM image point parameter for which this section is defined
Pnn_ID	No	System-generated internal parameter identification
Pnn_QUAL	No	System-generated internal parameter qualifier code
Pnn_DPAR	Yes	OPC Item Name to be mapped to the parameter in the above PAR field. This field is usually
Pnn_AD1	No	Not used.
Pnn_AD2	No	Not used.
Pnn_ISTA	No	System variable: State of the input value received
Pnn_IFNC:	No	<u>INPUT</u> Functions: Refer to the following individual items:
INPSCHED	Yes	Y= Only update the image point parameter when the associated PIT is scheduled or processed.
SLO	No	Y=Process only every nth cycle (based on SLO_DWN value in associated PIT)
ECP	No	Set to N. Not used.
PPS	No	Y=when image point is PPS'd, update any input data marked with INPSCHED = Y.
SYN	No	Y=input this parameter to synchronize the database when the associated device is placed IN_SERV.
CN	No	Set to N. Not used..
SB	No	Y=set this numeric parameter value BAD when the quality value returned is not valid. Does not apply to non-analog parameters.
RBI	No	Set to N. Not used.
ONDTACHG	No	Y=allow unsolicited data to be collected from the server object.
SB_OOS	Yes	Y=set this value BAD when the associated device is placed out of service. Does not apply to non-analog parameters. N=retain the last value.
Pnn_ICNV	Yes	Input parameter conversion code: Refer to Section 4.3.

Continued on next page

A.1 Image Point Parameter Interface Templates, Continued

Template Fields Repeated for each Image Point Parameter, continued

Field Name	Visible	Description												
Pnn_IDST	No	Last device access status of the input parameter: Visible on the PITE detail display: <table> <tr> <td>0</td><td>GOOD</td><td>The parameter input was successful.</td></tr> <tr> <td>1</td><td>ERROR</td><td>The parameter input was successful, but the value may be questionable.</td></tr> <tr> <td>2</td><td>FAILED</td><td>The device is either not available or, available but the parameter is not.</td></tr> <tr> <td>3</td><td>UNKNOWN</td><td>The device is unavailable.</td></tr> </table>	0	GOOD	The parameter input was successful.	1	ERROR	The parameter input was successful, but the value may be questionable.	2	FAILED	The device is either not available or, available but the parameter is not.	3	UNKNOWN	The device is unavailable.
0	GOOD	The parameter input was successful.												
1	ERROR	The parameter input was successful, but the value may be questionable.												
2	FAILED	The device is either not available or, available but the parameter is not.												
3	UNKNOWN	The device is unavailable.												
Pnn_ITVO	No	System generated: Offset from this word to the last input data value received.												
Pnn_OSTA	No	System variable: State of the output data value last sent.												
Pnn_OFNC	No	<u>OUTPUT</u> Functions: Refer to the following individual items:												
OUTSCHED	No	Y= Only output to the server object data item when the associated PIT is scheduled or processed.												
SLO	No	Y= Output only every nth cycle (based on SLO_DWN value in associated PIT)												
ECP	No	Y=exception processing: Output via operator demanded change.												
PPS	No	Y=when image point is PPS'd, output this parameter data.												
SYN	No	Y=output this parameter to synchronize the database when the associated device is placed IN_SERV.												
RB	No	Set to N. Not used.												
CBO	No	Set to N. Not used.												
Pnn_OCNV	No	Output parameter conversion code: Refer to Section 4.3.												
Pnn_ODST	No	Last device access status of the output parameter: Visible on the PITE detail display: <table> <tr> <td>0</td><td>GOOD</td><td>The parameter output was successful</td></tr> <tr> <td>1</td><td>ERROR</td><td>The parameter output was successful, but the value may be questionable</td></tr> <tr> <td>2</td><td>FAILED</td><td>The device is available but the parameter was not accepted.</td></tr> <tr> <td>3</td><td>UNKNOWN</td><td>The device is unavailable</td></tr> </table>	0	GOOD	The parameter output was successful	1	ERROR	The parameter output was successful, but the value may be questionable	2	FAILED	The device is available but the parameter was not accepted.	3	UNKNOWN	The device is unavailable
0	GOOD	The parameter output was successful												
1	ERROR	The parameter output was successful, but the value may be questionable												
2	FAILED	The device is available but the parameter was not accepted.												
3	UNKNOWN	The device is unavailable												
Pnn_OTVO	No	System-generated: Offset from this word to the output data value.												

Appendix B – Errors

B.1 Image Point Parameter Errors

Item Errors

The following error codes may be displayed on the CLM PITE Parameter Display in the Device Status columns. The x in the Values below indicates any digit from 0 to 9:

Value	Description
3	The status of the data item value is unknown because the virtual CLMI , Server object, or remote device is not in service.
1000	Bad Quality for Data Item due to an unspecified reason
102x	Bad Quality for Data Item due to a configuration error; the OPC Server could not locate the Data Item.
102x	Bad Quality for Data Item due to the item not being connected to a value
103x	Bad Quality for Data Item due to a device failure
104x	Bad Quality for Data Item due to a sensor failure
105x	Bad Quality for Data Item due to lost communications with device; the value is the last known good value
106x	Bad Quality for Data Item due to lost communications with device; there is NO last known good value.
107x	Bad Quality for Data Item due to item being Out of Service; the Item is Inactive
2000	Uncertain Quality for Data Item due to an unspecified reason
201x	Uncertain Quality for Data Item because the value is stale; the value is the last known good value.
204x	Uncertain Quality for Data Item due to a sensor not being accurate; sensor needs to be calibrated
205x	Uncertain Quality for Data Item because value exceeds Engineering Unit limits.
206x	Uncertain Quality for Data Item that is Sub-Normal; value is derived from multiple sources and not all of the sources is good.

Continued on next page

B.1 Image Point Parameter Errors, Continued

Value	Description
4001	conversion code in PITE Parameter invalid
4002	conversion code for PITE Parameter input invalid (displayed in Output Status column)
4003	Conversion error on input value
4004	Invalid CLM database configuration detected
4005	CLM NDT entry is invalid or missing
4006	OPC Item ID syntactically invalid
4007	OPC Item ID not in server address space
4008	OPC Item has an invalid data type
4009	OPC Item access path not known by OPC Server
4010	OPC Item was stored but clamped
4011	OPC Item value out of range
4012	OPC Item can not be written
4013	OPC Item no longer available in OPC Server
4014	OPC Server does not conform to the OPC Data Access Specification. Look in the NT Event Log for information on specific features that are non-conforming. Look for events in the Application Event Log with <i>Source</i> of “CLMOPC” and <i>Category</i> of “System Error”
4015	OPC Unable to Add Item to OPC Server. Look in the NT Event Log for more information. Look for events in the Application Event Log with <i>Source</i> of “CLMOPC” and <i>Category</i> of “System Error”
4016	OPC Conversion code not supported
4050	Unknown OPC data type
4051	Unknown LCN data type
4098	Device has been placed out of service
4099	Unknown error

B.2 Windows 2003/2008 Event Log Messages

Windows 2003/2008 Event Log Messages

When problems are detected by the CLM OPC Client Application, it logs messages describing the problem to the Windows 2003 and Windows 2008 Application Event Log. The Event Logs may be viewed using the *Event Viewer* located under the *Administrative Tools (Common)* menu. Following are messages logged by the APP/CLM OPC Client Application:

Message ID	Description
33024	OPC Server <server_name> Started Successfully. Indicates that the connection from the CLM OPC Client Application to the OPC Server with a Prog-Id of <server_name> was successful.
33280	Could not connect to a Mailbox. Indicates that the CLM OPC Client Application has an internal communications problem. If the CLM is configured correctly, taking the emulated CLMI board associated with the OPC Application OUT_SERV and then back IN_SERV may correct this problem. If this fails, contact Honeywell Technical Assistance Center.
33282	An Unrecognized message was received CLM OPC Client Application. This is not unusual and normally causes no problems.
33536	Failed Reading from MailBox. Indicates that the CLM OPC Client Application has an internal communications problem. This may be a temporary problem that corrects itself.
33537	Failed Creating MailBox. Indicates that the CLM OPC Client Application has an internal communications problem. If the CLM is configured correctly, taking the emulated CLMI board associated with the OPC Application OUT_SERV and then back IN_SERV may correct this problem. If this fails, contact Honeywell Technical Assistance Center.
33538	Could not Connect to CLM Message Router MailBox. Indicates that the CLM OPC Client Application could not connect with the CLM Platform. Taking the Port associated with the OPC Application OUT_SERV and then back IN_SERV should correct this problem if the CLM is configured correctly. If not, contact Honeywell Technical Assistance Center.
33539	CLM Network Memory Database Error. Problem with the CLM database. This may require re-building the CLM database using the CLM Builder.
33540	The OPC Interface, <opc_interface>, is not supported by the OPC Server. The OPC Server must support OPC Data Access 2.0.
33541	The OPC Server could not establish a Data Callback connection to the CLM OPC Client for group "<pit_name>". The OPC Server must support OPC Data Access 2.0 and, the OPC Data Access 2.0 version of OPCProxy.dll must be registered on this system.

Continued on next page

B.2 Windows 2003/2008 Event Log Messages, Continued

Message ID	Description
33542	Attempt to add Items (PITE Parameters) to OPC Server FAILED. The error value displayed is defined in the Microsoft file winerr.h. It is displayed in decimal but defined in hexadecimal in winerr.h. Each PITE Parameter defined in a CLM OPC Device has a corresponding Item created for it in the OPC Server.
33543	Attempt to add a Group "<pit_name>" to the OPC Server FAILED. The error value displayed is defined in the Microsoft file winerr.h. It is displayed in decimal but defined in hexadecimal in winerr.h. Each PIT defined in a CLM OPC Device has a corresponding Group created for it in the OPC Server.
33544	Quality Key on Port NDT is Missing. The OPC Quality key contains information on how to handle Bad/Uncertain Quality on OPC Data Items. It is typically the 2 nd key on the Port NDT.
33545	The Requested Data Type <VT_Type> for Item <OPC_Data_Item> is not supported by this OPC Server. Indicates that the data type, <VT_Type>, of an OPC Data Item is not a type supported by the OPC Server. This may not be a problem since the OPC Client Application can automatically handle the conversion between many data types. However, it may be necessary or desirable to modify the PITE conversion codes to specify OPC data types supported by the OPC Server.
33546	Array Items Requested for <OPC_Data_Item> are not supported by this OPC Server. Indicates that the OPC Server does not support arrays of the data type specified for this OPC Data Item.
33547	An OLE Exception has been caught. The error value displayed is defined either in the Microsoft file winerr.h or in the <i>OPC Data Access Custom Interface Specification, 2.0</i> . This exception was almost certainly generated by the OPC Server.
33548	An unknown Exception has been caught. This exception may have been generated by the OPC Server or by the CLM OPC Client Application.
33549	Error Occurred. This is a generic error message. It should contain additional information that describes the error that occurred.

Appendix C – Application Registry Entries

C.1 HKEY_LOCAL_USER

The following Windows Registry entries are required for the APP/CLM OPC Application. They should be initialized during execution of the installation script.

Key	Value	Type	Initial Value
HKEY_LOCAL_MACHINE/Software/ Honeywell/ TDC Emulators/ TdcBoard0/CLM	EventLevel	REG_DWORD	<unchanged>
	SlotNumber	REG_DWORD	<unchanged>
	Start	REG_DWORD	<unchanged>
	ThreadDll	REG_SZ	<unchanged>
	ApplDll	REG_SZ	<concatenate> clmopc.dll
	ApplName	REG_SZ	<concatenate> OPC
	ApplServer	REG_SZ	<concatenate>0
HKEY_LOCAL_MACHINE/Software/ Honeywell/ SourceName/ClmOPC	SubSystemID	REG_SZ	093
	ApplicationVersion	REG_SZ	0.0
HKEY_LOCAL_MACHINE/System/ CurrentControlSet/ Services/EventLog/ Application/ClmOPC	CategoryCount	REG_DWORD	0x08
	TypesSupported	REG_DWORD	0x07
	CategoryMessageFile	REG_SZ	\Program Files\ Honeywell\TPS\ Emulators\ hsicattbl.dll
	EventMessageFile	REG_SZ	\Program Files\ Honeywell\TPS\ Clm\opcmsg.dll
HKEY_LOCAL_MACHINE/Software/ Honeywell/ TDC Emulators/ TdcBoard0/CLM/ClmApp1	ApplName	REG_SZ	OPC
	EventLevel	REG_DWORD	0x00000003
	Init	REG_DWORD	0x10000000
	RemoteServerName1	REG_SZ	<blank>
	RemoteServerName2	REG_SZ	<blank>
	RemoteServerName3	REG_SZ	<blank>
	RemoteServerName4	REG_SZ	<blank>
	ServerProgID1	REG_SZ	<blank>
	ServerProgID2	REG_SZ	<blank>
	ServerProgID3	REG_SZ	<blank>
	ServerProgID4	REG_SZ	<blank>
	Start	REG_DWORD	0x00000001
	ThreadDll	REG_SZ	clmopc.dll

Appendix D – Windows 2003/2008 Application Files

D.1 Application Files Added during the Install Script

The following components are added to the hard disk for the APP/CLM OPC Application. The Registration column indicates whether the component needs to be registered as a COM Server. Components that have an .exe suffix on their name are typically self-registering. Components that have a .dll suffix, must be registered using the *regsvr32* program. Most self-registering components register when they are invoked with no arguments. If arguments are required to self-register, the arguments will be shown in the Registration column.

Note: The following components need not be registered manually for CLM R200.1.

Component	Description	Installation Location	Registration
clmopc.dll	The OPC Client application – board 1	\Program Files\Honeywell\TPS\Emulators	No
clmopc2.dll	The OPC Client application - board 2	\Program Files\Honeywell\TPS\Emulators	No
opcmsg.dll	OPC Event Message .dll	\Program Files\Honeywell\TPS\CLM	No
opcproxy.dll	Com Proxy file for OPC Data Access 2.0 (should normally come with OPC Server)	\Windows\System32 \Windows\SysWOW64	Yes
opccomn_ps.dll	COM Proxy file for OPC IOPCCommon Interface. (should normally come with OPC Server)	\WINNT\system32	Yes
opcenum.exe	The OPCEnum.exe provided by the OPC Foundation supports the IOPCServerList interface via DCOM to allow clients to determine available OPC servers on remote machines. Register via “ <i>opcenum /regserver</i> ”. (should normally come with OPC Server)	\WINNT\system32	Yes

Appendix E – UNINSTALL

E.1 OPC Client – Windows 2003 Uninstall

Shutdown TDC Emulators Service

In the Window 2003 environment, the Honeywell TDC Emulators Service emulates the peripheral boards that exist in a traditional TPN node chassis. In the APP/CLM, this service emulates a virtual communications board called the CLMI. Before the uninstall of the Client application may proceed, this emulation service must be stopped.

Step	Action
E.1.1	On the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
E.1.2	From the Start menu select Control Panel and Administrative Tools.
E.1.3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure E.1-1 below.
E.1.4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Stop button.
E.1.5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message “<i>Could not stop TDC Emulators Service service</i>”. When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcaserver.exe process is no longer in the active process list.

E.1 OPC Client - Windows 2003 Uninstall, Continued

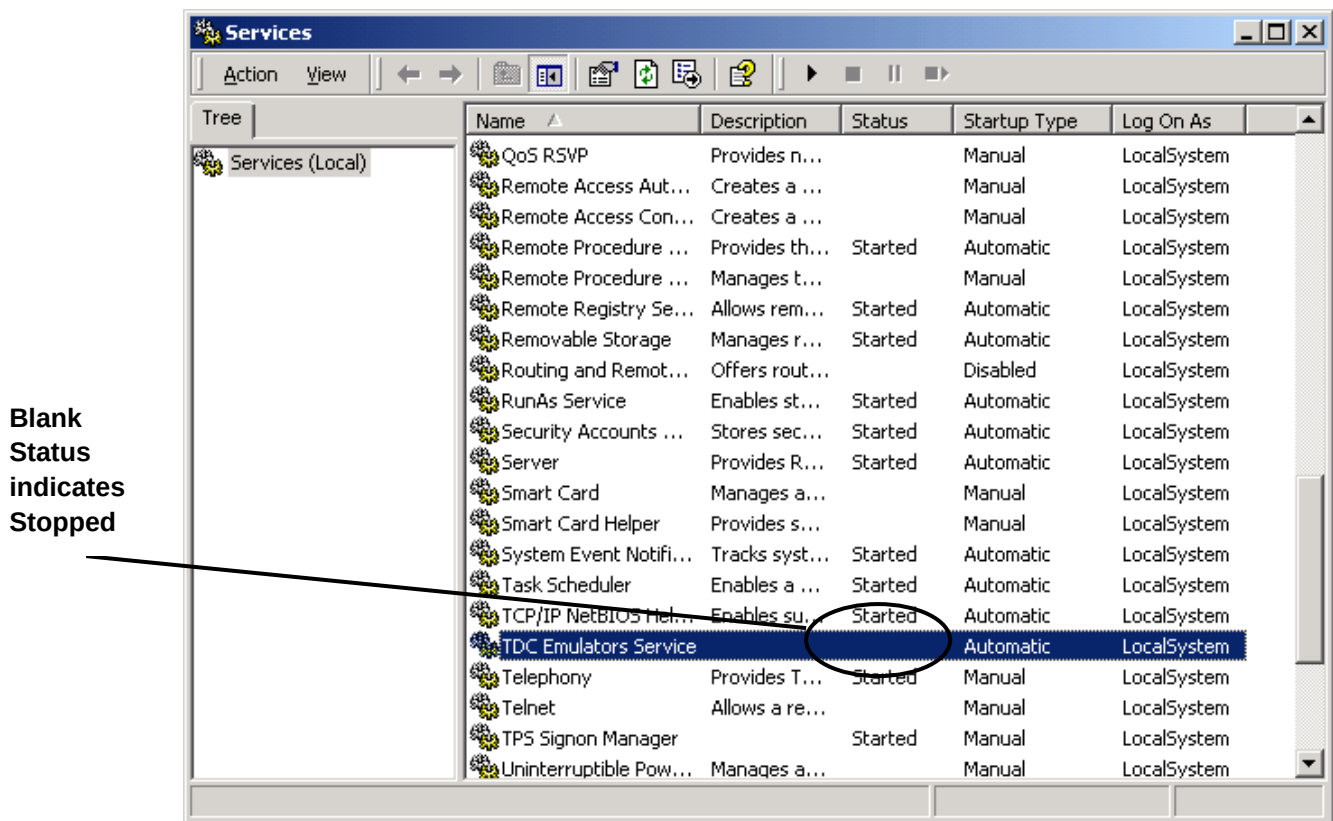


Figure E.1-1 TDC Emulators Service Stopped

Uninstall OPC Client Application

Now we are ready to uninstall the Client application from the TPS APP (Windows side of the APP/CLM).

Step	Action
E.1.6	If not logged in already on the TPS APP (Windows side of the APP/CLM), log onto Windows as Administrator.
E.1.7	Using Windows Explorer, go to the directory C:\Program Files\Honeywell\TPS\CLM
E.1.8	The listing of files there should include CLM OPC Uninstall.exe, which was placed there from the original installation process.
E.1.9	Double-Click on this file to initiate the uninstall script.
E.1.10	Follow the prompts to completion. Note: The uninstall process will not remove the CLM OPC Uninstall.exe file from the hard drive. You may remove this as you wish.

E.2 OPC Client – Windows 2008 Uninstall

Shutdown TDC Emulators Service

In the Window 2008 environment, the Honeywell TDC Emulators Service emulates the peripheral boards that exist in a traditional TPN node chassis. In the APP/CLM, this service emulates a virtual communications board called the CLMI. Before the uninstall of the Client application may proceed, this emulation service must be stopped.

Step	Action
E.2.1	On the Experion APP (Windows side of the APP/CLM), log onto Windows as Administrator.
E.2.2	From the Start menu select Control Panel and Administrative Tools.
E.2.3	Select Services and in the Services window, scroll down until the TDC Emulators Service appears. See Figure E.2-1 below.
E.2.4	If the Status for TDC Emulators Service is Started , select the TDC Emulators Service entry. From the drop-down menu, select Action, then the Stop button.
E.2.5	A message box appears indicating that the service is being stopped. This will take about 30 seconds. Note: Occasionally, the TDC Emulators Service does not stop in the required time and an Error Message box appears with the message “<i>Could not stop TDC Emulators Service service</i>”. When this happens, press the OK button in the message box. The TDC Emulators Service will normally stop within 10 additional seconds. This can be verified by pressing the Ctrl, Alt and Del keys simultaneously and invoking the Task Manager. From the Task Manager, view the Process tab and verify that the tdcaserver.exe process is no longer in the active process list.

E.2 OPC Client - Windows 2008 Uninstall, Continued

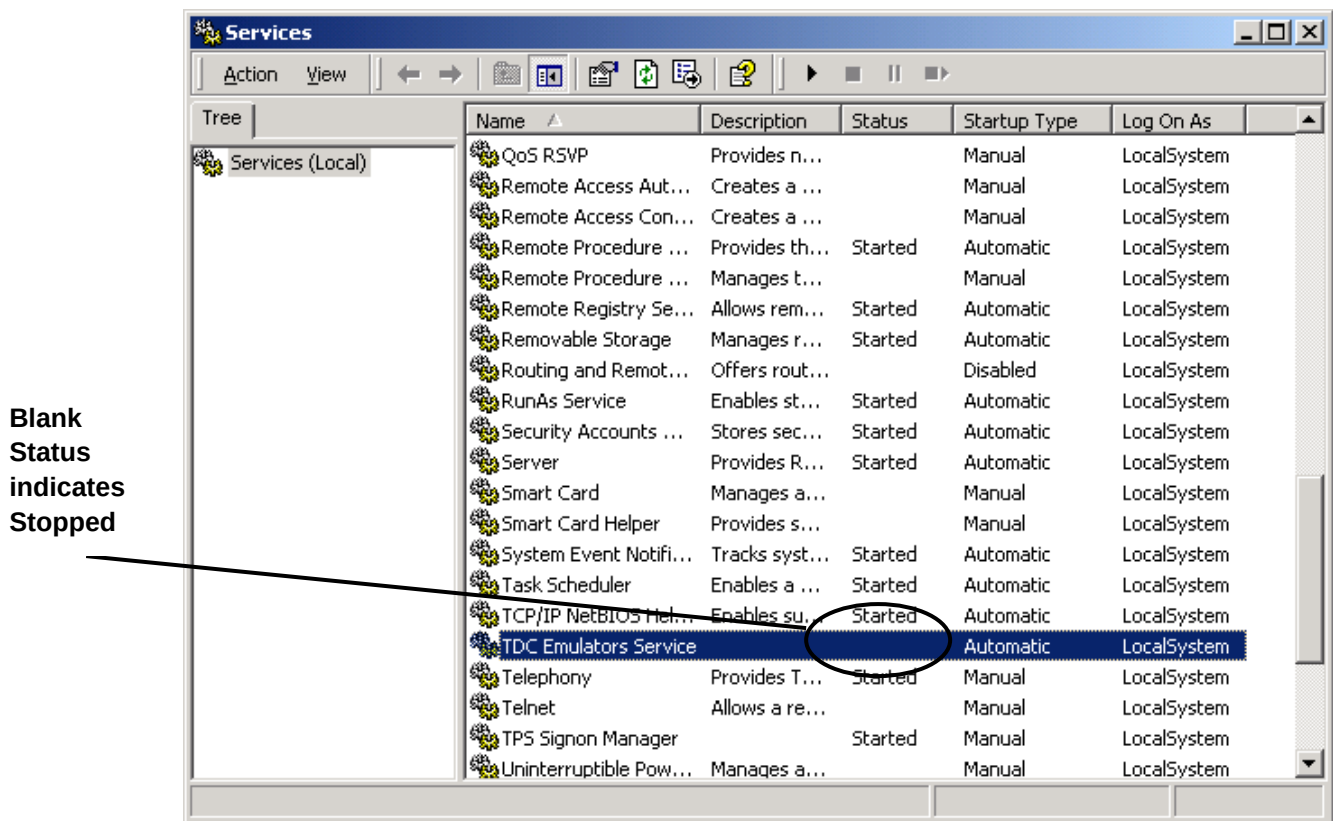


Figure E.2-1 TDC Emulators Service Stopped

Uninstall OPC Client Application

Now we are ready to uninstall the Client application from the Experion APP (Windows side of the APP/CLM).

Step	Action
E.2.6	Click Start and select Control Panel.
E.2.7	Click Uninstall a Program. The Uninstall and Change programs window opens up.
E.2.8	In the currently installed programs list, identify the program and Click Uninstall/Change.
E.2.9	Follow the on-screen messages to complete the installation procedure.

Appendix F – Servers that have Connected to the Client

F.1 List of Servers

Server Connection

Table F.1-1 below lists some, but not necessarily all of the servers that have been successfully connected to the Client.

Important! This Client/Server connection has been made either as a requirement to a project-specific need or in a development testing environment. The third-party Server is not fully certified unless the box is checked to indicate it. If this is the case, then a detailed report will be available that describes the testing accomplished, the specific functionality tested, the results of that testing and anomalies encountered. Contact your Honeywell representative for information on this certification service.

Table F.1-1 Connected Server List

Third-Party Server Name	Certified?
Matrikon OPC Server for Modicon Devices	No
Honeywell Hi-Spec ShadowPlant Server	No

F.2 Special Considerations for Connected Servers

While configuring the specific servers to communicate to the Client, certain situations or conditions were encountered that we considered relevant to document. This information is meant only to assist in the configuration of the Client to these specific servers.

F.2.1 Matrikon OPC Server for Modicon Devices

Modbus RTU

At this time, we have connected to the Matrikon server using the Modbus RTU protocol option.

Non-Supported Data Types

At this time, Matrikon does not support the following data types for the Modbus RTU option in the server.

- Strings (VT_BSTR)
- 32-Bit Unsigned Integers (VT_UI4)

Server Data Item initialization Lockout can cause long Synch Times

The Client synchronizes all of the mapped data items when the OPC device is placed in service. This means that a fresh status and value is brought up and stored into the mapped LCN parameter for each configured data item. After synchronization is complete, the Client enables subscription. At this point, the server sends all of the data items up with a bad quality status, forcing the Client to set all mapped parameters to bad values. The server then unlocks the values and sends them up with a good quality status, causing the Client to store all of the mapped parameters with the good values. This causes the entire synchronization process to do three full reads of the mapped item database when one read would have been sufficient.

F.2.2 Honeywell Hi-Spec ShadowPlant Server

**Interactive User
Required for DCOM**

Due to this server being started prior to the Client making the connection, a change was required in the DCOM configuration.

Step	Action
1.	From the START menu, select the RUN function.
2.	At the command line, enter DCOMCNFG.
3.	When the Applications tab appears, scroll down and select the ShadowPlant Server.
4.	Select the Properties tab at the bottom of the screen.
5.	From the Properties display, select the Identity tab.
6.	Select the radio button for The Interactive User and press OK.
7.	Select OK to exit from the Properties display.

Appendix G – When Do I need to Restart a Device?

G.1 CLMI Virtual Board

Restarting a server object	All server objects are controlled at the virtual board, or CLMI, level. This means that the server objects are established whenever the CLMI is placed logically In Service by the operator. If a server application has failed or has been shut down, the CLMI virtual board must be taken Out of Service and placed back In Service to re-establish a connection to that server object.
Adding a new server object	Anytime you add a new server object to the virtual board (via the CLM Configuration Builder), the CLMI virtual board must be taken Out of Service and placed back In Service to connect to the new server object.
All CLM Configuration Utility changes	Any changes made through the CLM Configuration Utility constitute a virtual board level change and requires a restart of the board.
Client/Server Communication Loss	A failure of communications between the Client and the Server object may require a restart of the CLMI virtual board in some instances. If the CLM Status Display shows the CLMI status/state as FAILED/FAILED , then the CLMI must be taken Out of Service and placed back In Service to restore communications.

G.2 Server Object (Port)

Re-synch all Devices on this Server object	Generally, the only time you will need to restart the server object (port) is when you want to re-synch all of the devices assigned to that particular server object. This will have the effect of re-establishing all groups and data items for each defined OPC device.
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G.3 OPC Device

Re-Synch all Groups and Data Items	You may re-synchronize all OPC groups and data items assigned through this OPC device by placing the device Out of Service and back In Service .
CLM Builder Changes	Any changes made to the device through the CLM Configuration Builder should require a restart of the specific device.

READER COMMENTS

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